



AKTU



B. Tech II-Year

Digital Electronics

Unit-2 in One Shot

Pdf Notes + AKTU PYQs + Imp. Questions



By Swati Ma'am



AKTU B.Tech II-Year

Subject : DIGITAL ELECTRONICS



Unit-2 : Syllabus

Combinational Logic: MSI devices like Magnitude comparator, Multiplexers, Demultiplexers, Decoders, Encoders. Multiplexed display, half and full adders, subtractors, serial and parallel adders, BCD adder

Combinational Logic : MSI devices

MSI (**Medium-Scale Integration**) devices in digital electronics are integrated circuits (ICs) containing a moderate number (around 100 to 10,000) of components.

Integration Level	Full Form of Integration Level	Typical components (Gate /Transistor)Count
SSI	Small –Scale Integration	Upto 100 (approx. 1-12 Gates)
✓ MSI	Medium Scale Integration	100 to 10,000 (approx. 13-99 Gates)
LSI	Large Scale Integration	10,000 to 100,000 (approx. 100-9999 Gates)
VLSI	^{Large} Very Scale Integration	100,000 to millions (approx. 10,000+ Gates)

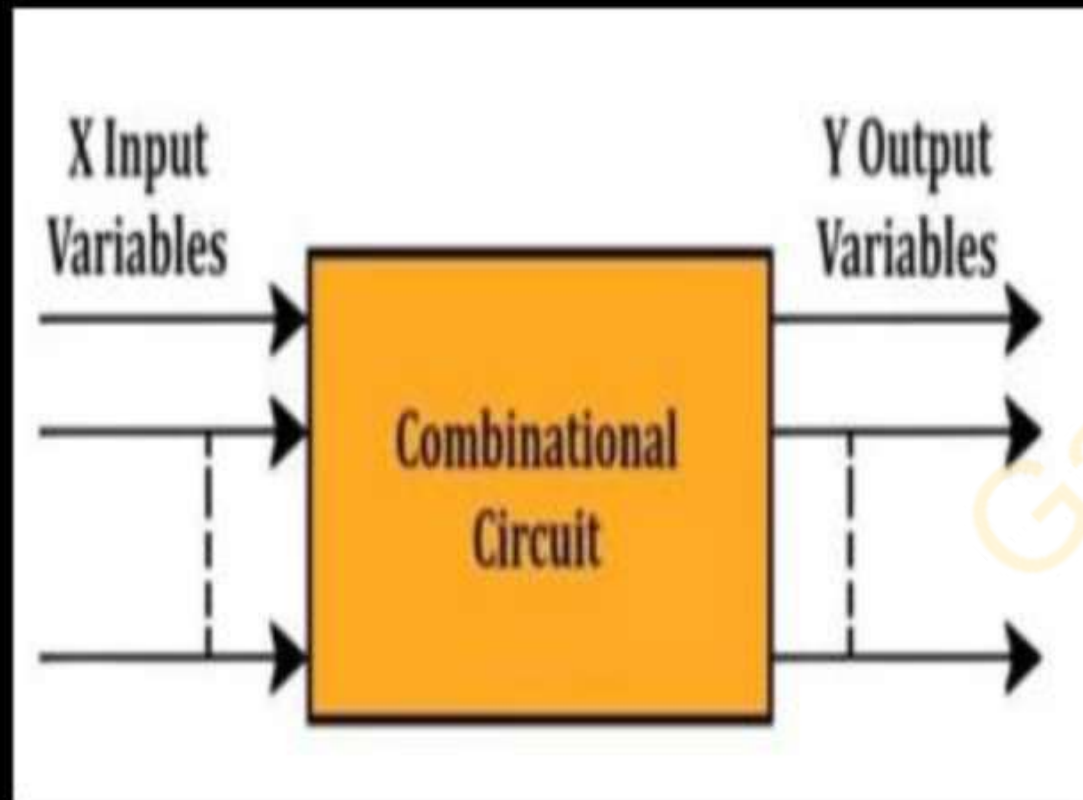
Examples :- Comparators, Adders, Encoders, Decoders, Multiplexers (Mux), Demultiplexers (Demux) ,Registers and Counters.

Digital Circuits

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Combinational Circuits

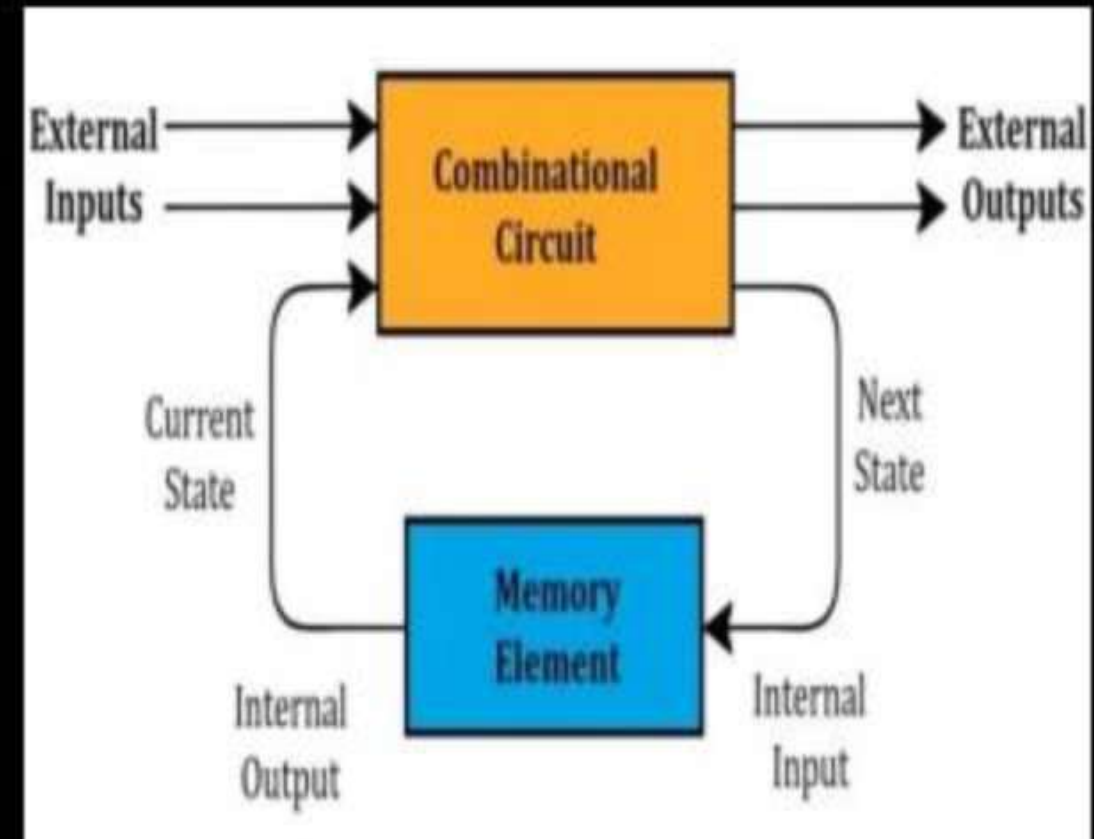
These circuits do not have memory i.e output of these circuits at any instant of time , depends only on present inputs.



Sequential Circuits

These circuits have memory i.e output of these circuits at any instant of time , depends on both current inputs and past states (previous outputs).

These can be achieved by incorporating memory elements like flip-flops and latches into a feedback loop with combinational logic.



Combinational Circuits

Examples –

Adders

Subtractors

Multiplexers

Demultiplexers

Encoder

Decoder etc.

Sequential Circuits

Examples –

Counters

Shift Registers

FSMs(Finite State Machine) etc.

Gateway Classes

Combinational Circuits

ANALYSIS

Logic Diagram (given)

Boolean Function for output of each gate

Obtain Boolean Expressions for all gates

Write Boolean Expressions for Final Output

Make Truth Table

DESIGN

Word Problem (given)

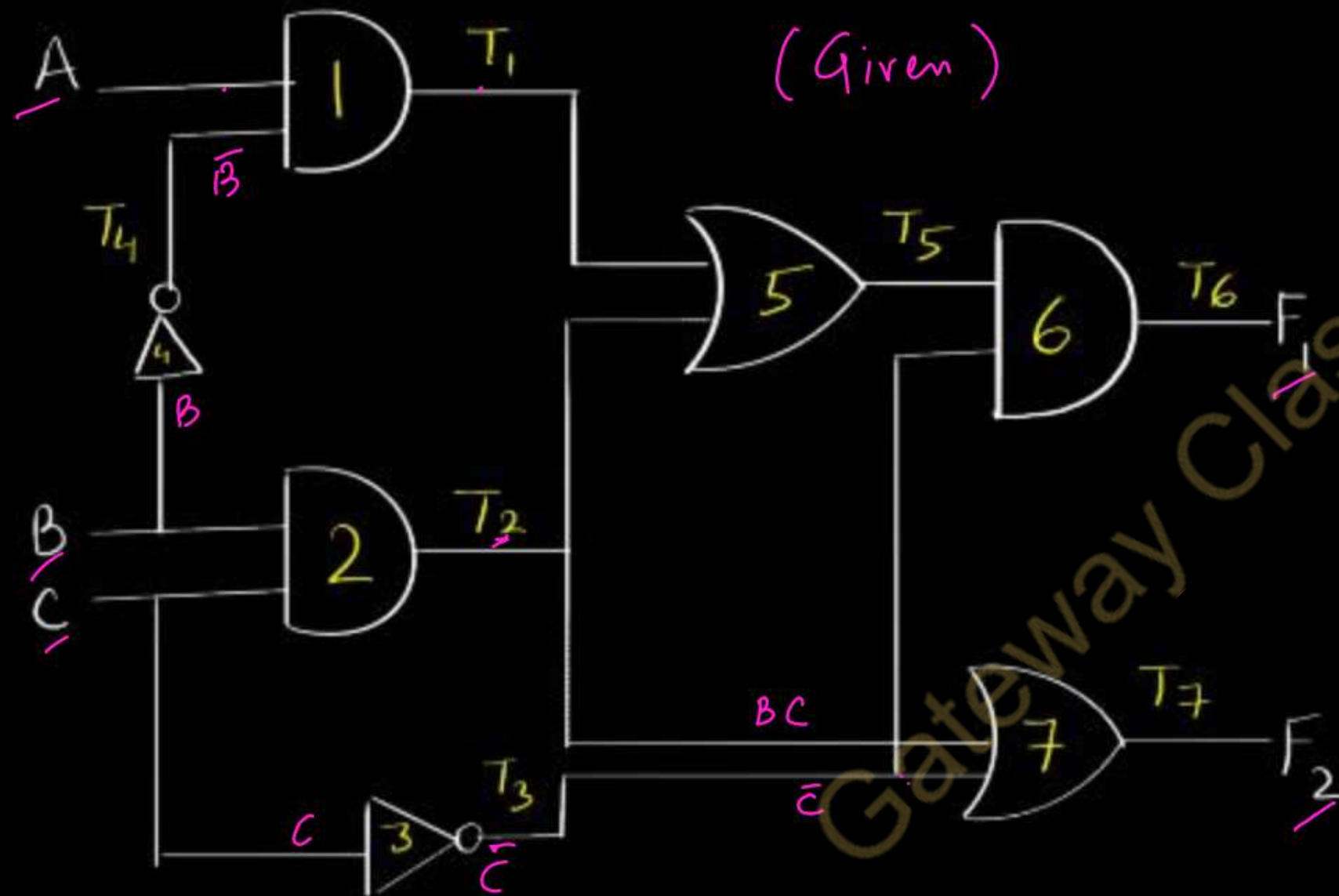
Assign Inputs and Outputs

Make Truth Table

Use K-Map & find expression for outputs.

Draw the Logic Diagram.

Que- Analyze the combinational circuit



Solⁿ

$$T_1 = A\bar{B} \quad \checkmark$$

$$T_2 = BC \quad \checkmark$$

$$T_3 = \bar{C} \quad \checkmark$$

$$T_4 = \bar{B} \quad \checkmark$$

$$T_5 = A\bar{B} + BC$$

$$\begin{aligned} T_6 = T_5 \cdot \bar{C} &= (A\bar{B} + BC)\bar{C} \\ &= A\bar{B}\bar{C} + \underbrace{BCC}_{0} \\ &= A\bar{B}\bar{C} \end{aligned}$$

$$T_7 = BC + \bar{C}$$

$$F_1 = T_6 = A\bar{B}\bar{C}$$

$$F_2 = T_7 = BC + \bar{C}$$

$$T_1 = A\bar{B}$$

$$T_2 = BC$$

$$T_3 = \bar{C}$$

$$T_4 = \bar{B}$$

$$T_5 = A\bar{B} + BC$$

$$\begin{aligned} T_6 = T_5 \cdot \bar{C} &= (A\bar{B} + BC)\bar{C} \\ &= A\bar{B}\bar{C} + \underbrace{BC\bar{C}}_{=0} \\ &= A\bar{B}\bar{C} \end{aligned}$$

$$T_7 = BC + \bar{C}$$

$$F_1 = T_6 = A\bar{B}\bar{C}$$

$$F_2 = T_7 = BC + \bar{C}$$

Truth Table

Inputs			Outputs						
⁴ A	² B	¹ C	¹ F ₁	² F ₂	¹ T ₁	T ₂	T ₃	T ₄	T ₅
0	0	0	0	1	0	0	1	1	0
0	0	1	0	0	0	0	0	1	0
0	1	0	0	1	0	0	1	0	0
0	1	1	0	1	0	1	0	0	1
1	0	0	1	1	1	0	1	1	1
1	0	1	0	0	1	0	0	1	1
1	1	0	0	1	0	0	1	0	0
1	1	1	0	1	0	1	0	0	1

Ans.

Que- Design a combinational circuit which has three inputs and one output, and the output is 1 if binary input number ≤ 3 .

Solⁿ Step 1 Three inputs A, B, C
Output Y

Step 2 Truth Table

	⁴ A	² B	¹ C	Y
0	0	0	0	1
1	0	0	1	1
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	0
7	1	1	1	0

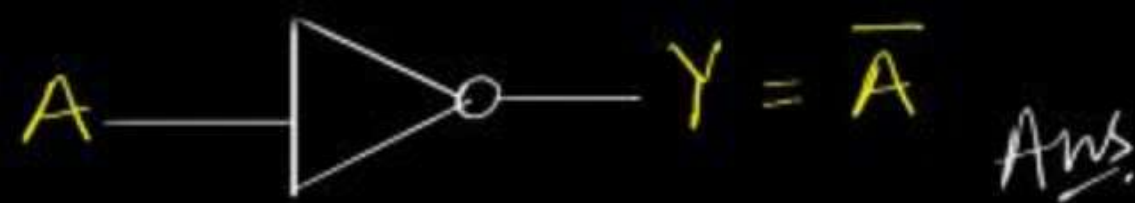
Step 3 K-Map for Y ^{output}

AB \ C	00	01	11	10
0	1	1		
1	1	1		

$\bar{A}$

$$\therefore Y = \bar{A}$$

Step 4 Logic Diagram



Que- The input to combinational logic circuit is a 4 bit binary number. Design the logic circuit with two outputs Y_1 & Y_2 for the following condition :

(i) $Y_1 = 1$ if the input binary number ≤ 5

(ii) $Y_2 = 0$ if input binary number ≥ 9 .

Step 1:- Inputs A, B, C, D Outputs Y_1, Y_2

Step 2:- Truth Table

	A	B	C	D	Y_1	Y_2
0	0	0	0	0	1	1
1	0	0	0	1	1	1
2	0	0	1	0	1	1
3	0	0	1	1	1	1
4	0	1	0	0	1	1
5	0	1	0	1	1	1
6	0	1	1	0	0	1
7	0	1	1	1	0	1
8	1	0	0	0	0	1
9	1	0	0	1	0	0
10	1	0	1	0	0	0
11	1	0	1	1	0	0
12	1	1	0	0	0	0
13	1	1	0	1	0	0
14	1	1	1	0	0	0
15	1	1	1	1	0	0

Step 3

K-Map for Y_1

AB \ CD	00	01	11	10
00	1	1		
01	1	1		
11	1			
10	1			

Groupings: $\bar{A}\bar{C}$ (top-left 2x2), $\bar{A}\bar{B}$ (left column)

$$Y_1 = \bar{A}\bar{B} + \bar{A}\bar{C}$$

K-Map for Y_2

AB \ CD	00	01	11	10
00	1	1		1
01	1	1		
11	1	1		
10	1	1		

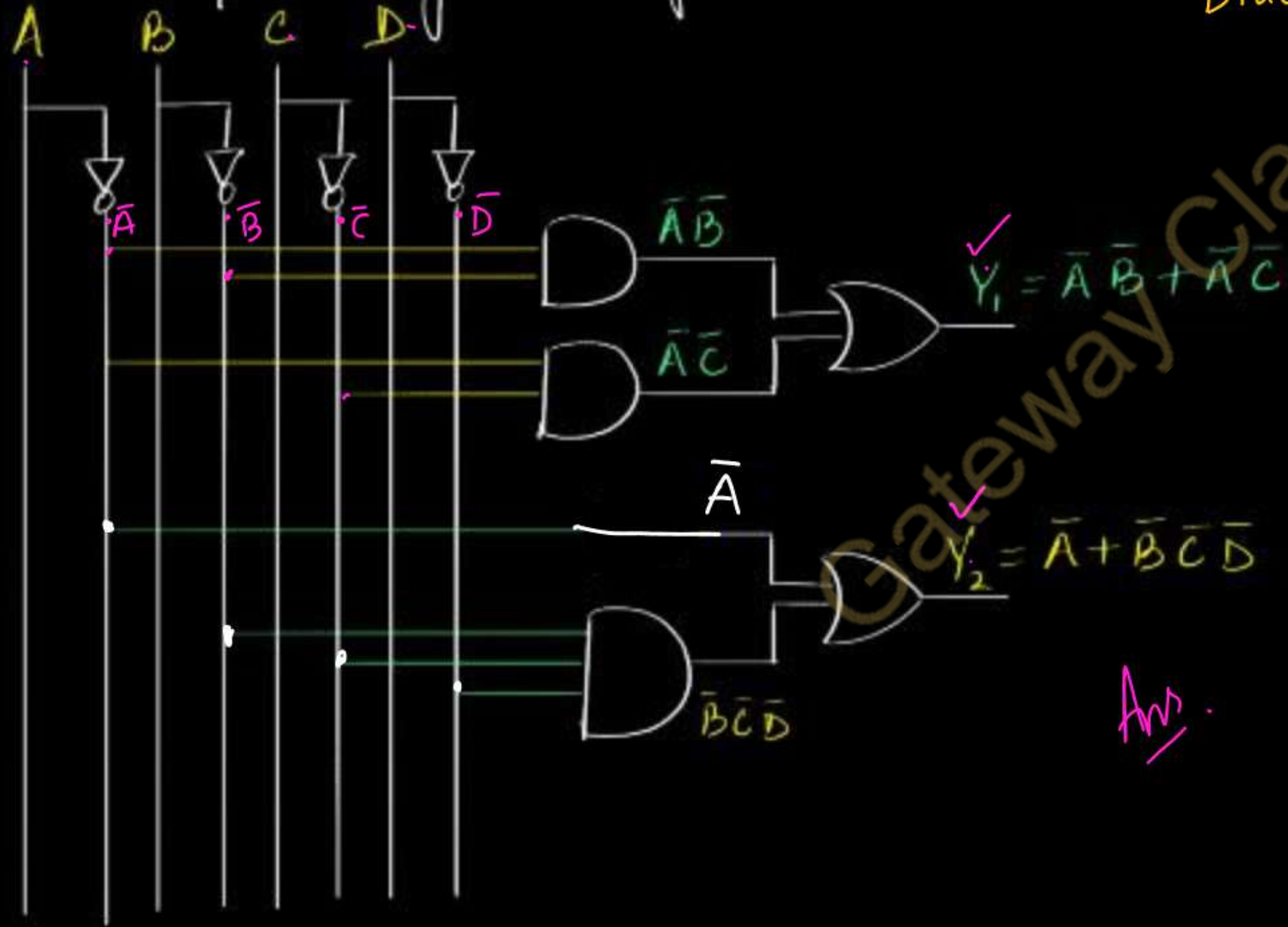
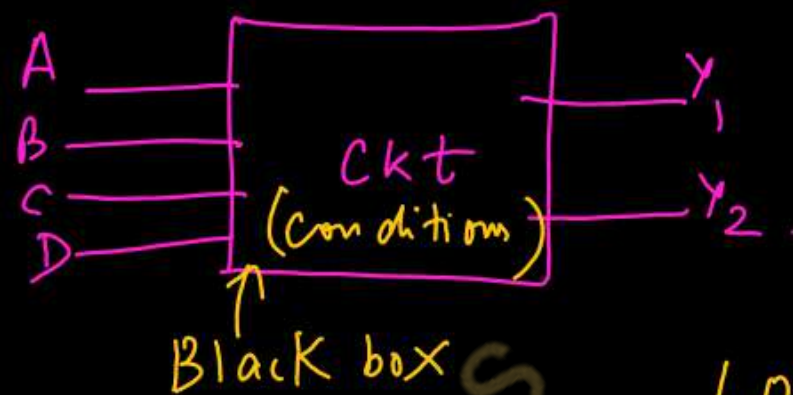
Groupings: $\bar{A}$ (left column), $\bar{B}\bar{C}\bar{D}$ (top-right 1)

$$Y_2 = \bar{A} + \bar{B}\bar{C}\bar{D}$$

$$Y_1 = \bar{A}\bar{B} + \bar{A}\bar{C}$$

$$Y_2 = \bar{A} + \bar{B}\bar{C}\bar{D}$$

Step 4 Logic Diagram



Logic
 T T
 K Map for o/p's
 expressions
 logic diag.

Design

Magnitude comparator

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It is a Combinational Circuit which is used to compare bits.

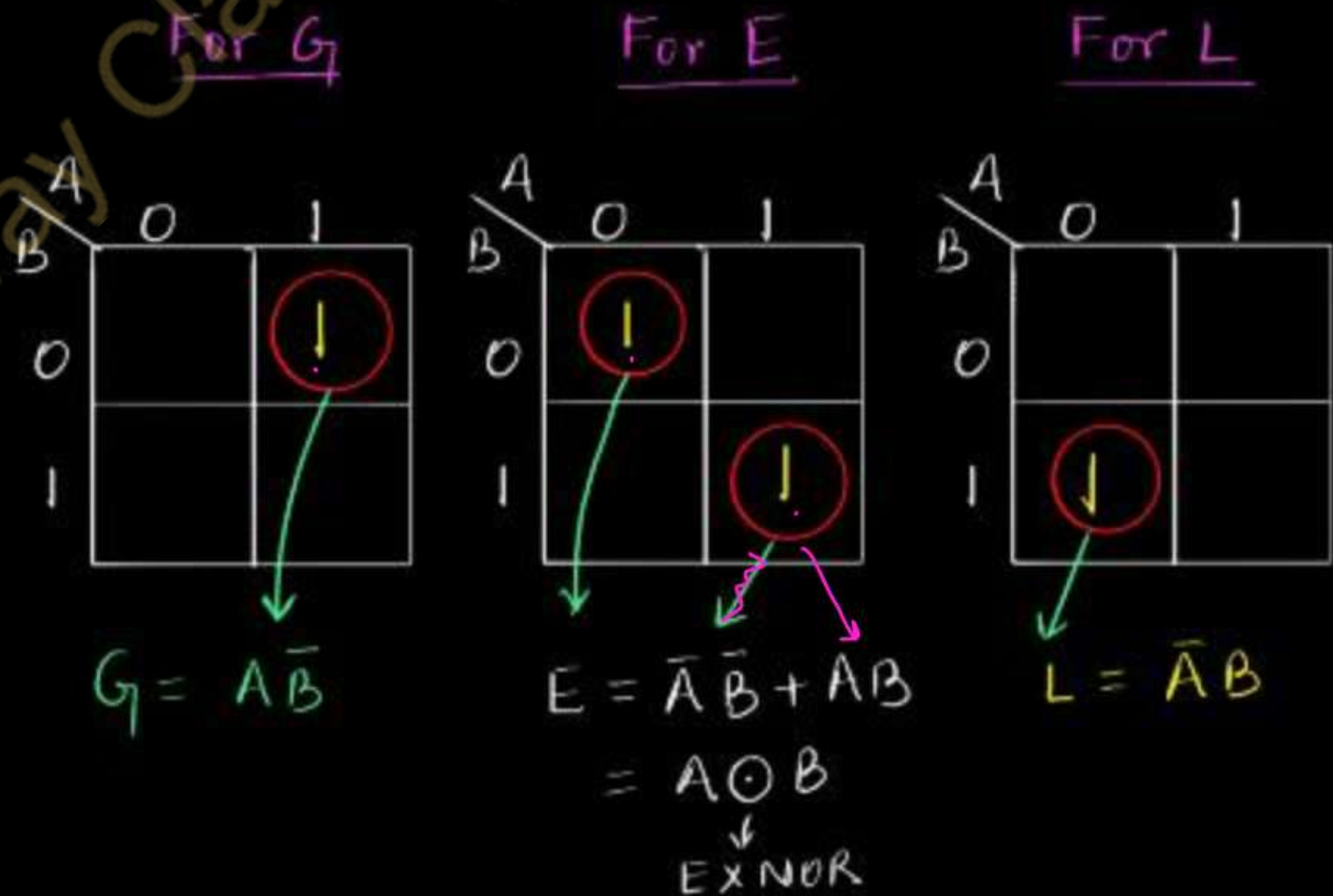
One bit Magnitude comparator



step 2 Truth Table

A	B	$A > B$ (G)	$A = B$ (E)	$A < B$ (L)
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

step 3 K-Map



Step 4 Logic Diagram

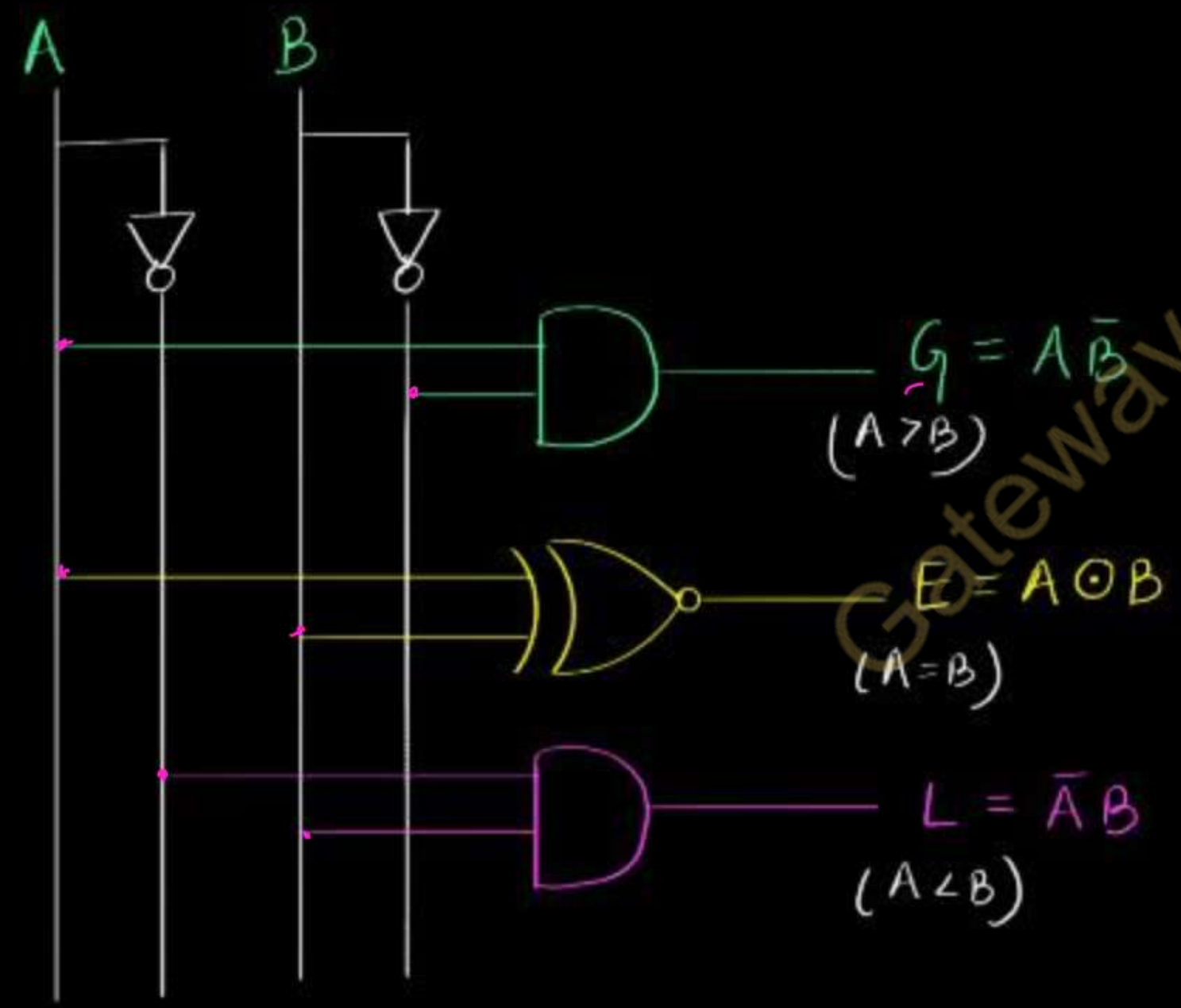
$$G = A\bar{B}$$

$$E = \bar{A}\bar{B} + AB$$

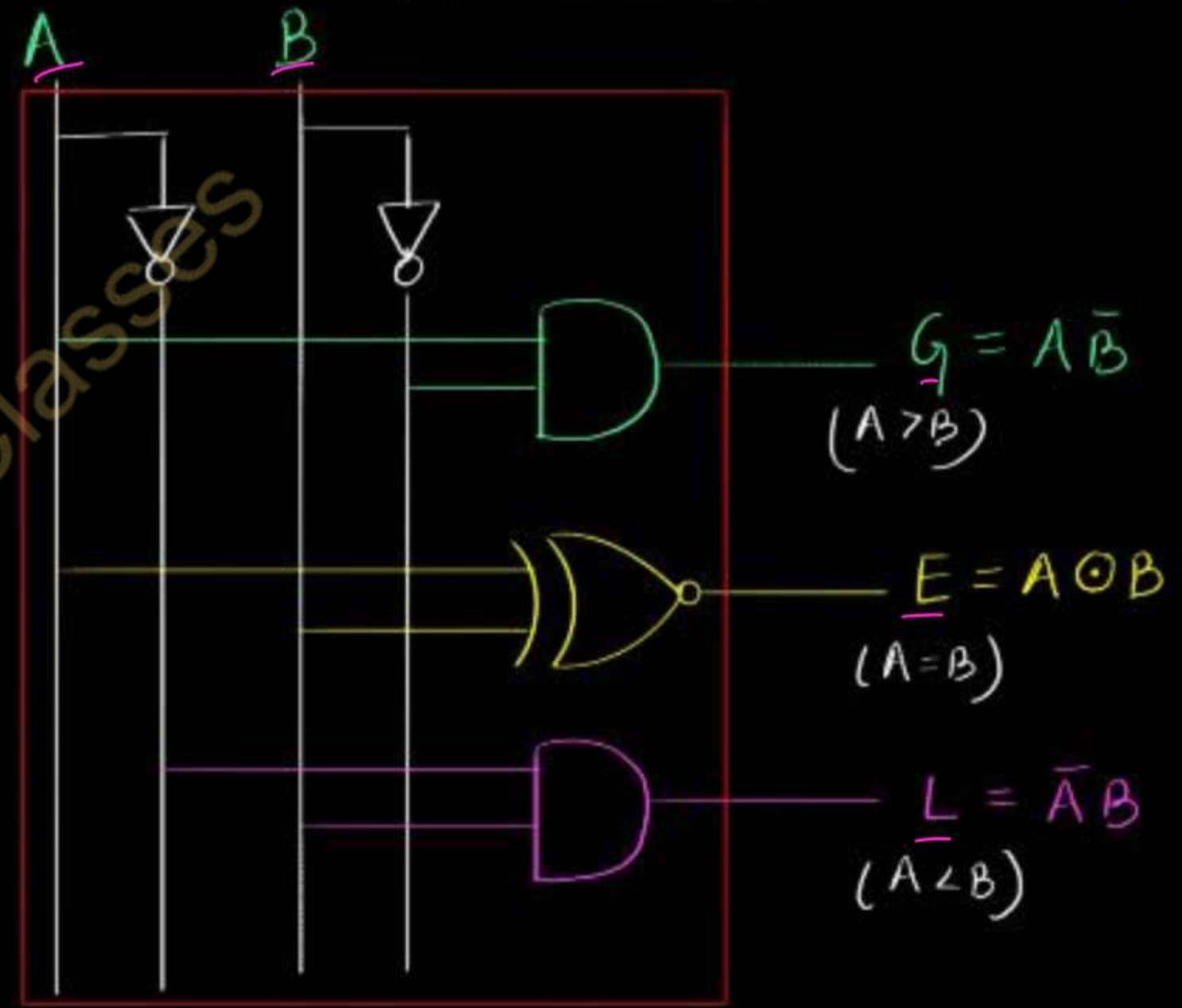
$$L = \bar{A}B$$

$$= A \odot B$$

EXNOR

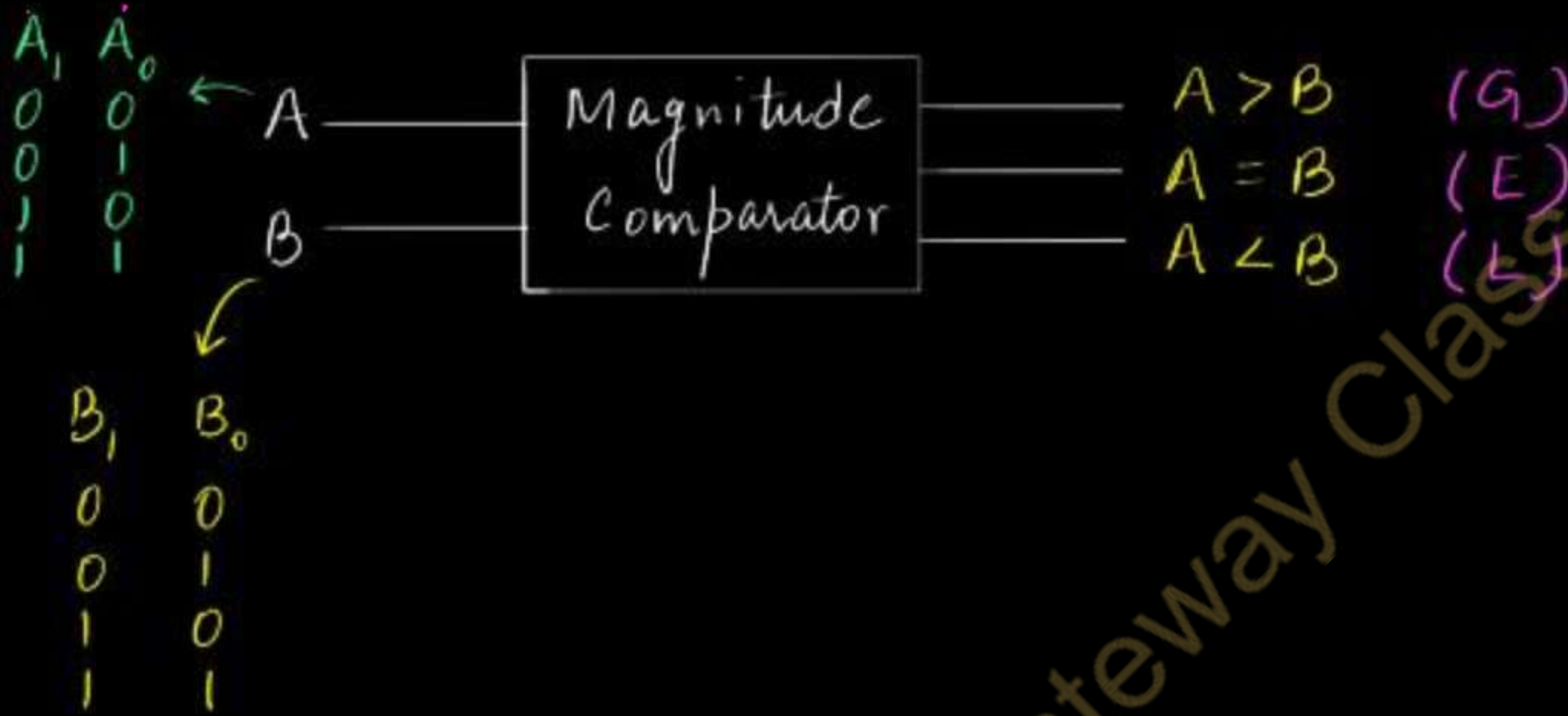


$\therefore$



1 bit Magnitude Comparator

Two bit Magnitude comparator

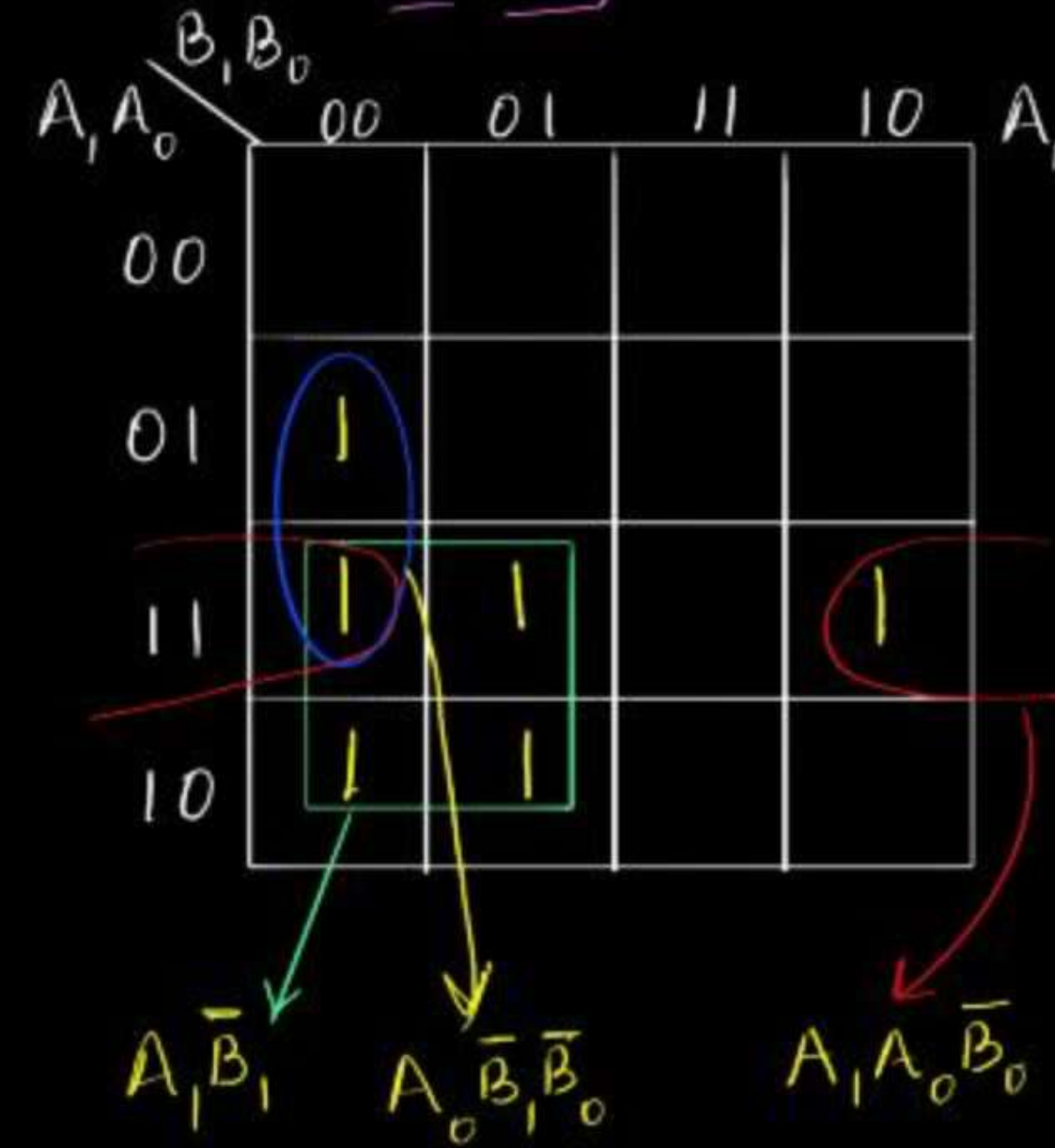


Truth Table

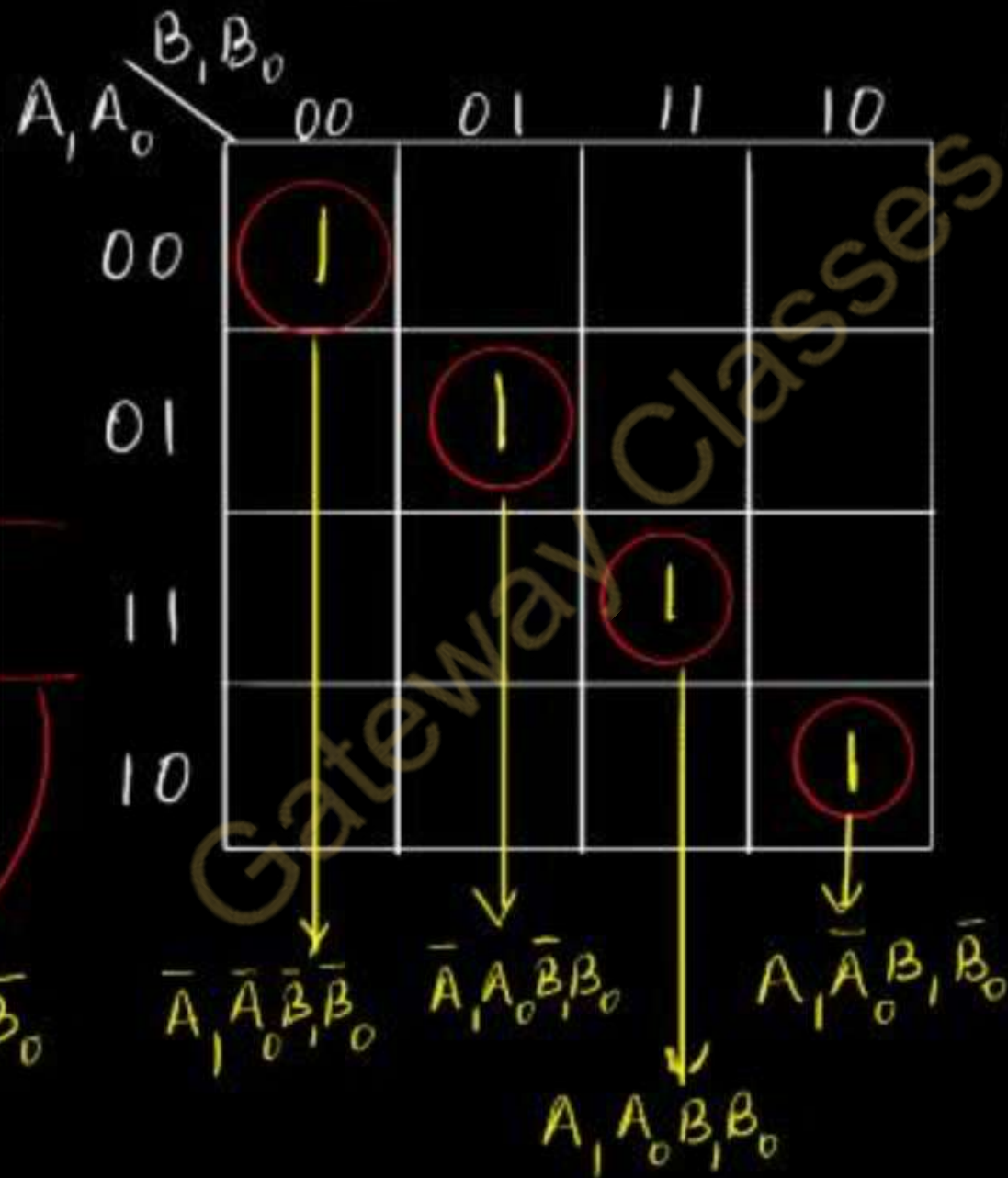
	A		B		Outputs		
	A_1	A_0	B_1	B_0	G_1 ($A > B$)	E ($A = B$)	L ($A < B$)
0	0	0	0	0	0	1	0
1	0	0	0	1	0	0	1
2	0	0	1	0	0	0	1
	0	0	1	1	0	0	1
	0	1	0	0	1	0	0
	0	1	0	1	0	1	0
	0	1	1	0	0	0	1
	0	1	1	1	0	0	1
	1	0	0	0	1	0	0
	1	0	0	1	1	0	0
	1	0	1	0	0	1	0
	1	0	1	1	0	0	1
	1	1	0	0	1	0	0
	1	1	0	1	1	0	0
	1	1	1	0	1	0	0
15	1	1	1	1	0	1	0

K-Map

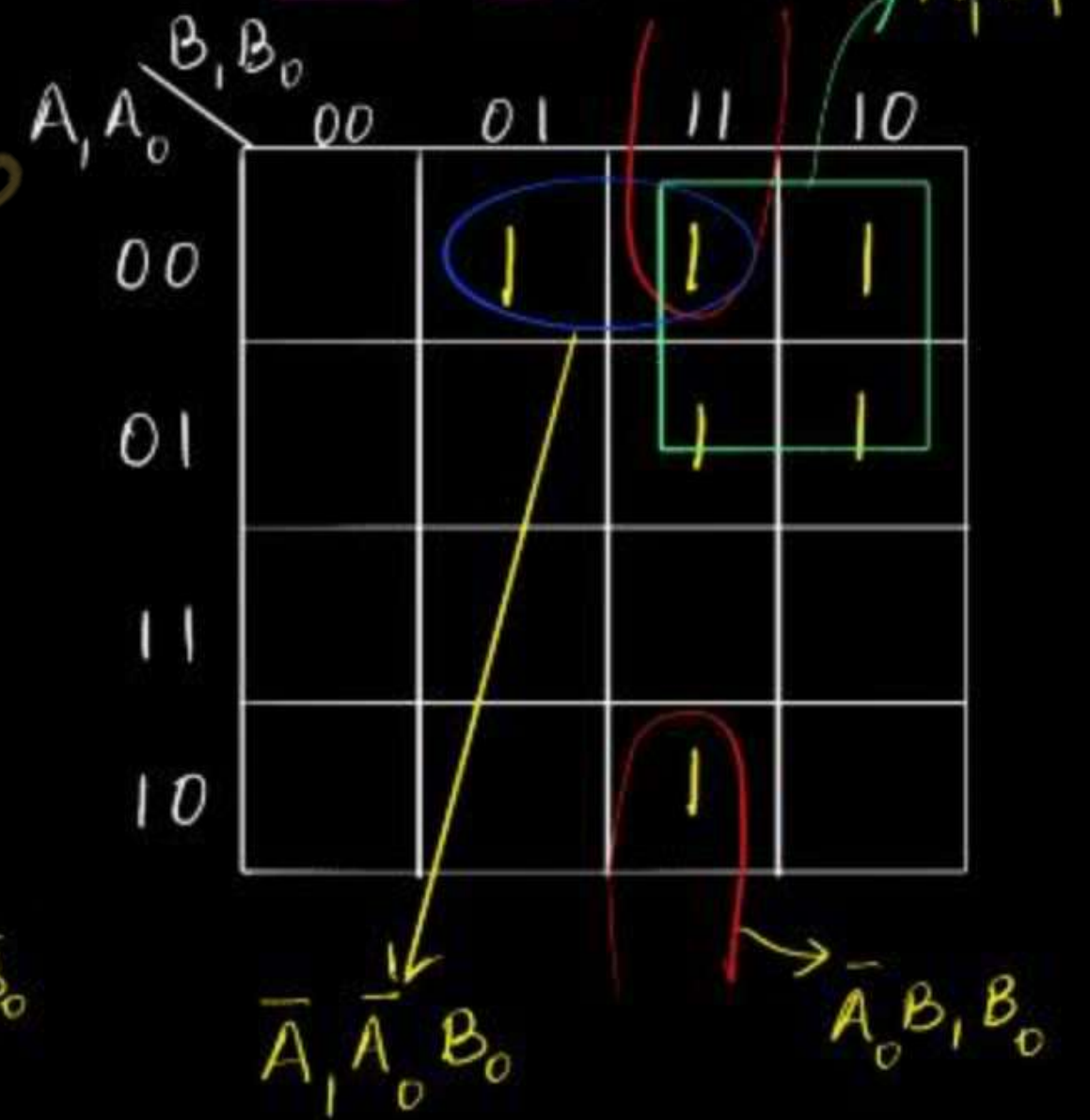
For G



For E



For L

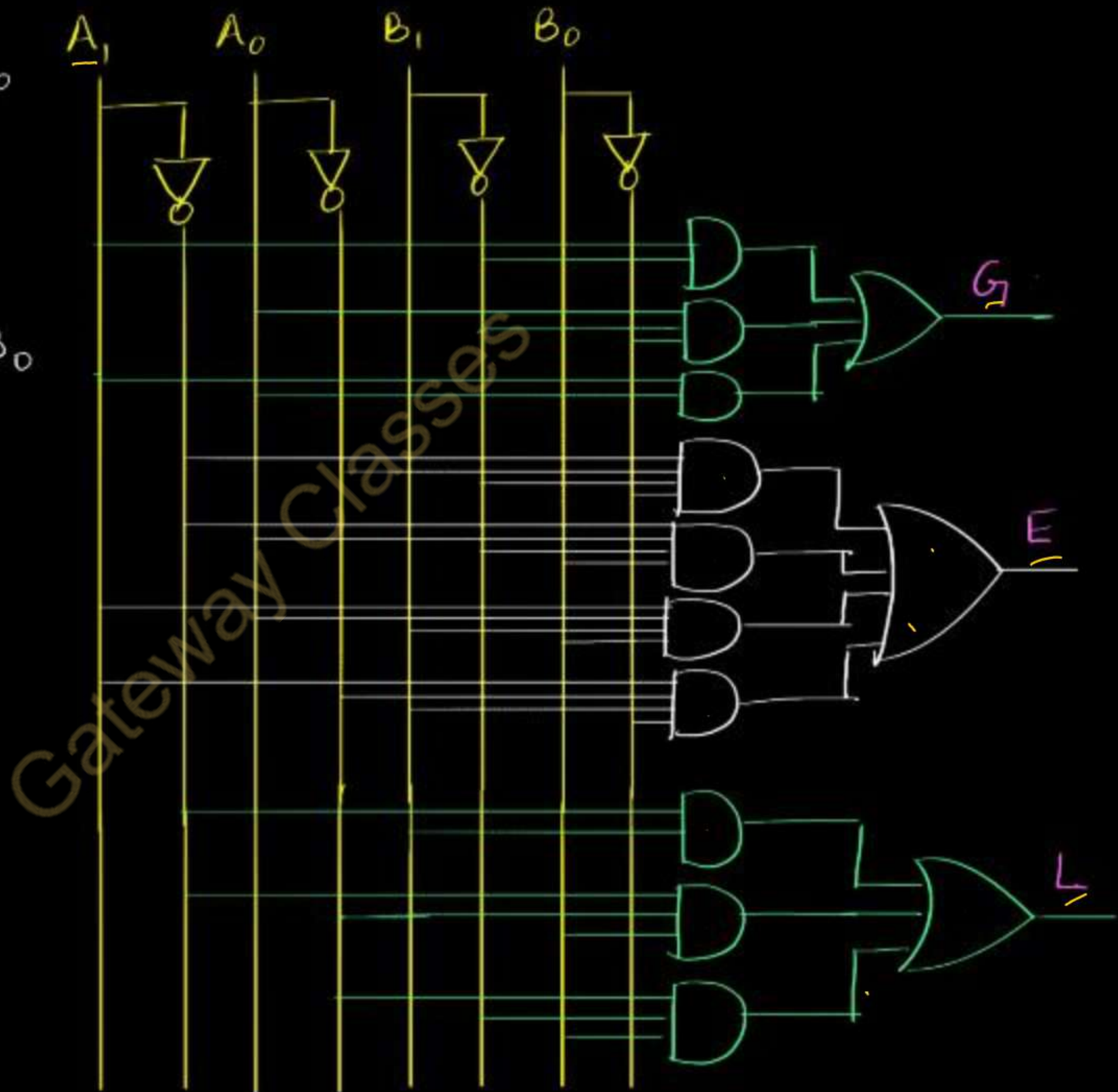


$$G = A_1 \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_0$$

$$E = \bar{A}_1 \bar{A}_0 \bar{B}_1 \bar{B}_0 + \bar{A}_1 A_0 \bar{B}_1 B_0$$

$$+ A_1 A_0 B_1 B_0 + A_1 \bar{A}_0 B_1 \bar{B}_0$$

$$L = \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0$$



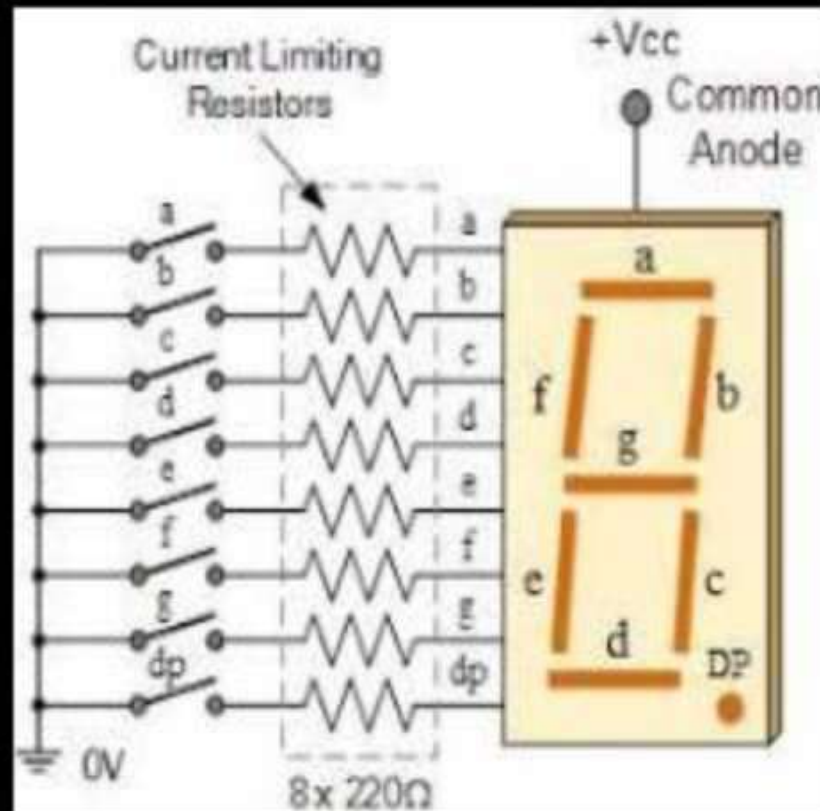
Multiplexers

Multiplexed display

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Multiplexed displays are electronic display devices where the entire display is not driven at one time

Example - **Multi digit 7-segment display**, like those found in digital clocks or calculators, where segments are shared across digits, and a microcontroller rapidly switches between displaying each digit.

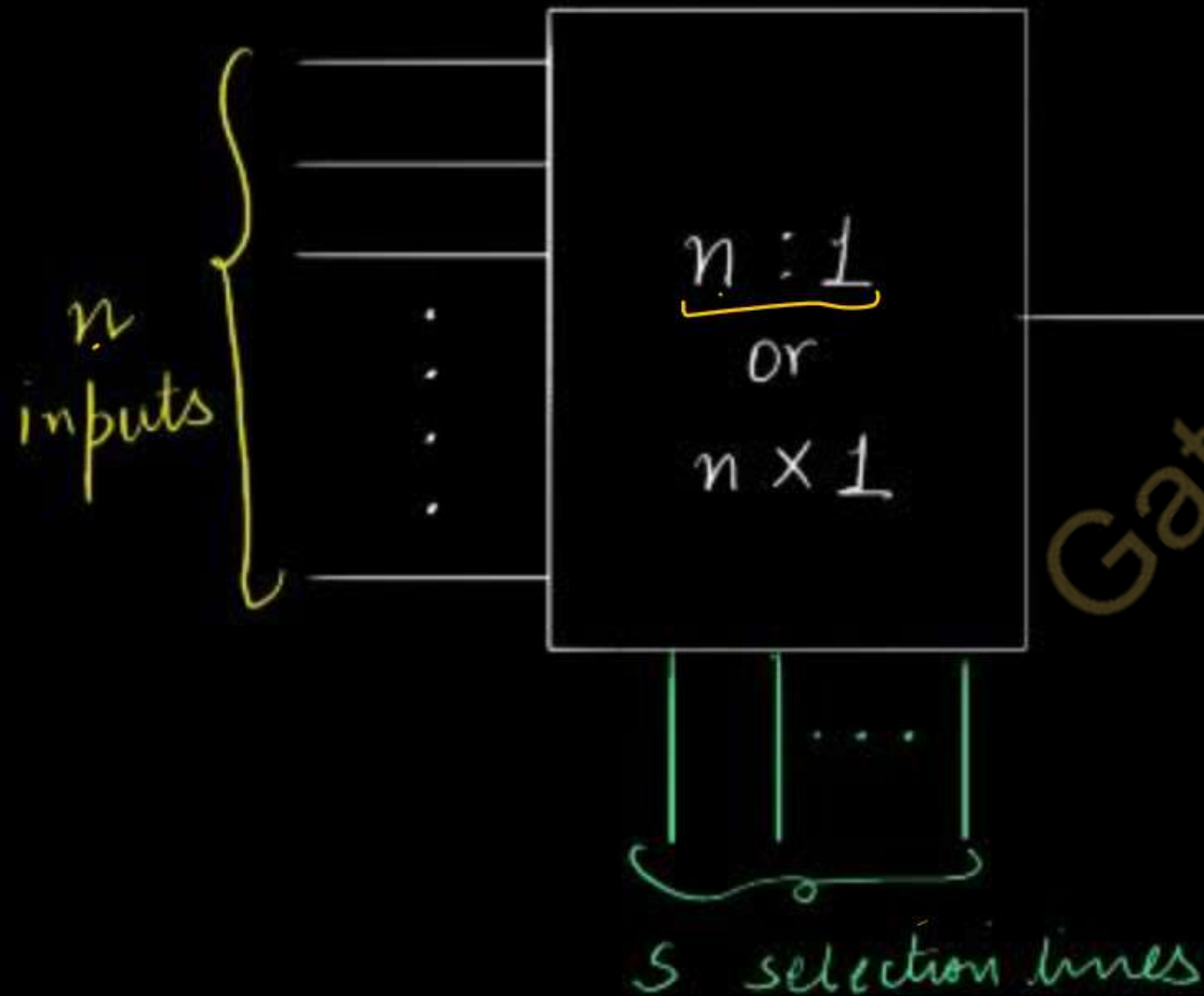


Multiplexers is a combinational logic circuit used to select only one input among several inputs, based on selection lines. It is also known as **Data Selector** or Digital Switch.

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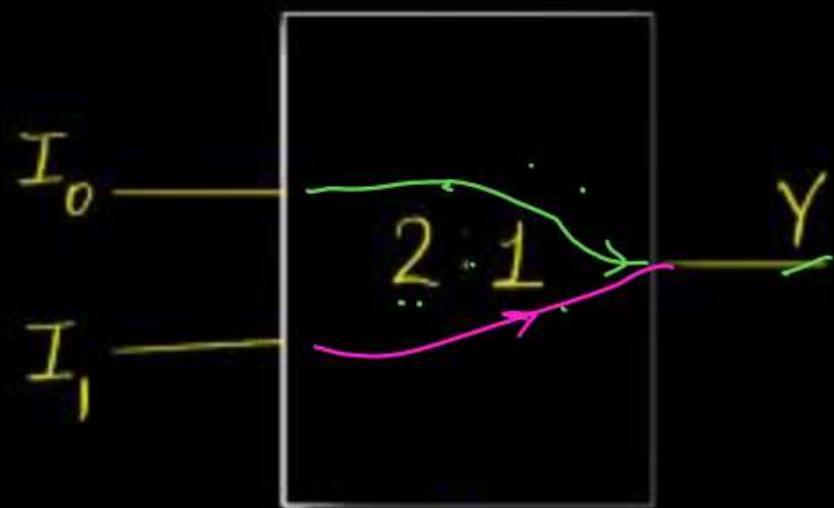


$$n = 2^S$$

where n is number of inputs
 S is number of selection lines.

- 1) $2:1$ $n=2$
 $2 = 2^S$
 $S=1$
- 2) $4:1$ $n=4$
 $4 = 2^2$
 $S=2$
- 3) $8:1$ $n=8$
 $8 = 2^3$
 $S=3$

2:1 MUX



A

A = 0 $I_0 \rightarrow Y$
A = 1 $I_1 \rightarrow Y$

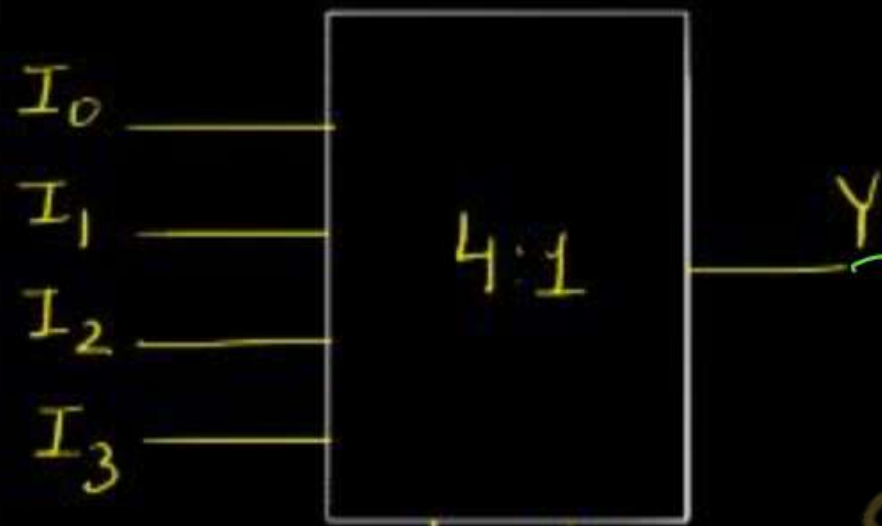
$$2^s = n$$

$$2^s = 2$$

$$s = 1$$

A	Y
0	I_0
1	I_1

4:1 MUX



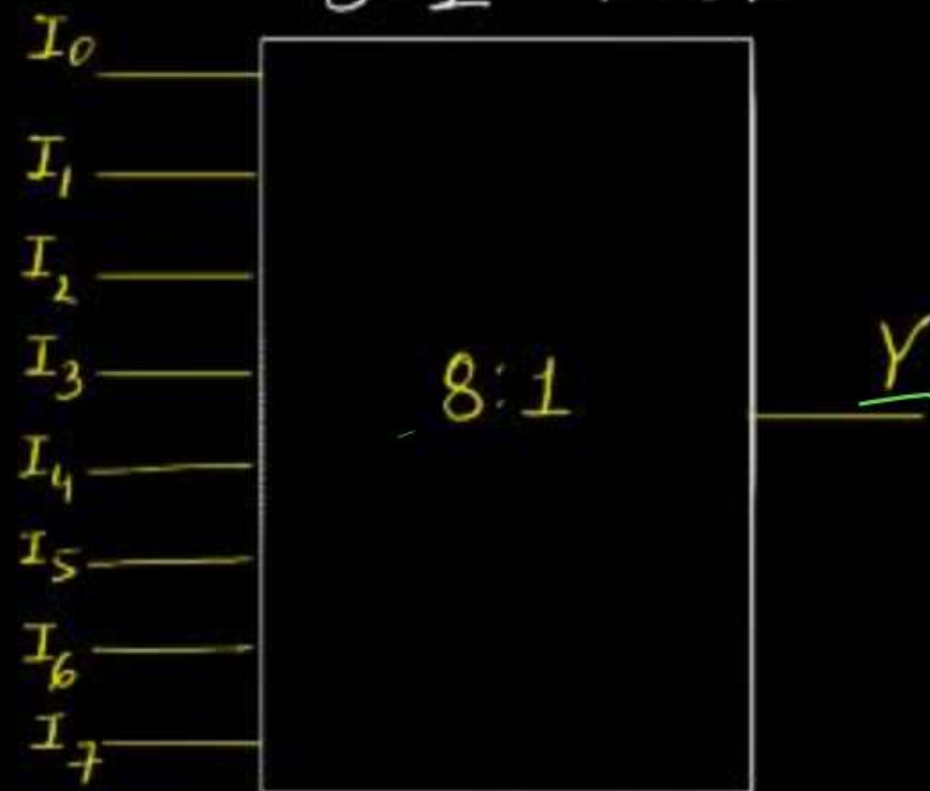
A

B

$$2^s = 4 \Rightarrow s = 2$$

A	B	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

8:1 MUX



A

B

C

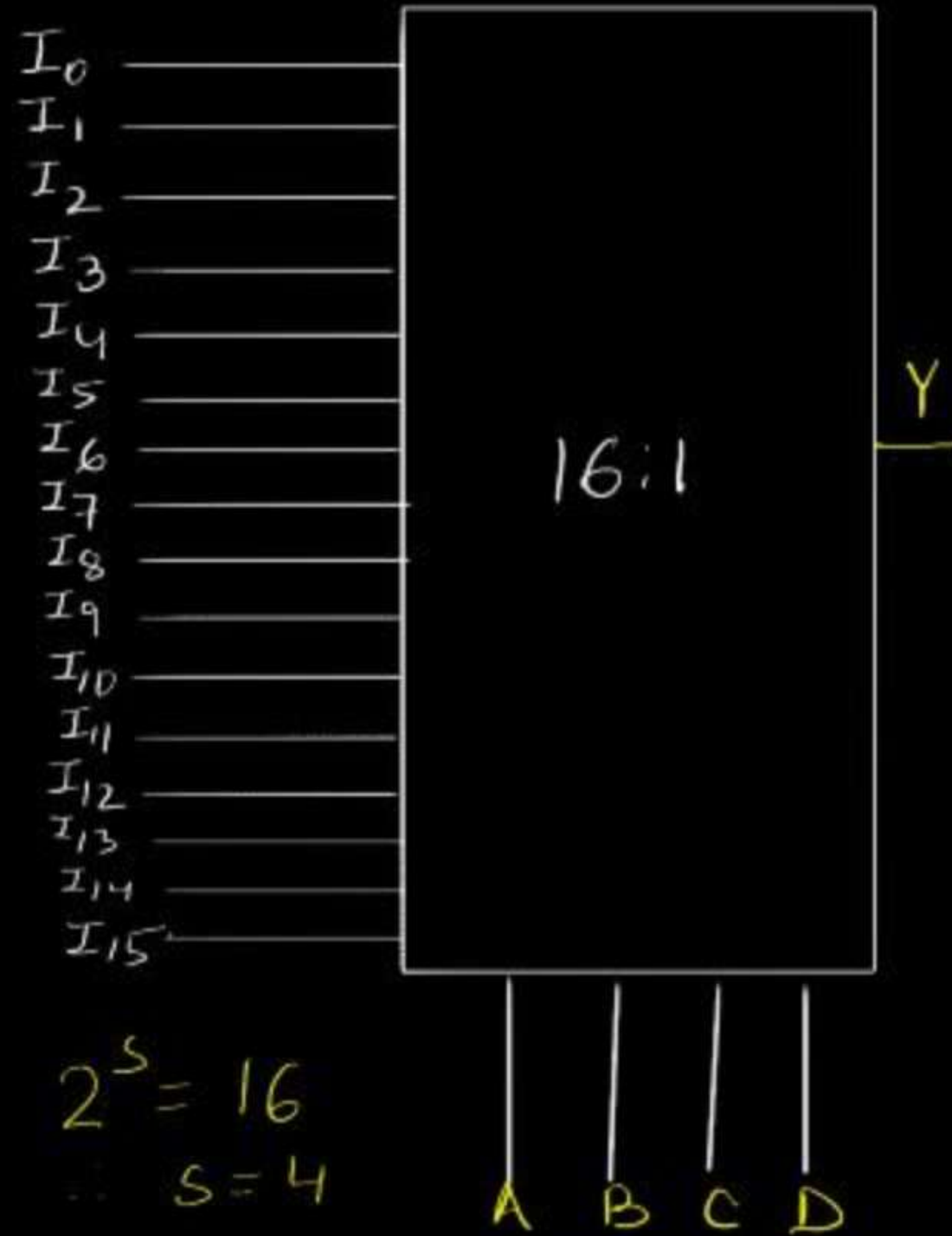
A	B	C	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

$$2^s = 8$$

$$2^s = 2^3$$

$$s = 3$$

16:1 MUX



Truth Table

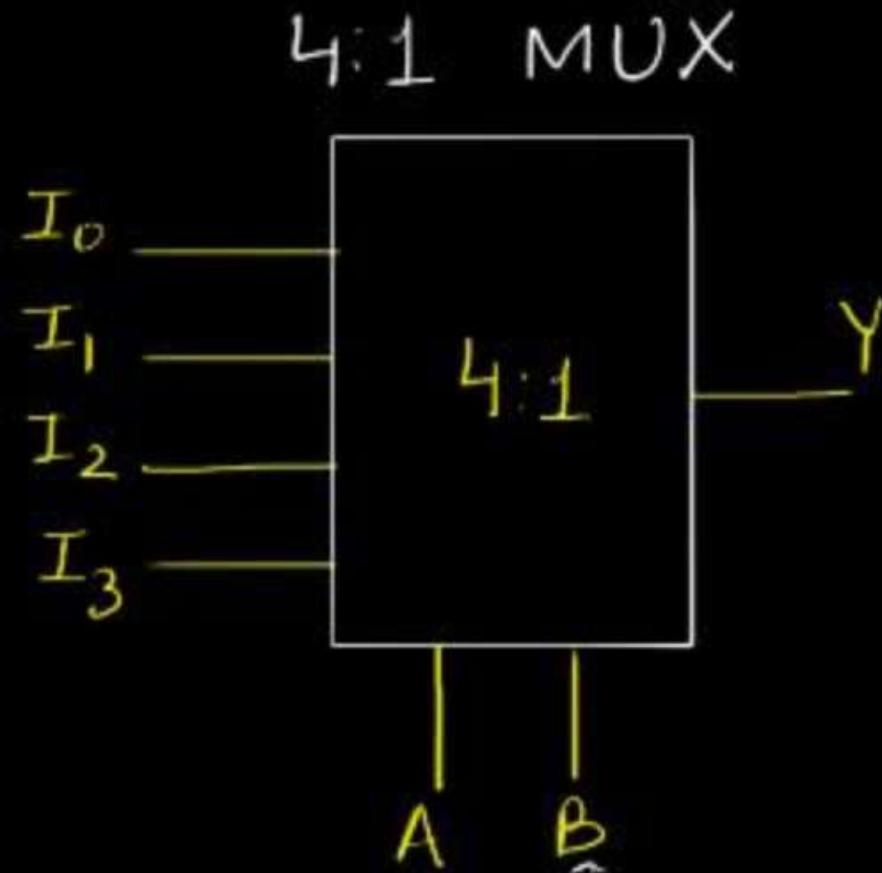
A	B	C	D	Y
0	0	0	0	I_0
0	0	0	1	I_1
0	0	1	0	I_2
0	0	1	1	I_3
0	1	0	0	I_4
0	1	0	1	I_5
0	1	1	0	I_6
0	1	1	1	I_7
1	0	0	0	I_8
1	0	0	1	I_9
1	0	1	0	I_{10}
1	0	1	1	I_{11}
1	1	0	0	I_{12}
1	1	0	1	I_{13}
1	1	1	0	I_{14}
1	1	1	1	I_{15}

Similarly we have 32:1 MUX, 64:1 MUX
 and so on.

Design 4:1 multiplexer using gates.

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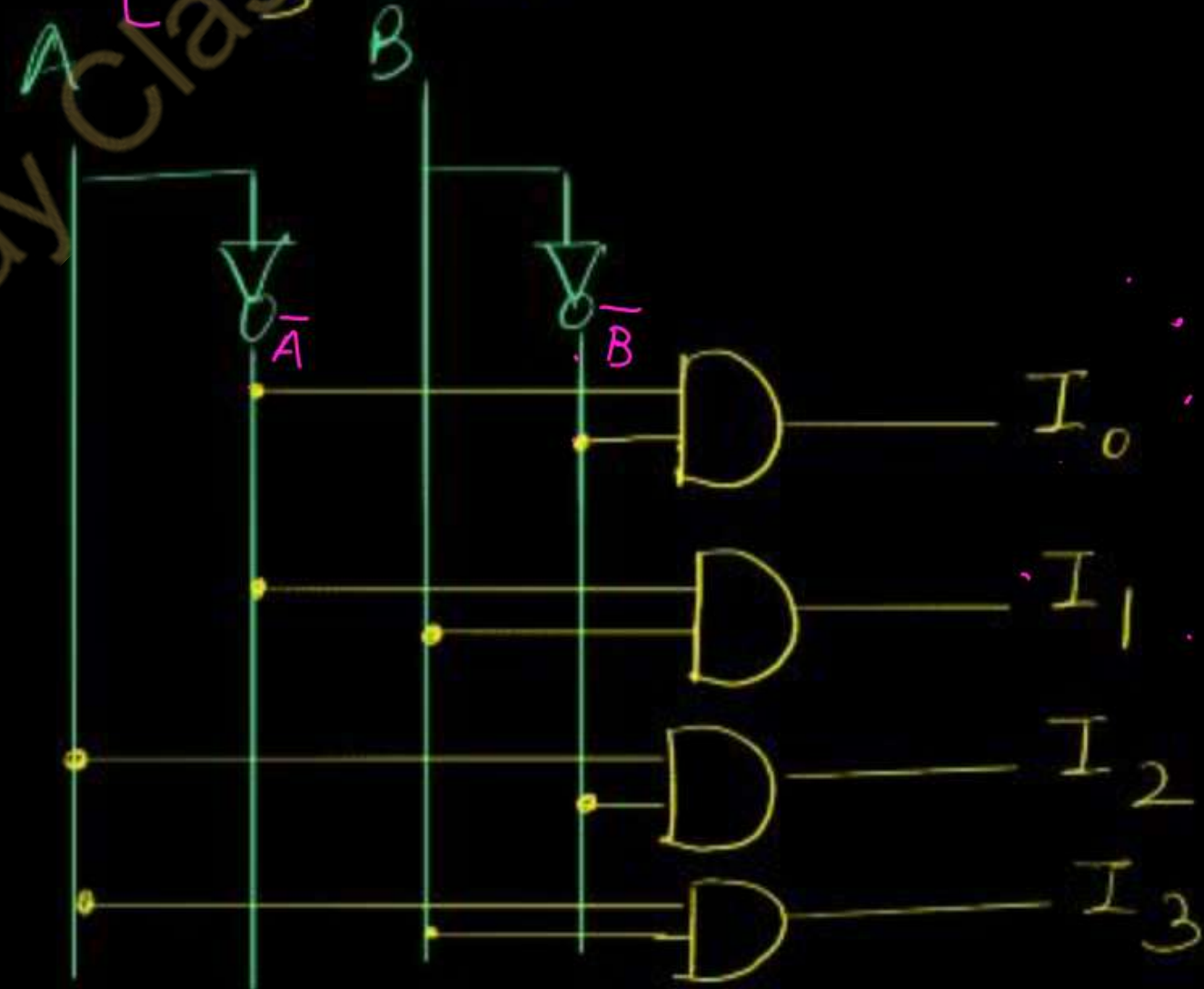
2024-25



$$2^s = 4 \Rightarrow s = 2$$

A	B	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

$$\begin{aligned} I_0 &= \bar{A} \bar{B} \\ I_1 &= \bar{A} B \\ I_2 &= A \bar{B} \\ I_3 &= A B \end{aligned}$$



Que Implement the following Boolean function

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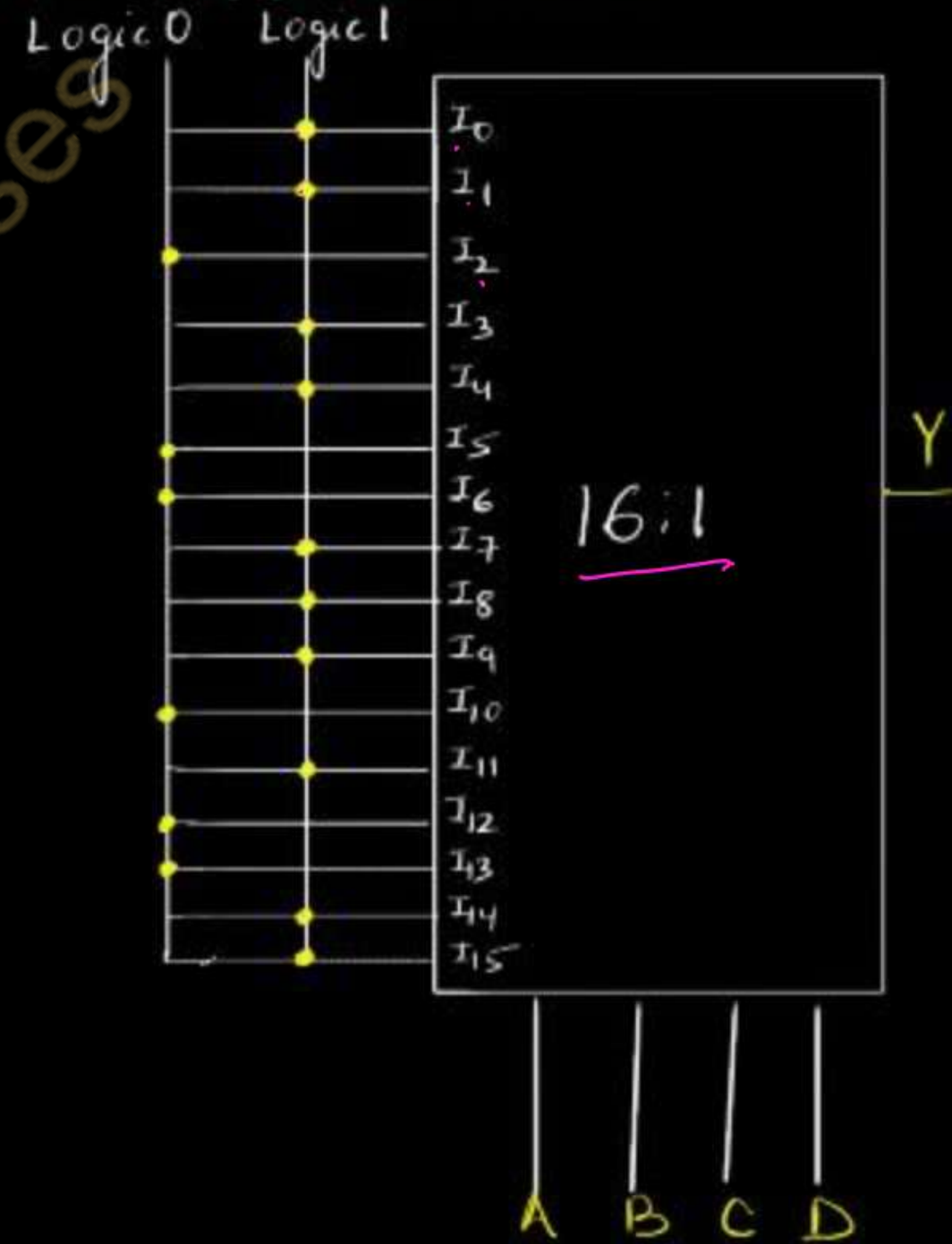
$$F(A, B, C, D) = \sum(0, 1, 3, 4, 7, 8, 9, 11, 14, 15) \text{ using}$$

(i) 16:1 MUX (ii) 8:1 MUX (iii) 4:1 MUX (iv) 2:1 MUX

i) 16:1 MUX

$$16 = 2^4 \text{ selection lines } A, B, C, D$$

A	B	C	D	Y	Cell no.	Given
0	0	0	0	I_0	0 ✓	Logic 1
0	0	0	1	I_1	1 ✓	Logic 1
0	0	1	0	I_2	2	Logic 0
0	0	1	1	I_3	3 ✓	Logic 1
0	1	0	0	I_4	4 ✓	Logic 1
0	1	0	1	I_5	5	Logic 0
0	1	1	0	I_6	6	Logic 0
0	1	1	1	I_7	7 ✓	Logic 1
1	0	0	0	I_8	8 ✓	Logic 1
1	0	0	1	I_9	9 ✓	Logic 1
1	0	1	0	I_{10}	10	Logic 0
1	0	1	1	I_{11}	11 ✓	Logic 1
1	1	0	0	I_{12}	12	Logic 0
1	1	0	1	I_{13}	13	Logic 0
1	1	1	0	I_{14}	14 ✓	Logic 1
1	1	1	1	I_{15}	15 ✓	Logic 1



(ii) 8:1 MUX

$$F(A, B, C, D) = \Sigma(0, 1, 3, 4, 7, 8, 9, 11, 14, 15)$$

$8 = 2^3 \rightarrow 3$ select lines

A, B, C

A, B, C, D
select lines $\rightarrow$ for input

A	B	C	D	Y	cell no.
0	0	0	0	I_0	0 ✓
0	0	0	1	I_1	1 ✓
0	0	1	0	I_2	2
0	0	1	1	I_3	3 ✓
0	1	0	0	I_4	4 ✓
0	1	0	1	I_5	5
0	1	1	0	I_6	6
0	1	1	1	I_7	7 ✓
1	0	0	0	I_8	8 ✓
1	0	0	1	I_9	9 ✓
1	0	1	0	I_{10}	10
1	0	1	1	I_{11}	11 ✓
1	1	0	0	I_{12}	12
1	1	0	1	I_{13}	13
1	1	1	0	I_{14}	14 ✓
1	1	1	1	I_{15}	15

$D=0$ at cell no. 0
 $D=1$ at cell no. 1
 $D=0$ at cell no. 2
 $D=1$ at cell no. 3
 $D=0$ at cell no. 4
 $D=1$ at cell no. 5
 $D=0$ at cell no. 6
 $D=1$ at cell no. 7
 $D=0$ at cell no. 8
 $D=1$ at cell no. 9
 $D=0$ at cell no. 10
 $D=1$ at cell no. 11
 $D=0$ at cell no. 12
 $D=1$ at cell no. 13
 $D=0$ at cell no. 14
 $D=1$ at cell no. 15

v. imp $\rightarrow I_0$ to I_7 : 8:1 MUX

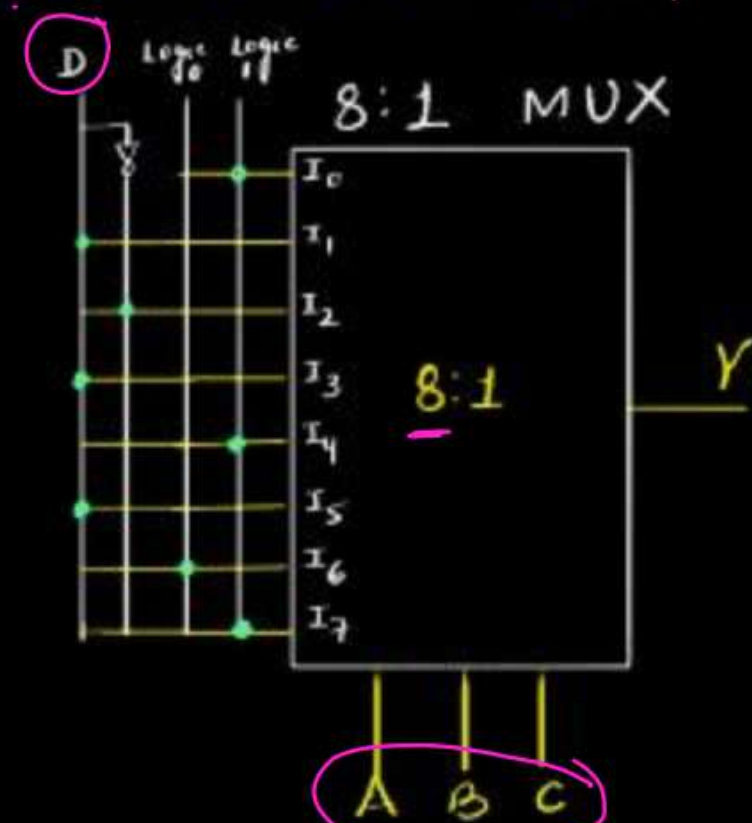
		I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
0	$\bar{D}$	0	2	4	6	8	10	12	14
1	D	1	3	5	7	9	11	13	15

$\Sigma(0, 1, 3, 4, 7, 8, 9, 11, 14, 15) \rightarrow$ circle these

		I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
0	$\bar{D}$	0	2	4	6	8	10	12	14
1	D	1	3	5	7	9	11	13	15

$D + \bar{D}$ $\bar{D}$ $\bar{D}$ D $D + \bar{D}$ D 0 $D + \bar{D} = 1$

$I_0=1, I_1=0, I_2=1, I_3=0, I_4=1, I_5=0, I_6=0, I_7=1$



(III) 4:1 MUX

$$F(A, B, C, D) = \Sigma(0, 1, 3, 4, 7, 8, 9, 11, 14, 15)$$

$4 = 2^2 \rightarrow$ 2 select lines
A, B

C, D for input

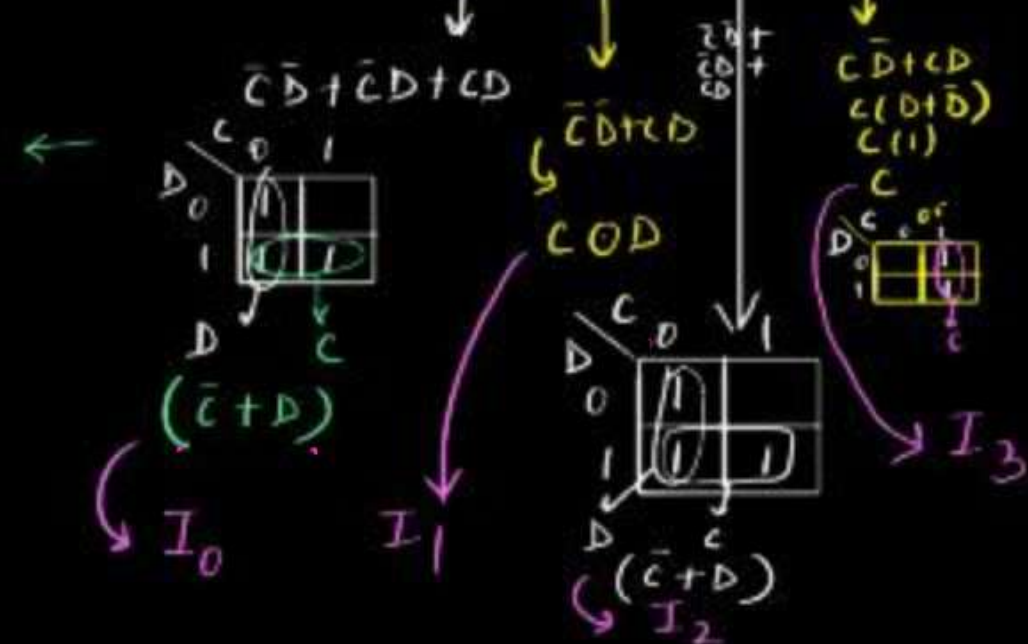
A	B	C	D	✓ Y	chk no.
0	0	0	0	I_0	0 ✓
0	0	0	1	I_1	1 ✓
0	0	1	0	I_2	2
0	0	1	1	I_3	3 ✓
0	1	0	0	I_4	4 ✓
0	1	0	1	I_5	5
0	1	1	0	I_6	6
0	1	1	1	I_7	7 ✓
1	0	0	0	I_8	8 ✓
1	0	0	1	I_9	9 ✓
1	0	1	0	I_{10}	10
1	0	1	1	I_{11}	11 ✓
1	1	0	0	I_{12}	12
1	1	0	1	I_{13}	13
1	1	1	0	I_{14}	14 ✓
1	1	1	1	I_{15}	15

inp ✓ I_0 to I_3
4:1 MUX

	I_0	I_1	I_2	I_3
00 $\bar{C}\bar{D}$	0	4	8	12
01 $\bar{C}D$	1	5	9	13
10 $C\bar{D}$	2	6	10	14
11 CD	3	7	11	15

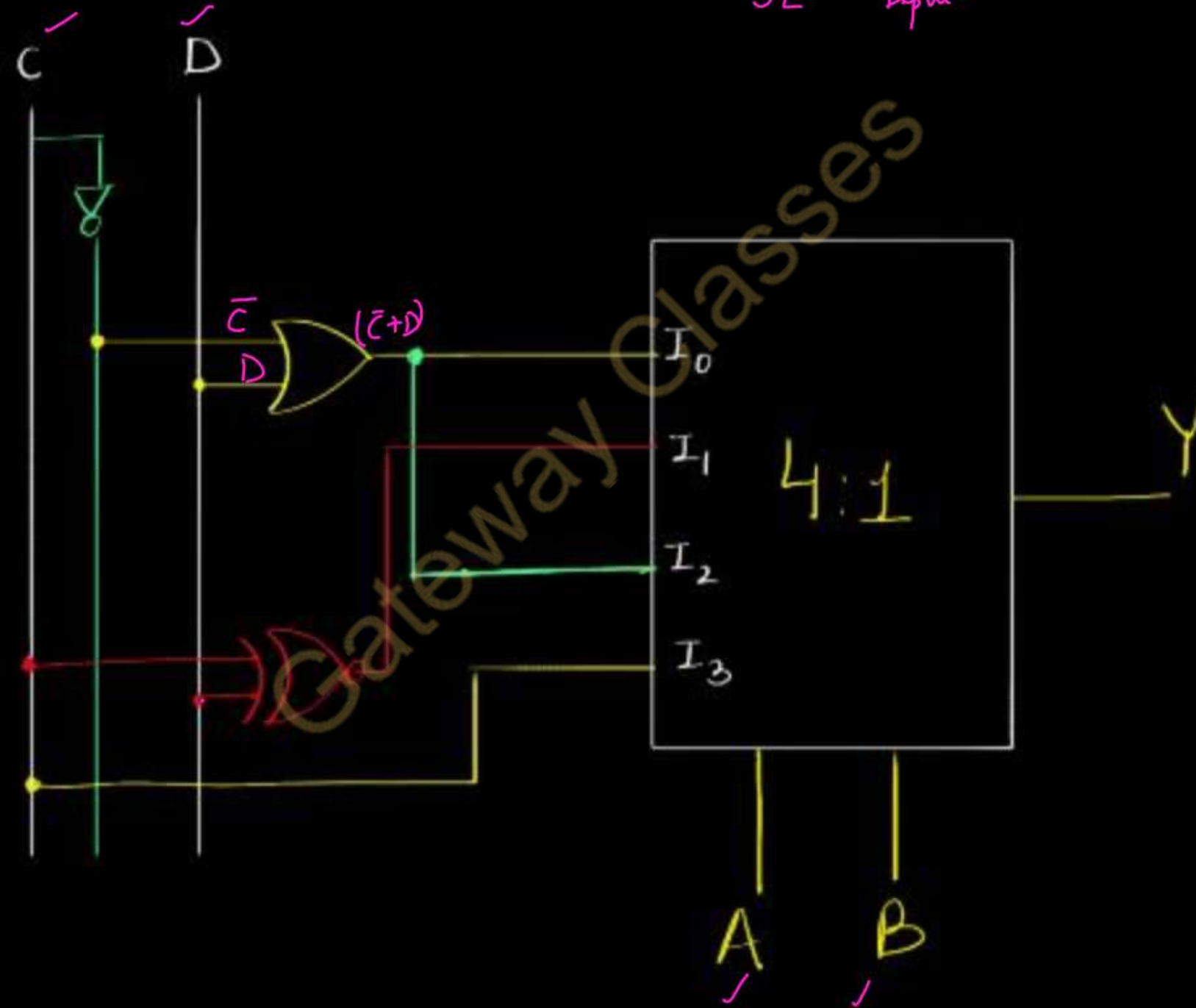
$\Sigma(0, 1, 3, 4, 7, 8, 9, 11, 14, 15) \rightarrow$ Circle these

	I_0	I_1	I_2	I_3
$\bar{C}\bar{D}$	0	4	8	12
$\bar{C}D$	1	5	9	13
$C\bar{D}$	2	6	10	14
CD	3	7	11	15



$$\begin{cases} I_0 = I_2 = \bar{C} + D \\ I_1 = C \odot D \\ I_3 = C \end{cases}$$

$\underbrace{A, B}_{SL} \quad \underbrace{C, D}_{\text{Input}}$



(IV) 2:1 MUX

$$F(A, B, C, D) = \Sigma(0, 1, 3, 4, 7, 8, 9, 11, 14, 15)$$

$2 = 2^1 \rightarrow$ 1 select line
A

A { B, C, D }

B, C, D for input

A	B	C	D	Y	Cell no.
0	0	0	0	I_0	0 ✓
0	0	0	1	I_1	1 ✓
0	0	1	0	I_2	2
0	0	1	1	I_3	3 ✓
0	1	0	0	I_4	4 ✓
0	1	0	1	I_5	5
0	1	1	0	I_6	6
0	1	1	1	I_7	7 ✓
1	0	0	0	I_8	8 ✓
1	0	0	1	I_9	9 ✓
1	0	1	0	I_{10}	10
1	0	1	1	I_{11}	11 ✓
1	1	0	0	I_{12}	12
1	1	0	1	I_{13}	13
1	1	1	0	I_{14}	14 ✓
1	1	1	1	I_{15}	15

inp ✓

		I_0	I_1
000	$\bar{B}\bar{C}\bar{D}$	0	8
001	$\bar{B}\bar{C}D$	1	9
010	$\bar{B}C\bar{D}$	2	10
011	$\bar{B}CD$	3	11
100	$B\bar{C}\bar{D}$	4	12
101	$B\bar{C}D$	5	13
110	$BC\bar{D}$	6	14
111	BCD	7	15

$\Sigma(0, 1, 3, 4, 7, 8, 9, 11, 14, 15) \rightarrow$ Circle these

		I_0	I_1
000	$\bar{B}\bar{C}\bar{D}$	0	8
001	$\bar{B}\bar{C}D$	1	9
010	$\bar{B}C\bar{D}$	2	10
011	$\bar{B}CD$	3	11
100	$B\bar{C}\bar{D}$	4	12
101	$B\bar{C}D$	5	13
110	$BC\bar{D}$	6	14
111	BCD	7	15

Draw K-Map for I_0

Draw K-Map for I_1

For I_0

BC \ D	00	01	11	10
0	1			1
1	1	1	1	

$$I_0 = \bar{B}D + CD + \bar{C}\bar{D}$$

$$= \bar{B}D + (C \odot D)$$

For I_1

BC \ D	00	01	11	10
0	1		1	
1	1	1	1	

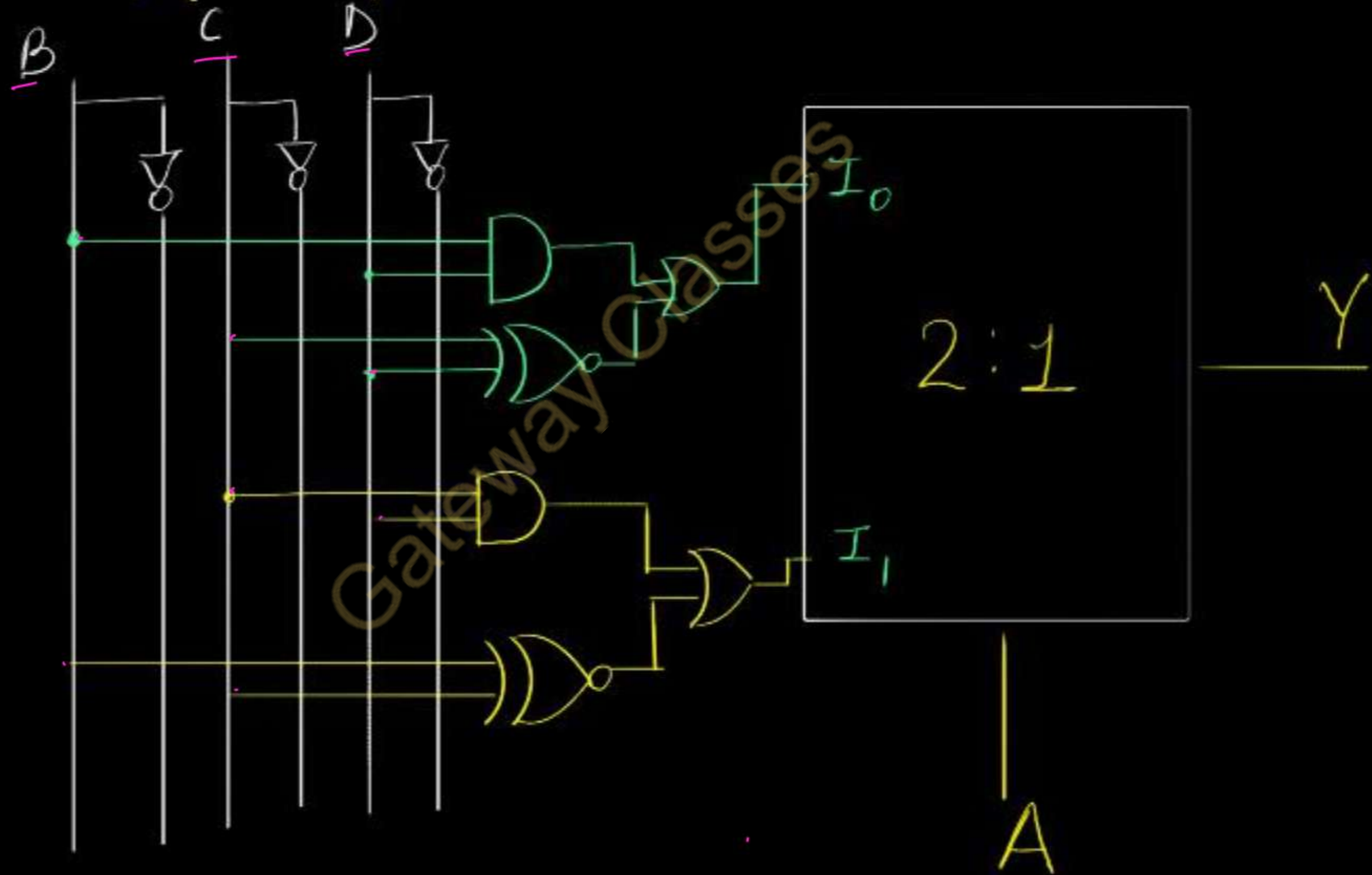
$$I_1 = CD + BC + \bar{B}\bar{C}$$

$$= CD + (B \odot C)$$

$$I_0 = BD + (C \odot D)$$

$$I_1 = CD + (B \odot C)$$

$\underbrace{A}_{SL}, \underbrace{B, C, D}_{Input}$



Using 4:1 multiplexers, implement the following function $F(A, B, C) = \sum m(0, 2, 3, 5, 7)$

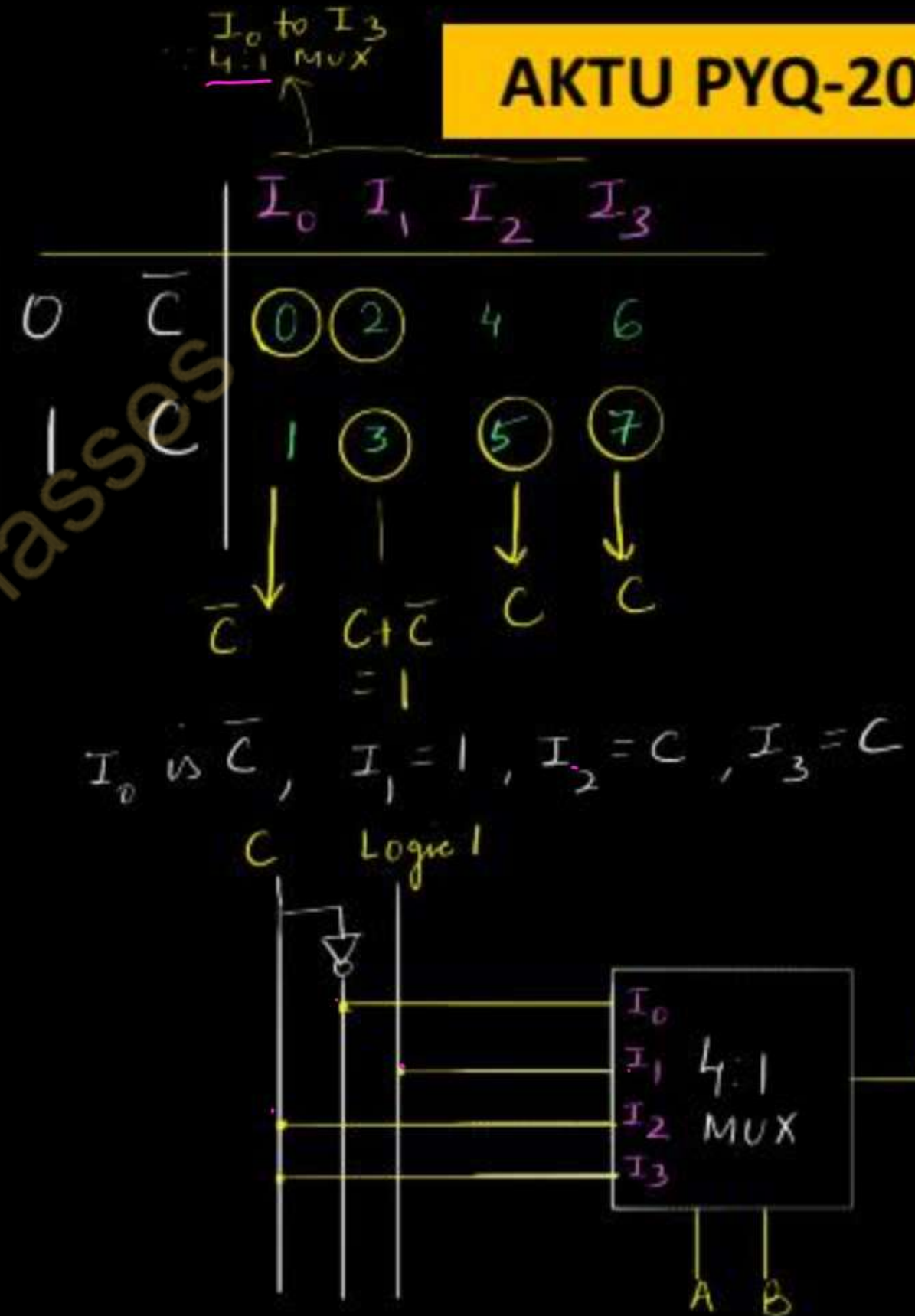
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4:1 MUX

$4 = 2^2$ select lines

A, B, C
Select lines for input

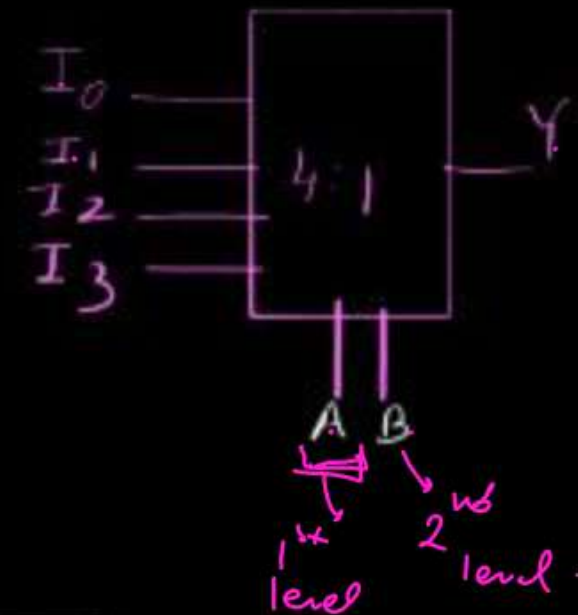
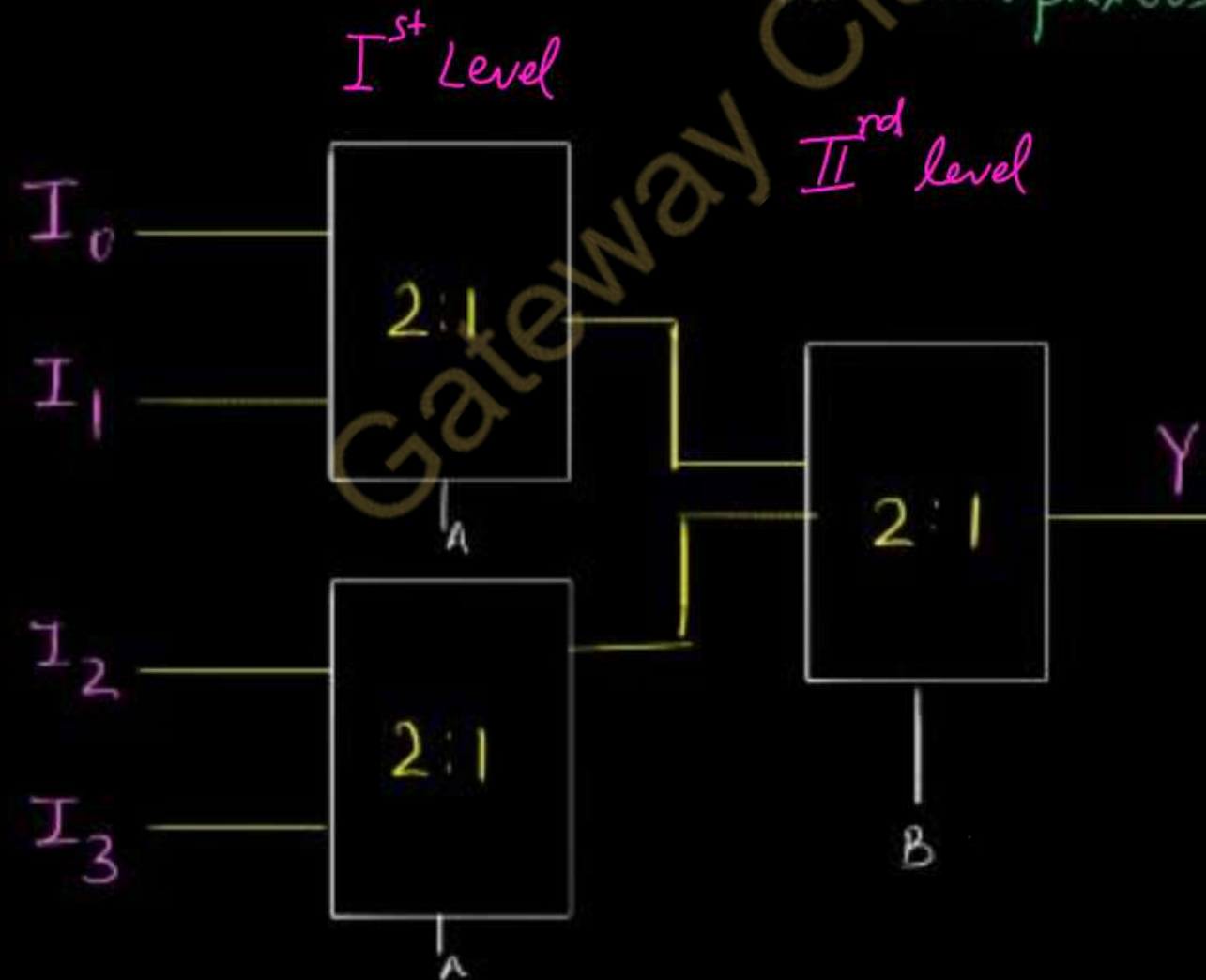
A	B	C	F	cell no.
0	0	0	I_0 ✓	0
0	0	1	I_1	1
0	1	0	I_2 ✓	2
0	1	1	I_3 ✓	3
1	0	0	I_4	4
1	0	1	I_5 ✓	5
1	1	0	I_6	6
1	1	1	I_7 ✓	7



Implement a 4:1 multiplexer using 2:1 multiplexer.

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I^{st} level $\frac{4}{2} = 2$
 II^{nd} level $\frac{2}{2} = 1$
 $2 + 1 = 3$
 Three 2:1 Multiplexers are used to implement 4:1 multiplexers.



A	B	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Que

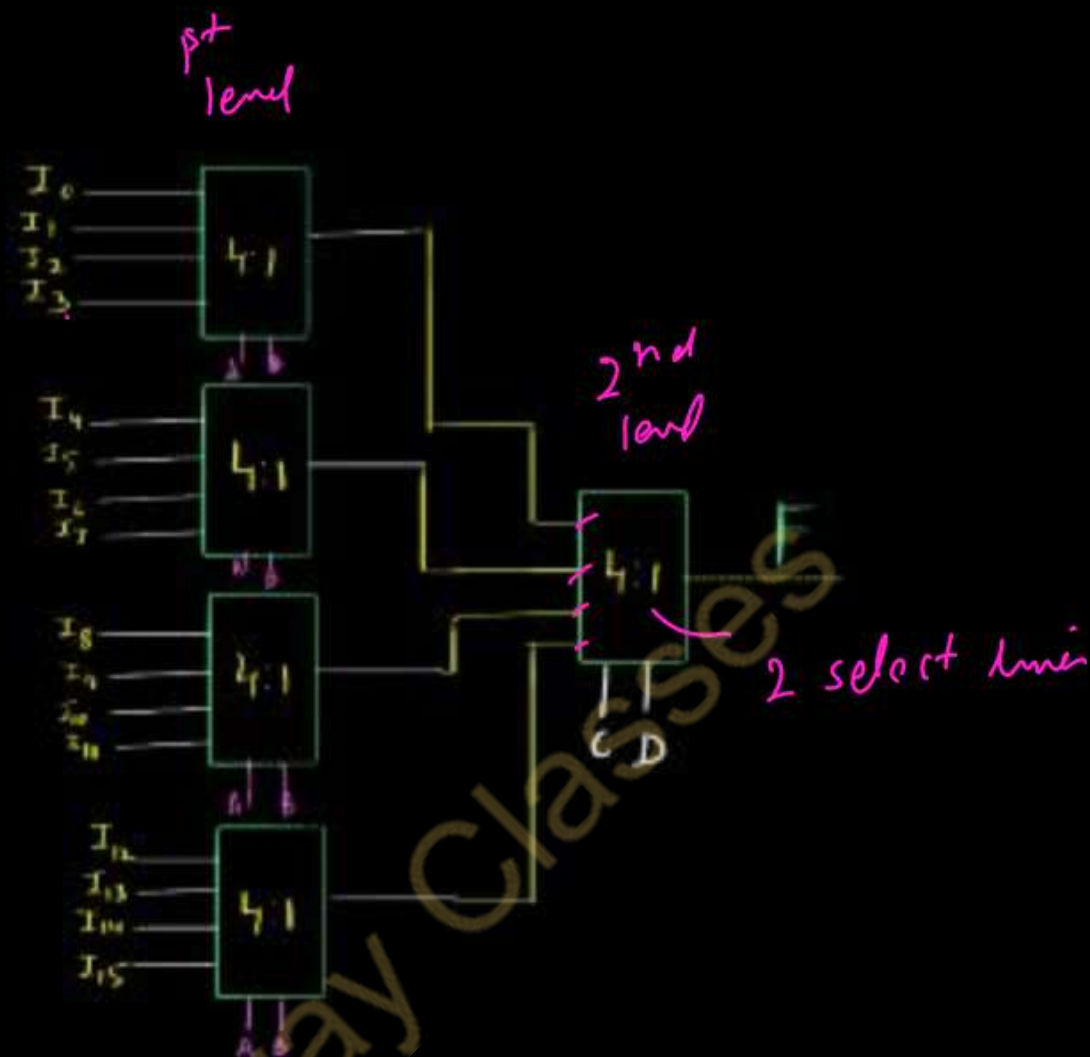
16:1 MUX using 4:1 MUX

$$\left. \begin{array}{l} \text{1st level} \\ \frac{16}{4} = 4 \\ \text{2nd level} \\ \frac{4}{4} = 1 \end{array} \right\} 4+1=5$$

16:1

$$16 = 2^4 \Rightarrow 4 \text{ select lines}$$

A, B, C, D
1st level 2nd level



How many 4 X 1 multiplexers are required to implement 64 X 1 multiplexer.

I^{st} level $\frac{64}{4} = 16$

II^{nd} level $\frac{16}{4} = 4$

III^{rd} level $\frac{4}{4} = 1$

$64:1 \leftarrow 4:1$

$16 + 4 + 1$

$= 21$

4:1 MUX are required to implement 64:1 MUX.

32:1 MUX using 4:1 MUX

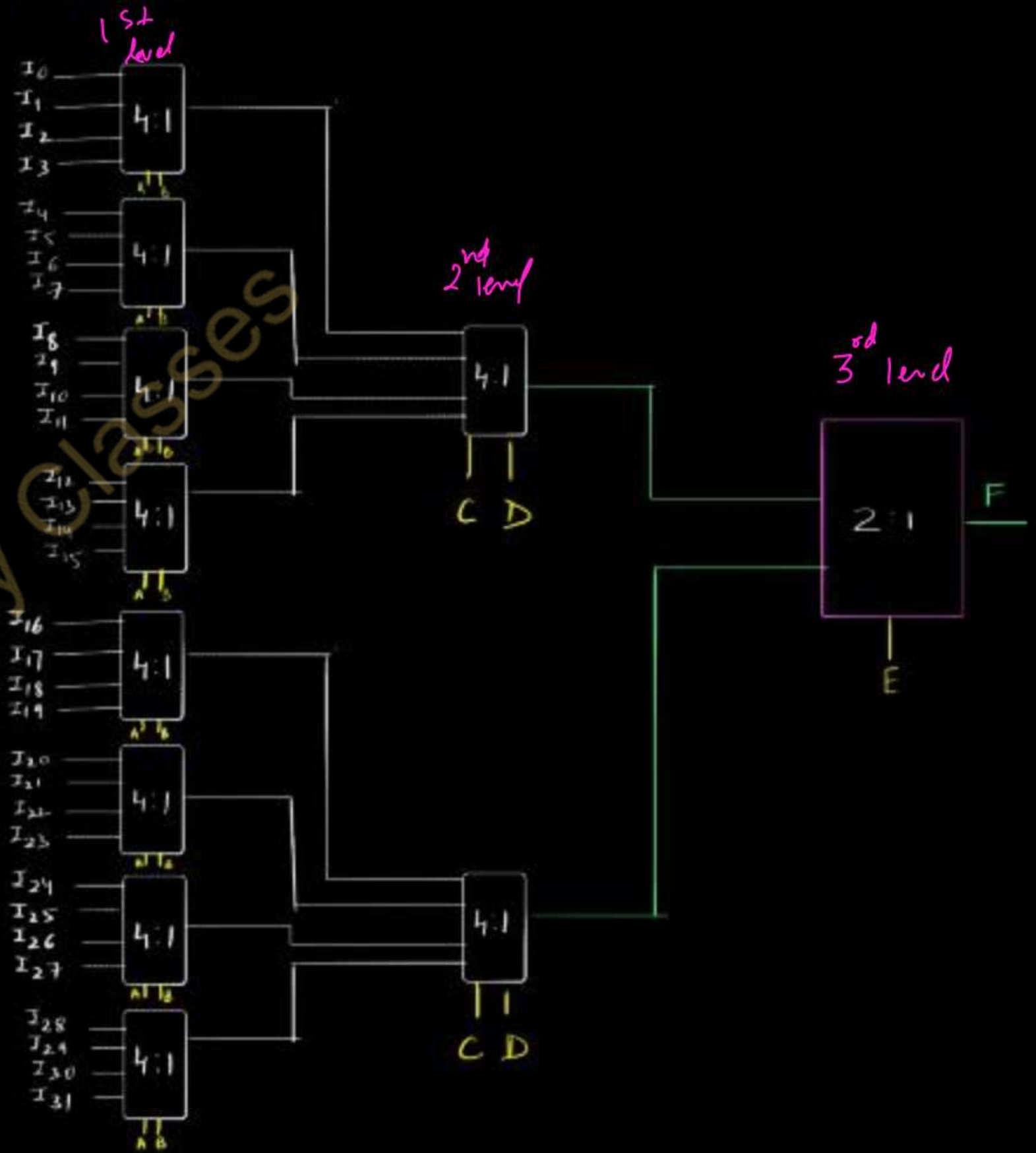
Que

1st level →
 32:1 ← 4:1
 2nd level →
 3rd level →

$$\begin{array}{l} \frac{32}{4} = 8 \\ \frac{8}{4} = 2 \\ \frac{2}{2} = 1 \end{array} \left. \begin{array}{l} 8+2 \\ 10 \\ \text{Ten} \end{array} \right\} 4:1 \text{ MUX}$$

→ one 2:1 MUX

32:1
 32 = 2⁵ → select lines
 A, B, C, D, E
 1st 2nd 3rd

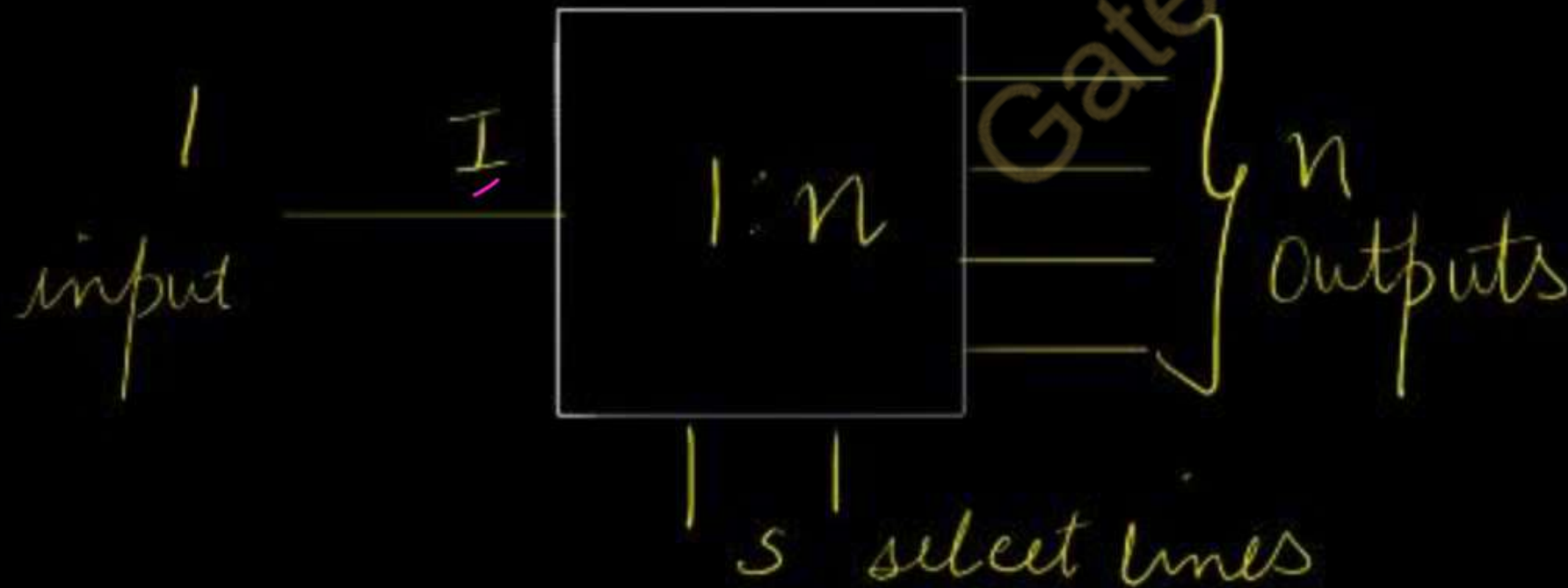


Demultiplexers

Demultiplexer is a combinational logic circuit that accepts a single input and distributes it over several output lines.

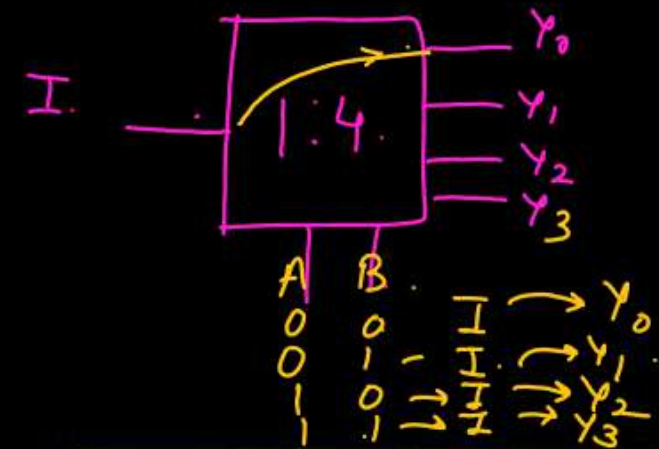
Demultiplexer is also termed as DEMUX in short.

As Demultiplexer is used to transmit the same data to different destinations, hence it is also known as **data distributor**.



1:4 De-mux.

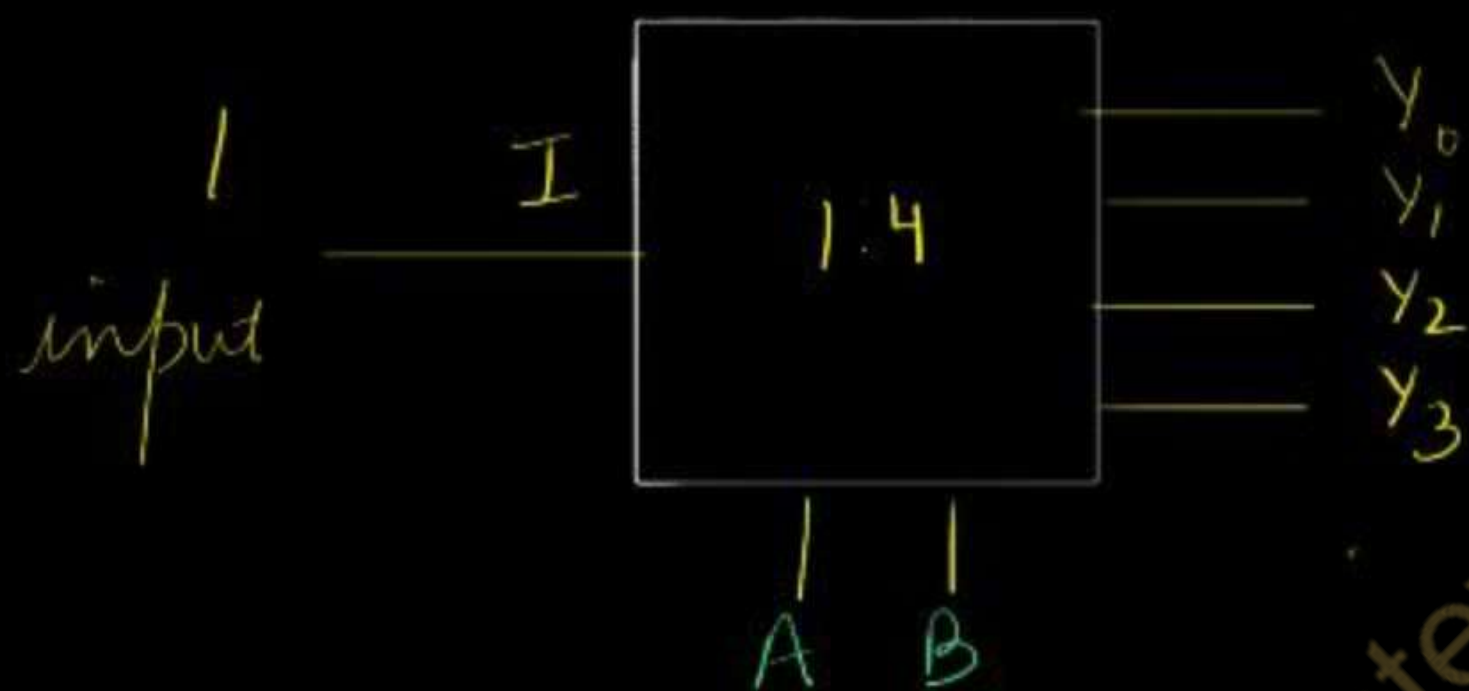
$$4 = 2^s$$
$$s = 2$$



$$n = 2^s$$

1.4 De-mux

$4 = 2^{\textcircled{2}} \rightarrow$ 2 select lines A, B



A	B	Output
0	0	Y_0
0	1	Y_1
1	0	Y_2
1	1	Y_3

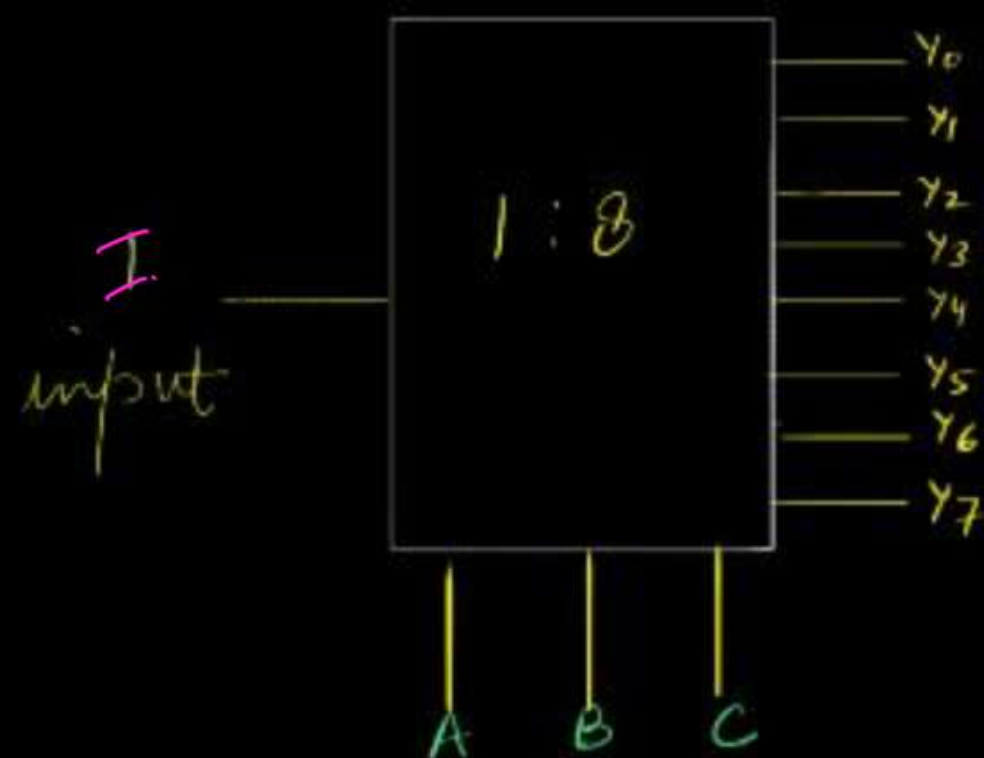
When $A=0, B=0$ I connects to Y_0

When $A=0, B=1$ I connects to Y_1

When $A=1, B=0$ I connects to Y_2

When $A=1, B=1$ I connects to Y_3

1.8 De-mux
 $8 = 2^3 = 3$ select lines
 A, B, C

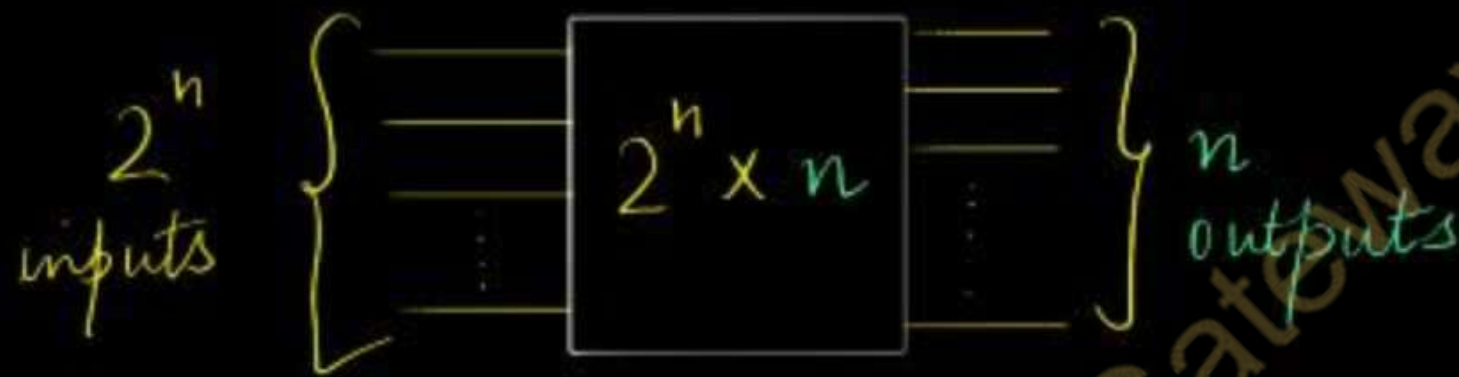


A	B	C	Output
0	0	0	Y_0
0	0	1	Y_1
0	1	0	Y_2
0	1	1	Y_3
1	0	0	Y_4
1	0	1	Y_5
1	1	0	Y_6
1	1	1	Y_7

Encoders

An encoder is a digital combinational circuit that converts multiple input signals into a binary code.

It typically has **one active input** at a time and generates a **binary output**.

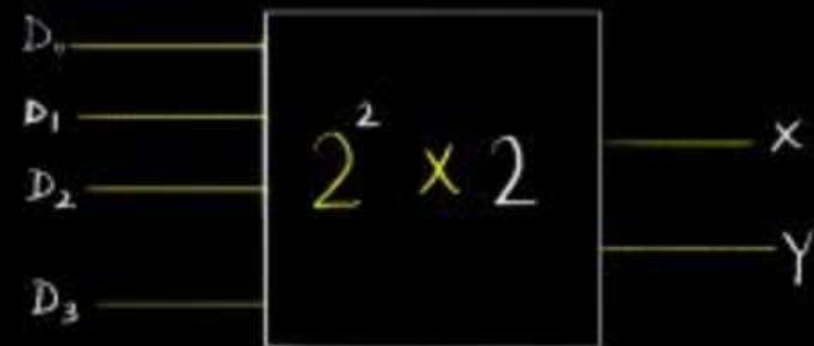


It is used at *transmitting* end of Communication System.

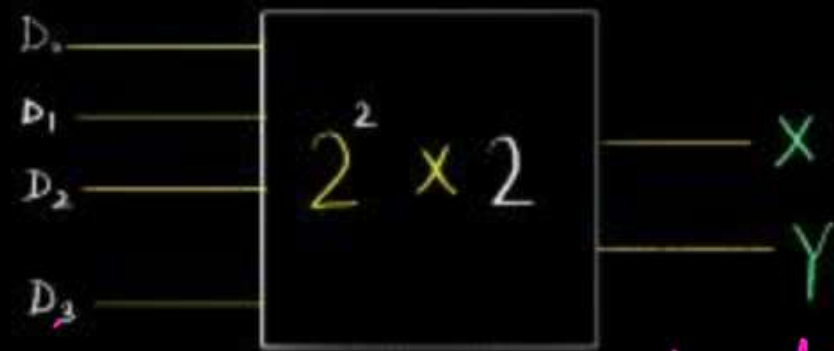
Encoders are useful in digital systems for data compression and efficient transmission.

eg $n = 2$

$2^2 = 4$
inputs



$2^2 = 4$
inputs



only one input is active at a single time

D_3	D_2	D_1	D_0	X	Y
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

$X = 1$ when $D_2 = 1$ or $D_3 = 1$
(Don't read 0)

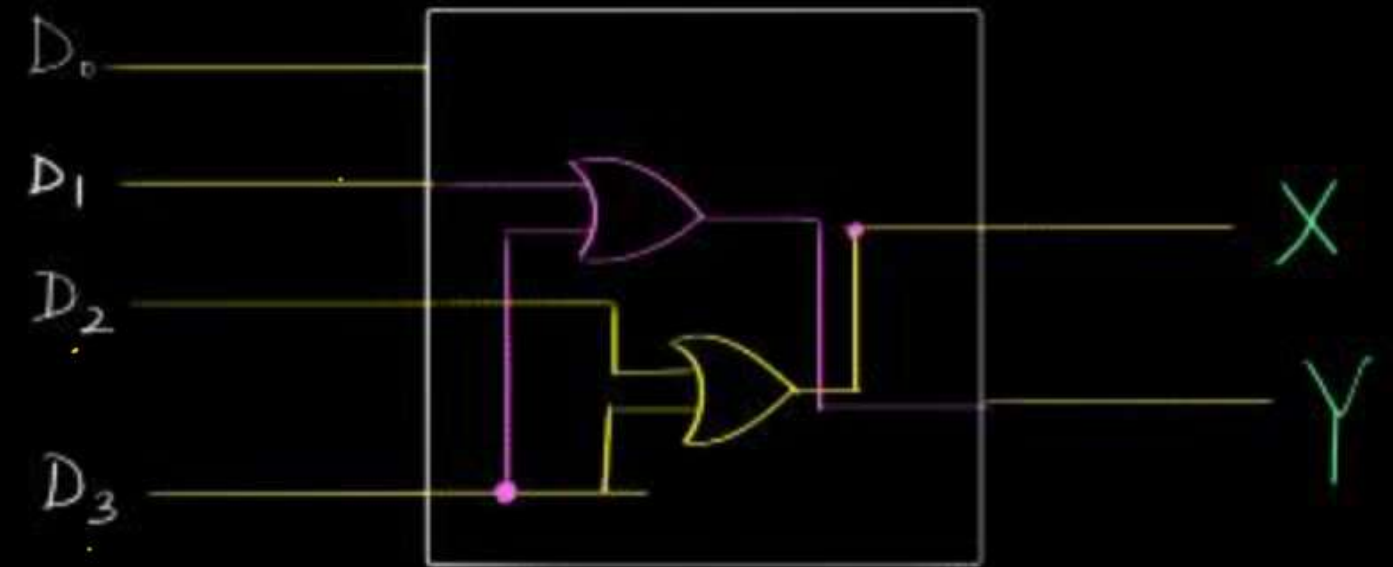
When D_0 is active, output is 00
 When D_1 is active, output is 01
 When D_2 is active, output is 10
 When D_3 is active, output is 11

$$X = D_2 + D_3$$

$$Y = D_1 + D_3$$

X is 1 when
 $D_2 = 1$
 or
 $D_3 = 1$

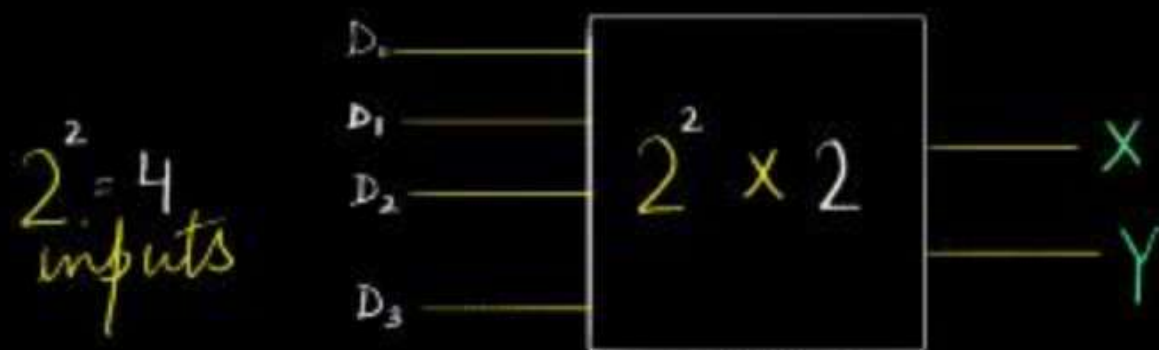
Y is 1 when
 $D_1 = 1$
 or
 $D_3 = 1$



(Simple Encoder)

Priority Encoder

A priority encoder produces output code based on the highest priority input.



But Priority order
 $\rightarrow$ Highest Priority

D_3
 D_2
 D_1
 $\downarrow$ D_0

	D_3	D_2	D_1	D_0	X	Y
4 th	0	0	0	1	0	0
3 rd	0	0	1	X	0	1
2 nd	0	1	X	X	1	0
1 st	1	X	X	X	1	1

Read 0
or

X $\rightarrow$ 0
or
1

Now output X is

$$X = \bar{D}_3 D_2 + D_3$$

$$X = D_3 + \bar{D}_3 D_2 \quad A + \bar{A}B = A + B$$

$$X = D_3 + D_2$$

$$Y = \bar{D}_3 \bar{D}_2 D_1 + D_3$$

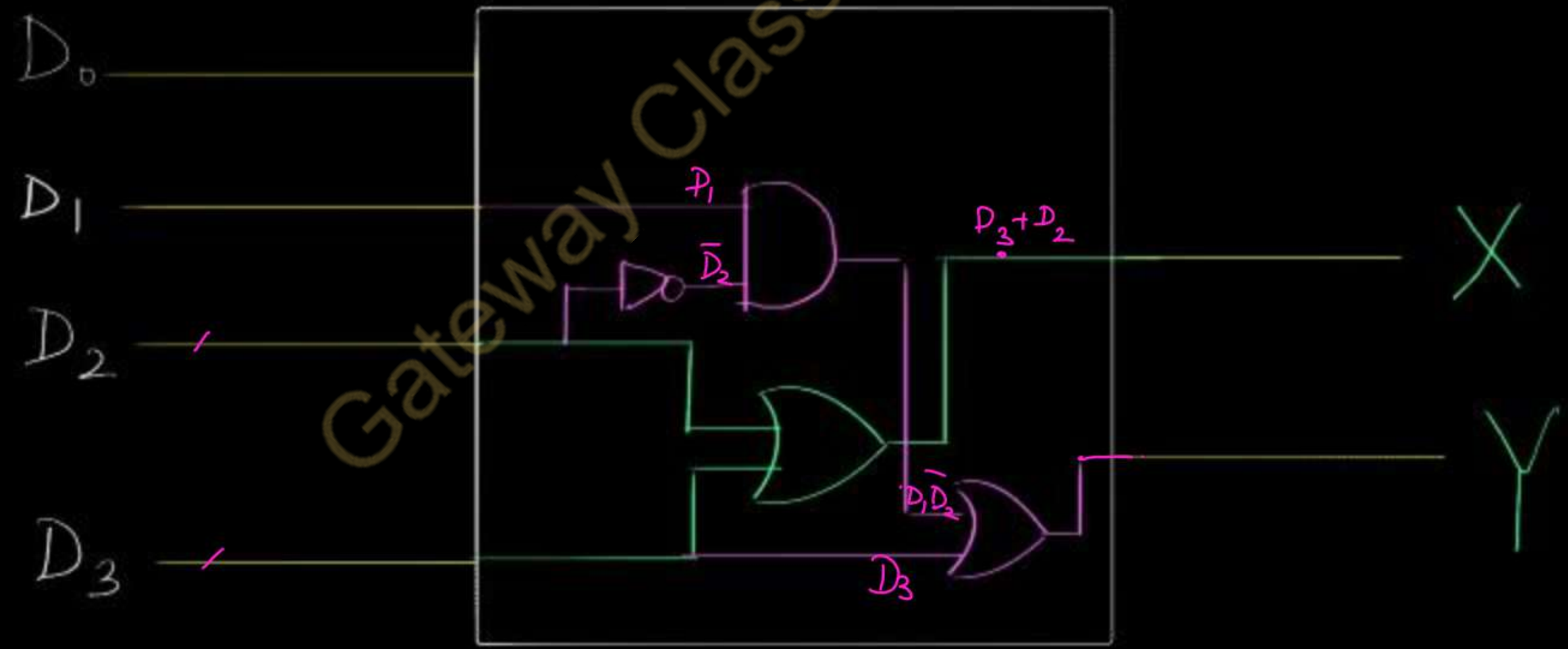
$$= D_3 + \bar{D}_3 \bar{D}_2 D_1$$

$$Y = D_3 + D_1 \bar{D}_2$$

$$X = D_3 + D_2$$

and

$$Y = D_3 + D_1 \bar{D}_2$$



Priority Encoder

Decoders

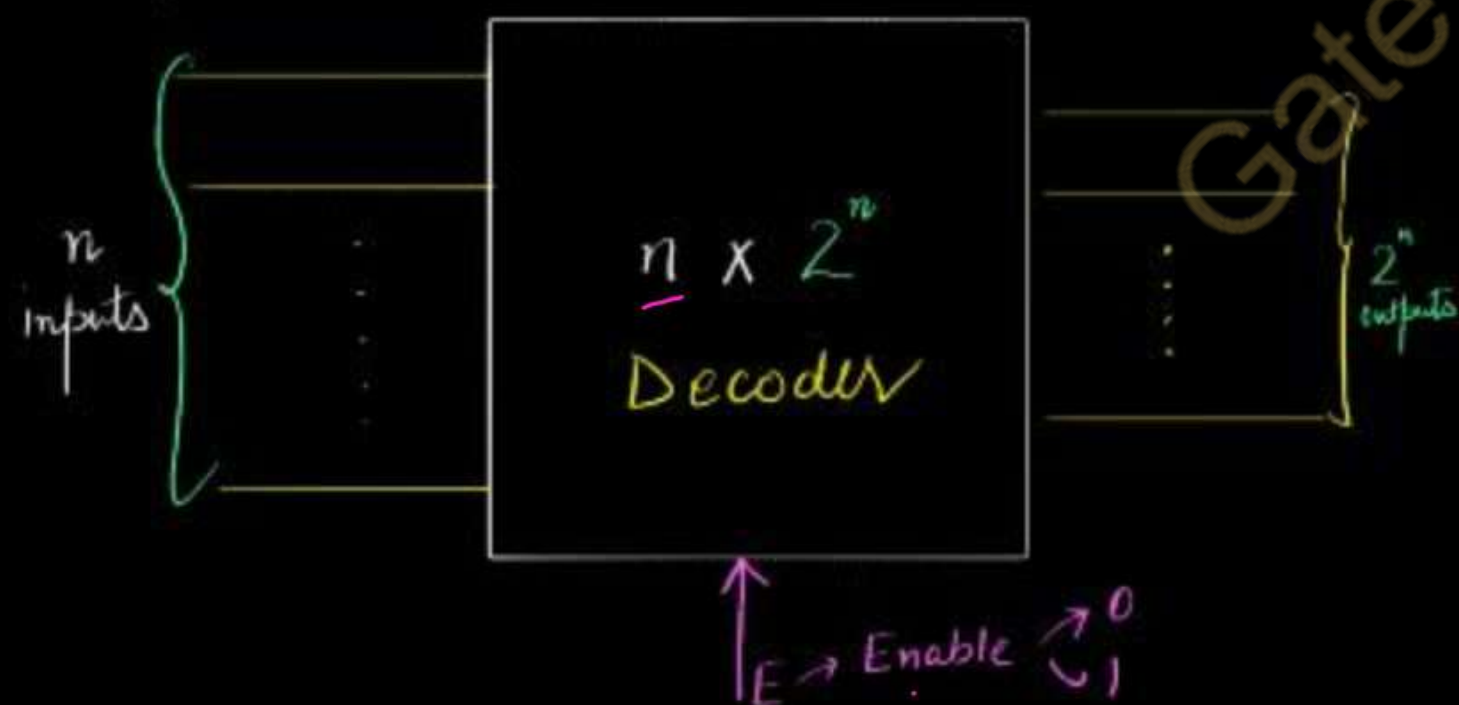
It is a combinational multi-input, multi-output logic circuit which decodes n inputs into 2^n possible outputs.

It is used at receiving end of Communication System.

Q - Demultiplexer is decoder circuit with an additional enabling input. Do you agree with the above statement?

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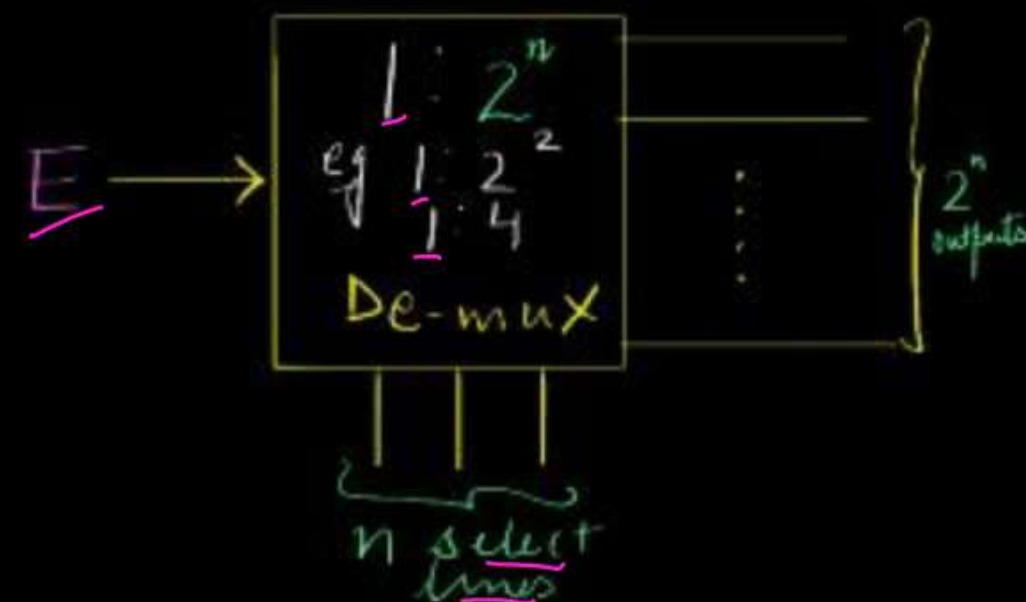
Yes, demultiplexer is decoder with $\bar{E}$.



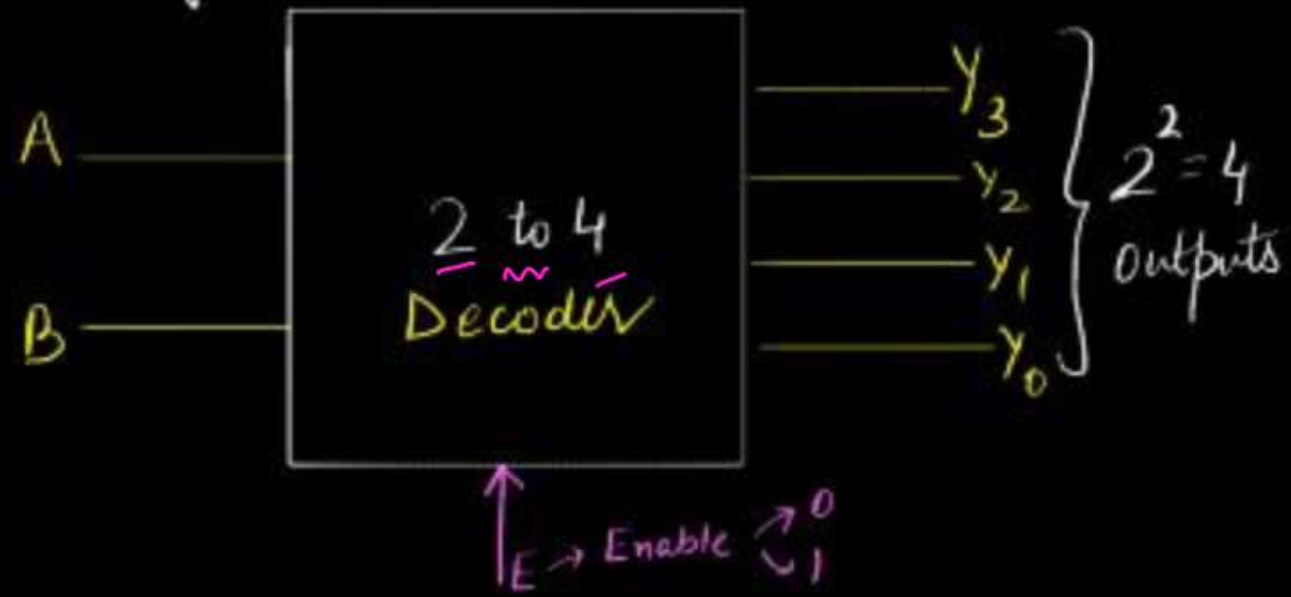
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When $E=0$, Decoder is disabled $\Rightarrow$ If inputs are given, output won't come.
When $E=1$, Decoder is enabled.

Treat $\bar{E}$ (Enable input) of Decoder as Data Input of De-mux.
and
Decoder inputs as Select lines of De-mux.



eg 2 to 4 Decoder



Truth Table

E	A	B	Y_3	Y_2	Y_1	Y_0
0	x	x	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

→ Decoder will provide 1 at Y_0

→ Decoder will provide 1 at Y_1

→ Decoder will provide 1 at Y_2

→ Decoder will provide 1 at Y_3

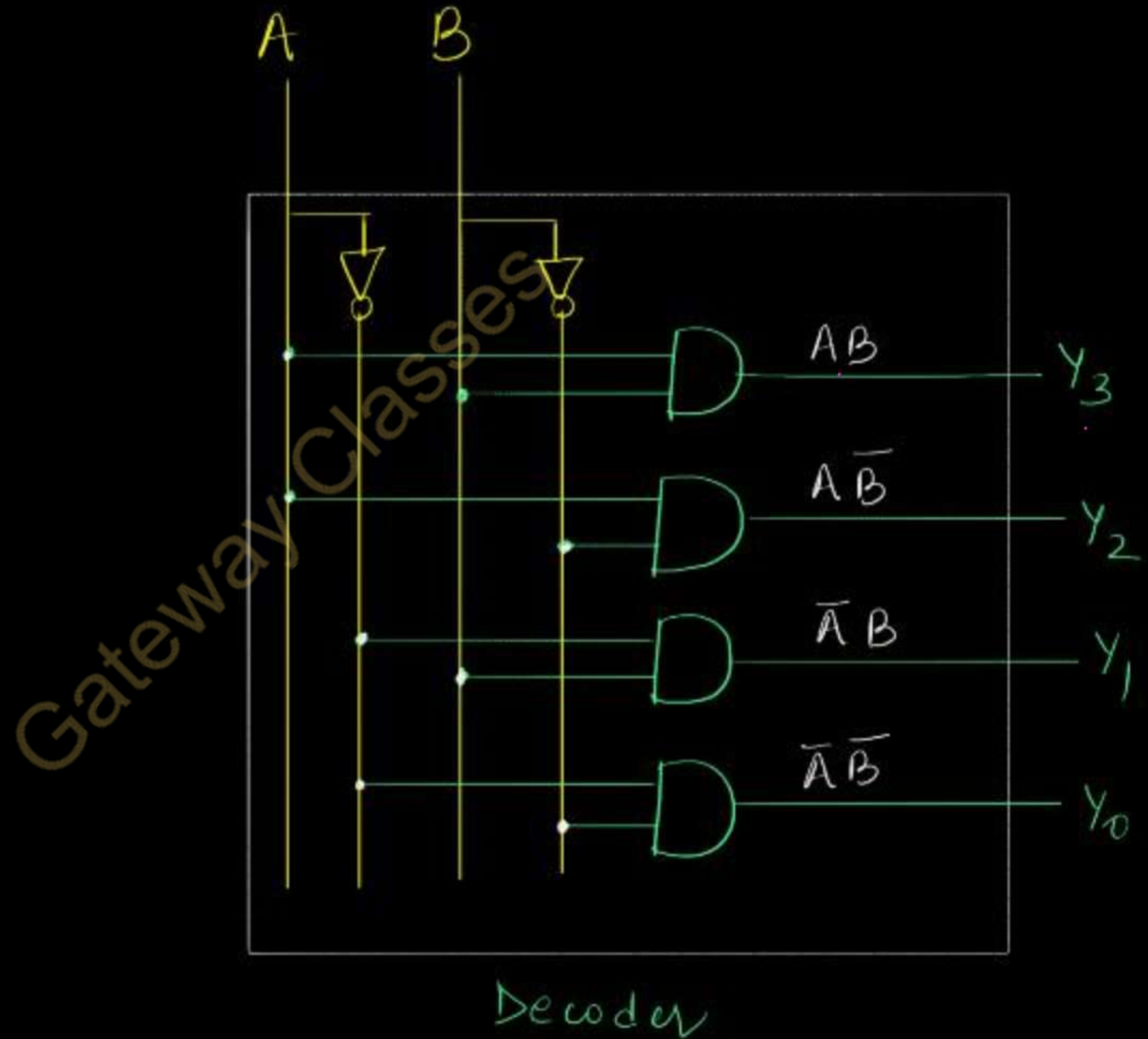
$$Y_3 = AB$$

$$Y_2 = A\bar{B}$$

$$Y_1 = \bar{A}B$$

$$Y_0 = \bar{A}\bar{B}$$

$$\begin{aligned}
 Y_3 &= AB \\
 Y_2 &= A\bar{B} \\
 Y_1 &=\bar{A}B \\
 Y_0 &=\bar{A}\bar{B}
 \end{aligned}$$



Design a 3:8 Decoder circuit using gates.

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3:8 Decoder

$n \times 2^n$

$n = 3$

3 inputs, 8 outputs

A, B, C

$y_7, y_6, y_5, y_4, y_3, y_2, y_1, y_0$

A	B	C	y_7	y_6	y_5	y_4	y_3	y_2	y_1	y_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

$$y_0 = \bar{A} \bar{B} \bar{C}$$

$$y_1 = \bar{A} \bar{B} C$$

$$y_2 = \bar{A} B \bar{C}$$

$$y_3 = \bar{A} B C$$

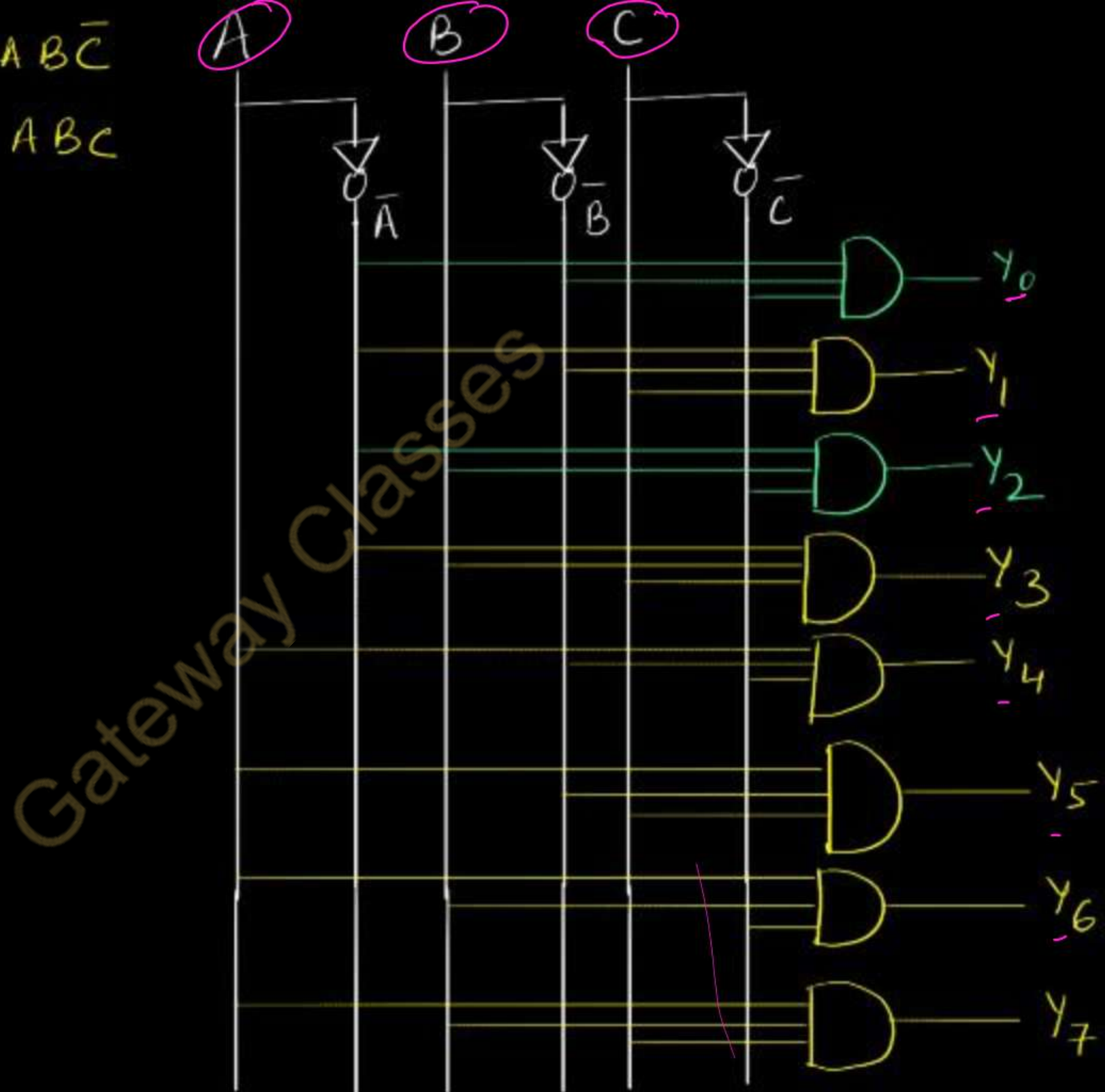
$$y_4 = A \bar{B} \bar{C}$$

$$y_5 = A \bar{B} C$$

$$y_6 = A B \bar{C}$$

$$y_7 = A B C$$

$$\begin{aligned}
 y_0 &= \bar{A}\bar{B}\bar{C}, & y_3 &= \bar{A}BC, & y_6 &= ABC\bar{C} \\
 y_1 &= \bar{A}B\bar{C}, & y_4 &= A\bar{B}\bar{C}, & y_7 &= ABC \\
 y_2 &= \bar{A}BC, & y_5 &= A\bar{B}C
 \end{aligned}$$



HALF ADDER

Half Adder is a combinational logic circuit designed to add **Two Single bit numbers**.

It has two inputs (**A**, **B**) and two outputs (Sum **S**, Carry **C**)



$$\begin{array}{r} A \rightarrow 0 \\ B \rightarrow 0 \\ \hline S \rightarrow 0 \\ C \rightarrow 0 \end{array} \quad \begin{array}{r} 0 \\ + 1 \\ \hline 1 \\ 0 \end{array} \quad \begin{array}{r} 1 \\ + 0 \\ \hline 1 \\ 0 \end{array} \quad \begin{array}{r} 1 \\ + 1 \\ \hline 0 \\ 1 \end{array} \quad \left. \vphantom{\begin{array}{r} 0 \\ + 1 \\ \hline 1 \\ 0 \end{array}} \right\}$$

Truth Table

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$0 + 0 = 0$$

$$S = 0, C = 0$$

$$0 + 1 = 1$$

$$S = 1, C = 0$$

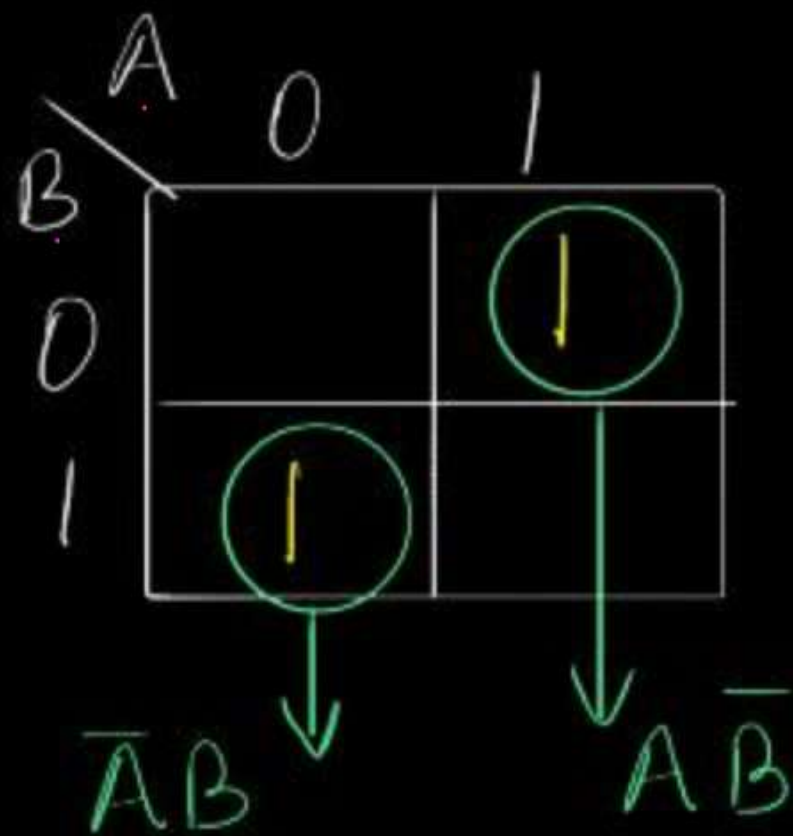
$$1 + 0 = 1$$

$$S = 1, C = 0$$

$$1 + 1 = 0 \text{ carry } 1$$

$$S = 0, C = 1$$

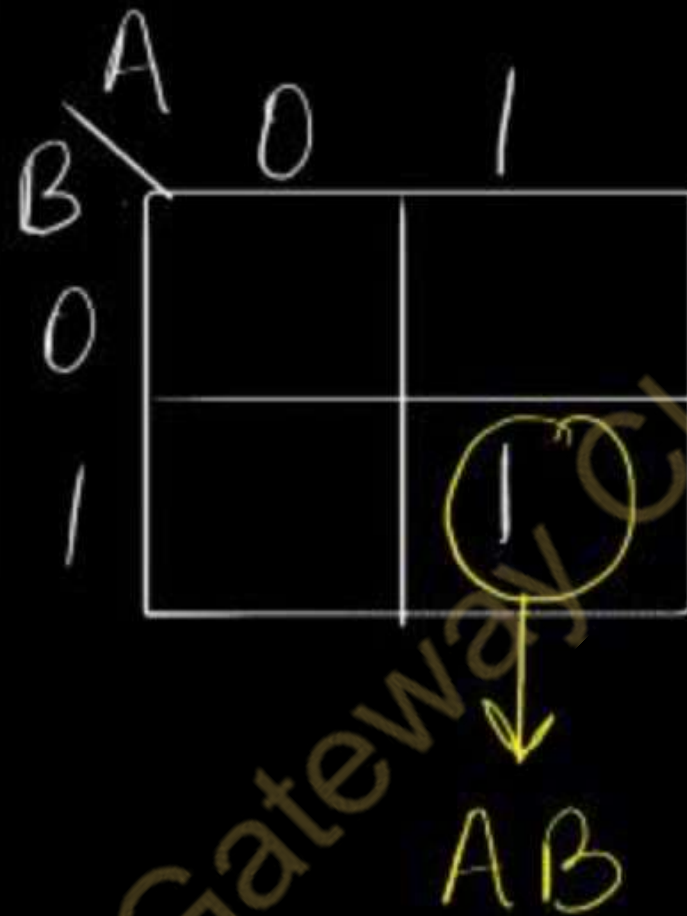
K-Map
For S



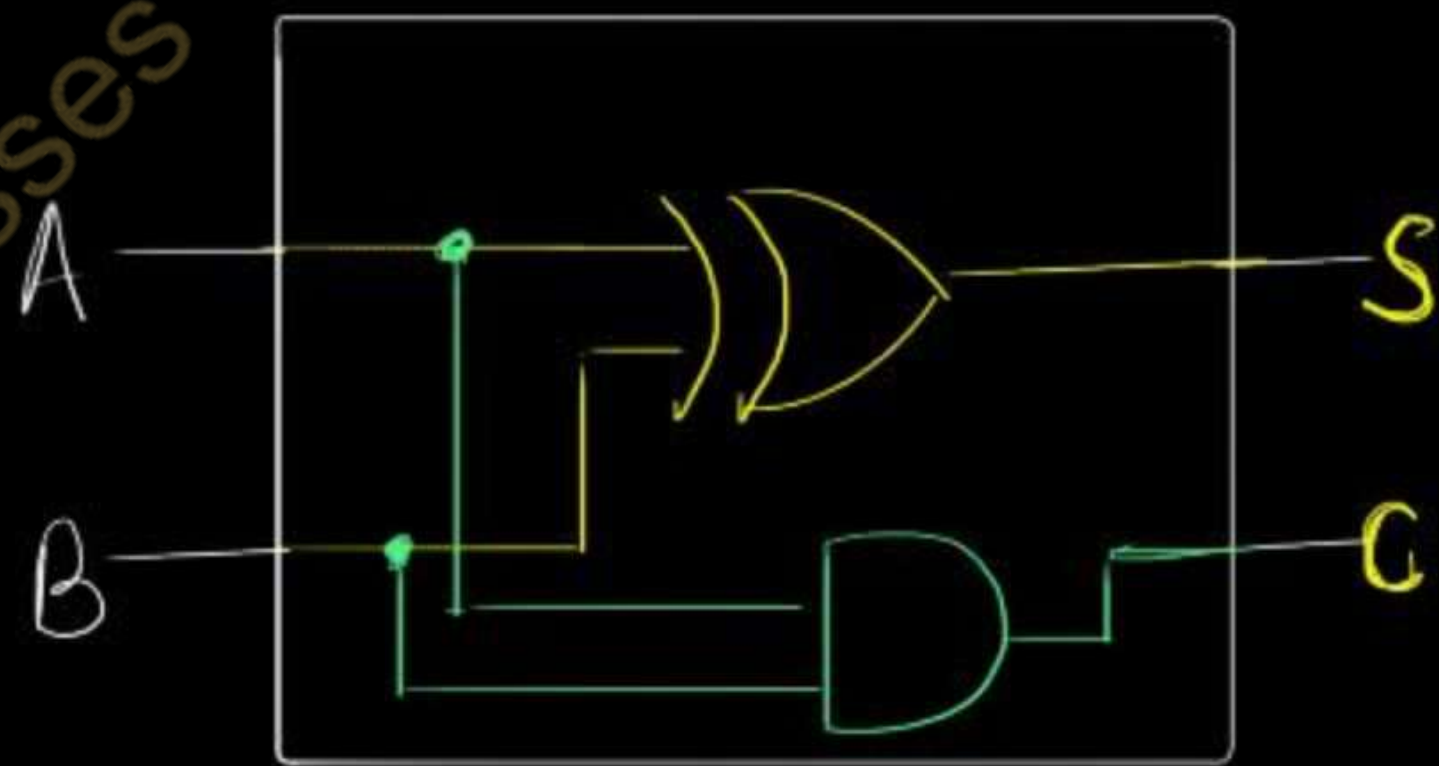
$$S = \bar{A}B + A\bar{B}$$

$$S = A \oplus B$$

For C



$$C = AB$$



Half Adder

FULL ADDER

Full Adder is a combinational circuit that adds three inputs and produces two outputs.

The first two inputs are A and B and the third input is an input carry as C_{in}.

The output carry is designated as C_{out} and the normal output is designated as S which is SUM.

When two digits number are added

Cases

Rules (Binary):
 $0+0=0$ | $1+1=0$ carry 1
 $0+1=1$ | $1+1+1=1$ carry 1
 $1+0=1$

$A \rightarrow \overset{A_1}{0} \overset{A_0}{0}$
 $B \rightarrow \overset{B_1}{+} \overset{B_0}{0} \overset{B_0}{0}$
 $\hline 00$

00	00	00
$+01$	$+10$	$+11$
$\hline 01$	$\hline 10$	$\hline 11$

$$\begin{array}{r} 01 \\ + 00 \\ \hline 01 \end{array}$$

$$\begin{array}{r} 10 \\ + 00 \\ \hline 10 \end{array}$$

$$\begin{array}{r} 11 \\ + 00 \\ \hline 11 \end{array}$$

$$\begin{array}{r} \overset{C_{in}}{1} \\ 01 \\ + 01 \\ \hline 10 \end{array}$$

$$\begin{array}{r} 01 \\ + 10 \\ \hline 11 \end{array}$$

$$\begin{array}{r} 10 \\ + 10 \\ \hline 100 \end{array}$$

$$\begin{array}{r} 1 \\ 11 \\ + 01 \\ \hline 100 \end{array}$$

$$\begin{array}{r} 1 \\ 01 \\ + 11 \\ \hline 100 \end{array}$$

$$\begin{array}{r} 10 \\ + 11 \\ \hline 101 \end{array}$$

$$\begin{array}{r} 1 \\ 11 \\ + 11 \\ \hline 110 \end{array}$$

Truth Table

A	B	C _{in}	S	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-Map

For S

AB \ C _{in}	00	01	11	10
0		1		1
1	1		1	

$\bar{A}\bar{B}C_{in}$ $\bar{A}B\bar{C}_{in}$ ABC_{in} $A\bar{B}\bar{C}_{in}$

$$\begin{aligned}
 S &= \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + ABC_{in} + A\bar{B}\bar{C}_{in} \\
 &= C_{in}(AB + \bar{A}\bar{B}) + \bar{C}_{in}(\bar{A}B + A\bar{B}) \\
 &= C_{in}(A \oplus B) + \bar{C}_{in}(A \oplus B)
 \end{aligned}$$

Let $C_{in} = X$, $A \oplus B = Y$

$\therefore \bar{C}_{in} = \bar{X}$, $A \oplus B = Y$

$$S = XY + \bar{X}Y = X \oplus Y \Rightarrow S = C_{in} \oplus A \oplus B$$

$$S = A \oplus B \oplus C_{in}$$

For C_{out} ,

$C_{in} \backslash AB$	00	01	11	10
0			1	
1		1	1	1

Annotations:
- A pink oval around the '1' in row 0, column 11 points to AB .
- A yellow oval around the '1' in row 1, column 01 points to BC_{in} .
- A red oval around the '1's in row 1, columns 11 and 10 points to AC_{in} .

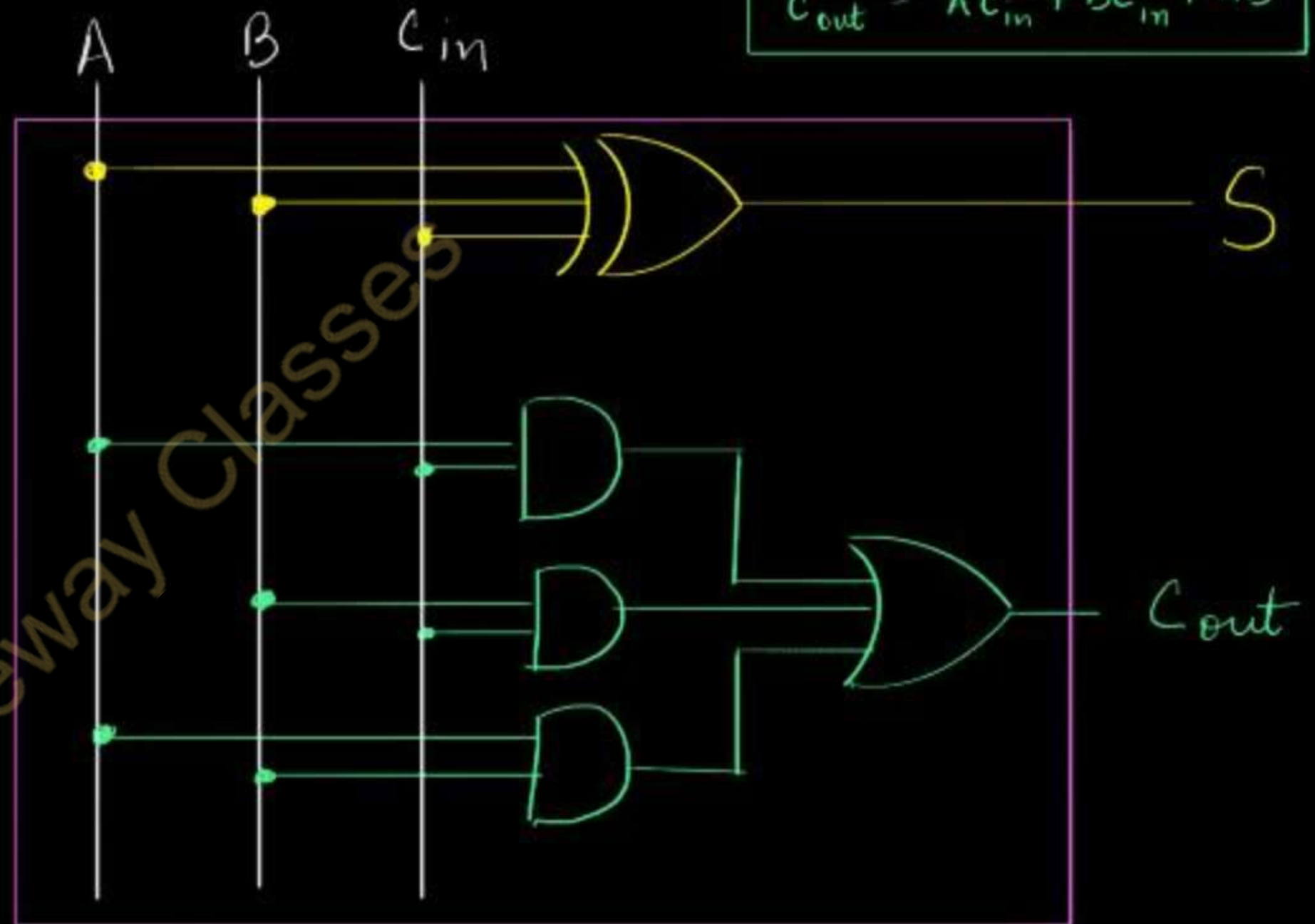
✓

$$C_{out} = AC_{in} + BC_{in} + AB$$

Logic Diagram

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = AC_{in} + BC_{in} + AB$$



FULL ADDER USING HALF ADDERS

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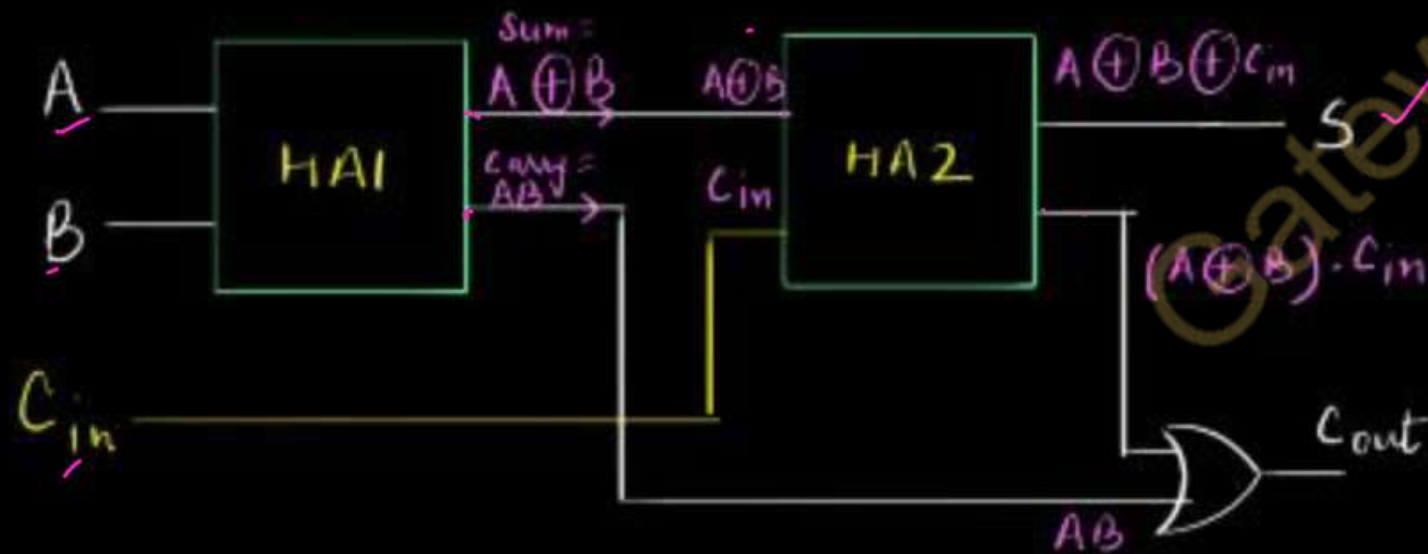
Expressions for Full Adder

$$S = A \oplus B \oplus C_{in} \quad - (1)$$

$$C_{out} = AC_{in} + BC_{in} + AB \quad - (2)$$

Now, Full Adder can be made from 2 Half Adders and 1 OR Gate

Sum = (Input 1) $\oplus$ (Input 2)
Carry = (Input 1) $\cdot$ (Input 2)



$$S = A \oplus B \oplus C_{in}$$

(Same as (1))

Now,

$$C_{out} = (A \oplus B) \cdot C_{in} + AB$$

$$= (\bar{A}\bar{B} + \bar{A}B)C_{in} + AB$$

$$= \bar{A}\bar{B}C_{in} + \bar{A}BC_{in} + AB$$

$$= \bar{A}\bar{B}C_{in} + B(A + \bar{A}C_{in})$$

$$= \bar{A}\bar{B}C_{in} + B(A + C_{in})$$

$x + \bar{x}y = x + y$

$$= \bar{A}\bar{B}C_{in} + AB + BC_{in}$$

$$= A(B + \bar{B}C_{in}) + BC_{in}$$

$$= A(B + C_{in}) + BC_{in}$$

$$C_{out} = AB + AC_{in} + BC_{in} \quad (\text{Same as (2)})$$

FULL ADDER USING NAND GATES

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$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = AC_{in} + BC_{in} + AB$$

$$(I) S = A \oplus B \oplus C_{in}$$

or

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + AB\bar{C}_{in} + A\bar{B}C_{in}$$

Double inversion

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + AB\bar{C}_{in} + A\bar{B}C_{in}$$

$$= \bar{A}\bar{B}C_{in} \cdot \bar{A}B\bar{C}_{in} \cdot AB\bar{C}_{in} \cdot A\bar{B}C_{in}$$

stop

$$(II) C_{out} = AB + AC_{in} + BC_{in}$$

Double Inversion.

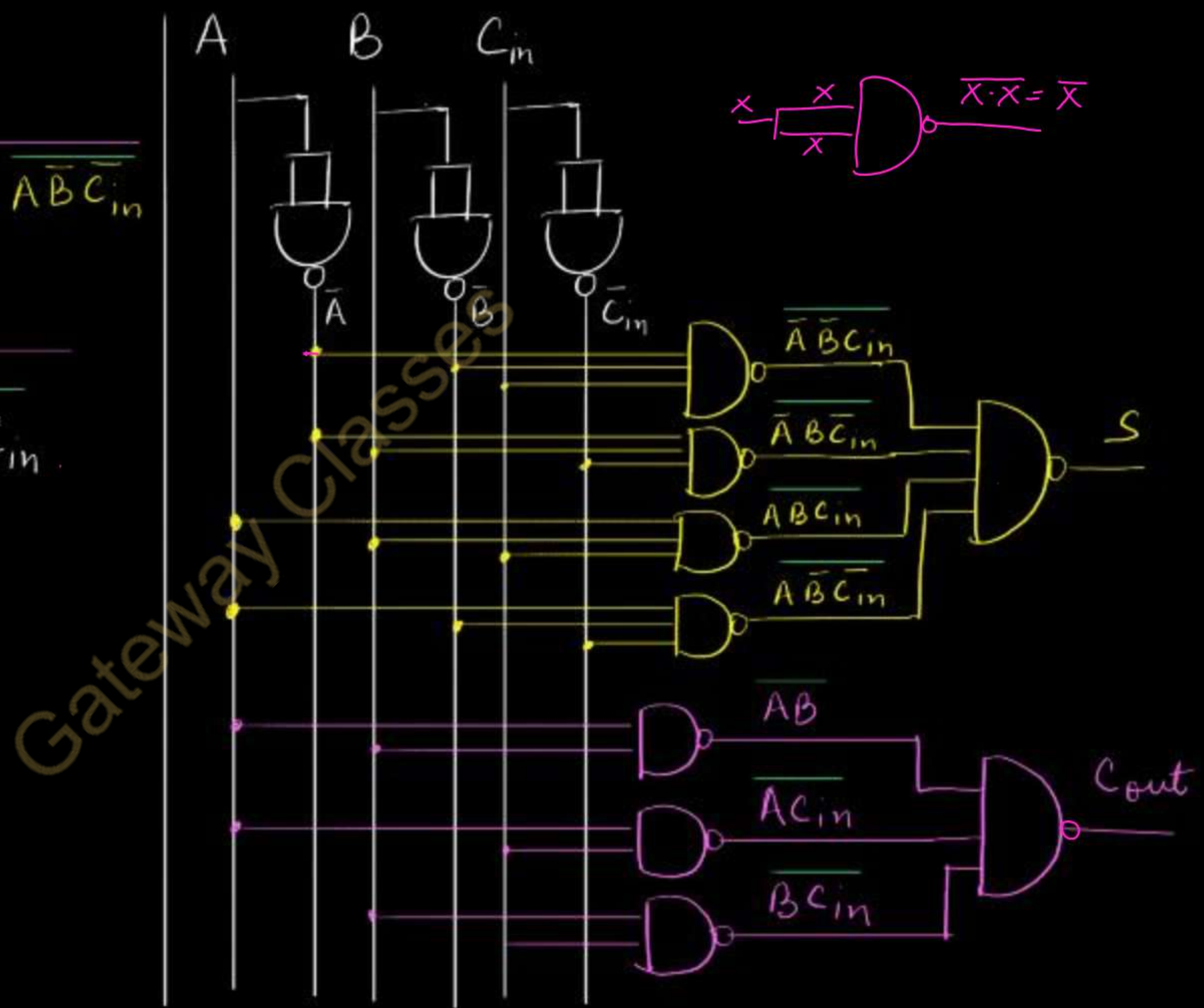
$$C_{out} = AB + AC_{in} + BC_{in}$$

$$= \bar{A}\bar{B} \cdot \bar{A}C_{in} \cdot \bar{B}C_{in}$$

stop

$$S = \overline{A}\overline{B}C_{in} \cdot \overline{A}B\overline{C}_{in} \cdot ABC_{in} \cdot A\overline{B}\overline{C}_{in}$$

$$C_{out} = \overline{AB} \cdot \overline{AC_{in}} \cdot \overline{BC_{in}}$$



HALF SUBTRACTOR

AKTU PYQ-2024-25

A half subtractor is a digital logic circuit that performs the binary subtraction of two single-bit binary numbers.

It has **two inputs**, **A** and **B**, and two outputs, **Difference** and **Borrow**.

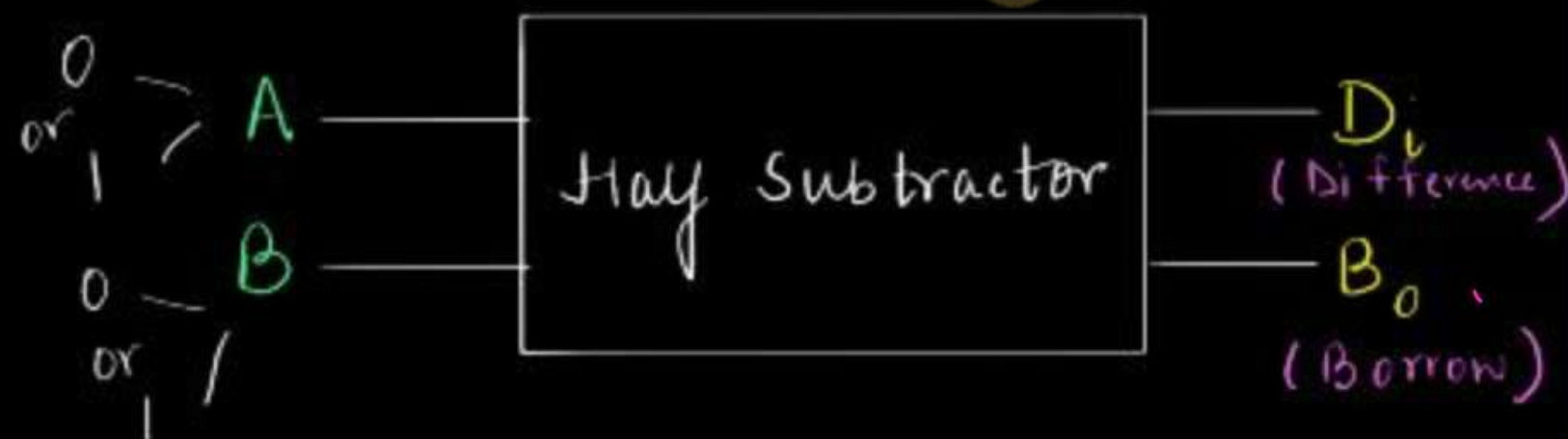
The Difference output represents the result of subtracting B from A, while the Borrow output indicates whether a borrow is needed when A is smaller than B.

$$\begin{array}{r} A \rightarrow 0 \\ B \rightarrow - 0 \\ \hline D_i \rightarrow 0 \\ B_o \rightarrow 0 \end{array}$$

$$\begin{array}{r} 0 \\ - 1 \\ \hline 1 \\ 1 \end{array} \quad \begin{array}{r} 1 \\ - 0 \\ \hline 1 \\ 0 \end{array} \quad \begin{array}{r} 1 \\ - 1 \\ \hline 0 \\ 0 \end{array}$$

Truth Table

A	B	D_i	B_o
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



$$\left\{ \begin{array}{l} 0 - 0 = 0 \\ 0 - 1 = 1 \\ 1 - 0 = 1 \\ 1 - 1 = 0 \end{array} \right. \quad \begin{array}{l} \text{no borrow} \\ \text{borrow} \\ \text{no borrow} \\ \text{no borrow} \end{array}$$

A	B	D_i	B_0
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

K-Map for D_i

A \ B	0	1
0		1
1	1	

$\rightarrow A\bar{B}$ (for cell 0,1)
 $\rightarrow \bar{A}B$ (for cell 1,0)

K-Map for B_0

A \ B	0	1
0		
1	1	

$\rightarrow \bar{A}B$ (for cell 1,0)

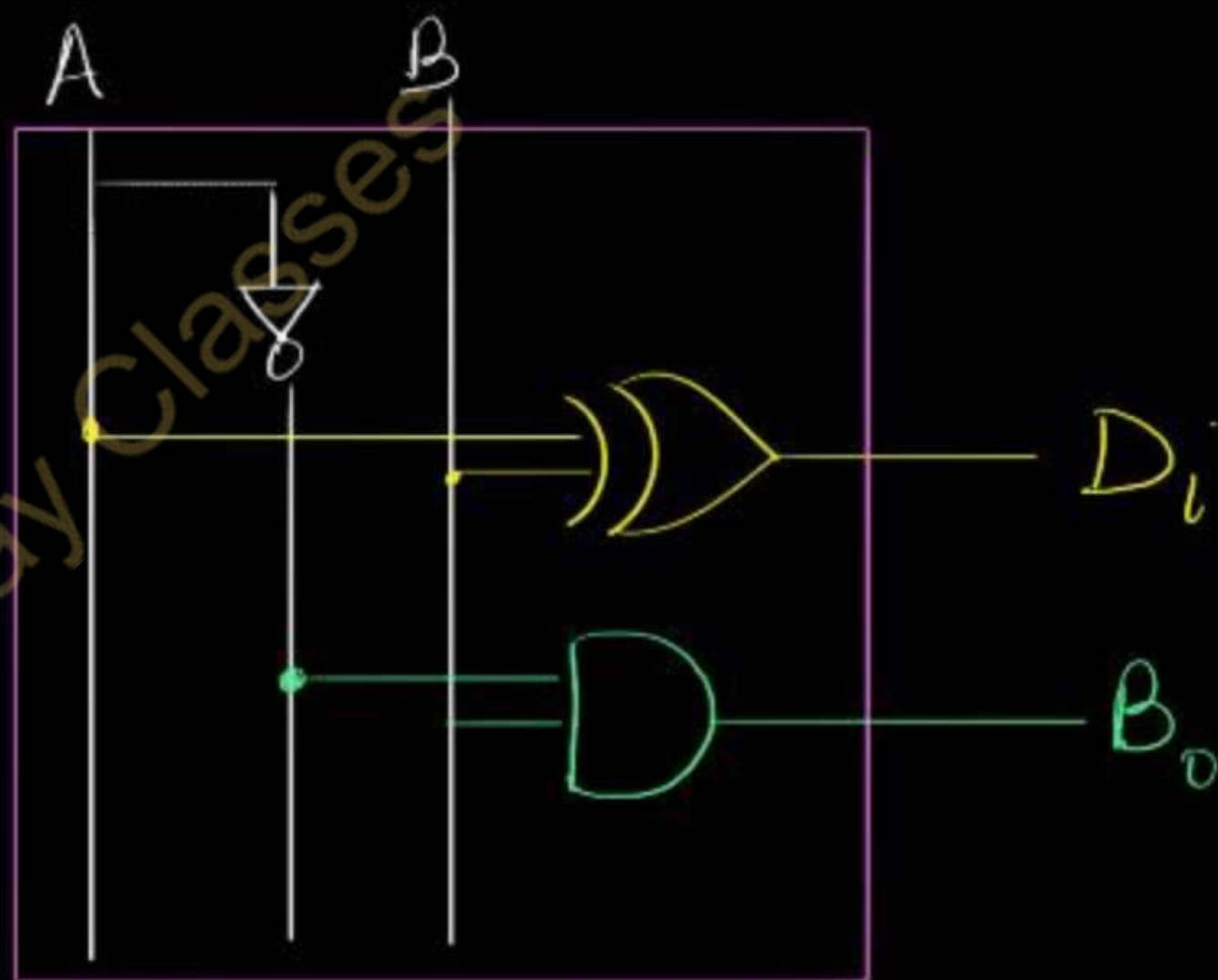
$$D_i = A\bar{B} + \bar{A}B$$

$$= A \oplus B$$

$$B_0 = \bar{A}B$$

$$D_i = A \oplus B$$

$$B_0 = \bar{A}B$$



(1-bit subtractor)

FULL SUBTRACTOR

A Full Subtractor is a combinational circuit used to perform binary subtraction.

It has three inputs:

A (Minuend)

B (Subtrahend)

B-in (Borrow-in from the previous stage)

It produces two outputs:

Difference (D): The result of the subtraction.

Borrow-out (Bout): Indicates if a borrow is needed for the next stage.



A	B	B _{in}	D _i	B _o
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

A	B	B_{in}	D_i	B_o
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

K-Map for D_i

B_{in} \ AB	00	01	11	10
0		1		1
1	1		1	

$\bar{A}\bar{B}\bar{B}_{in}$ (pointing to 01, 0)
 $A\bar{B}\bar{B}_{in}$ (pointing to 10, 0)
 $\bar{A}B\bar{B}_{in}$ (pointing to 01, 1)
 $AB\bar{B}_{in}$ (pointing to 11, 1)

K-Map for B_o

B_{in} \ AB	00	01	11	10
0		1		
1	1	1	1	

$\bar{A}B$ (pointing to 01, 0)
 $\bar{A}B_{in}$ (pointing to 00, 1)
 $B B_{in}$ (pointing to 01, 1 and 11, 1)

$$\begin{aligned}
 D &= \bar{A}\bar{B}\bar{B}_{in} + A\bar{B}\bar{B}_{in} + \bar{A}B\bar{B}_{in} + AB\bar{B}_{in} \\
 &= \bar{B}_{in}(A\bar{B} + \bar{A}\bar{B}) + \bar{B}_{in}(A\bar{B} + \bar{A}B) \\
 &= \bar{B}_{in}(A \odot B) + \bar{B}_{in}(A \oplus B)
 \end{aligned}$$

Let $X = B_{in}$ and $Y = A \oplus B$
 $\therefore \bar{X} = \bar{B}_{in}$ and $\bar{Y} = A \odot B$

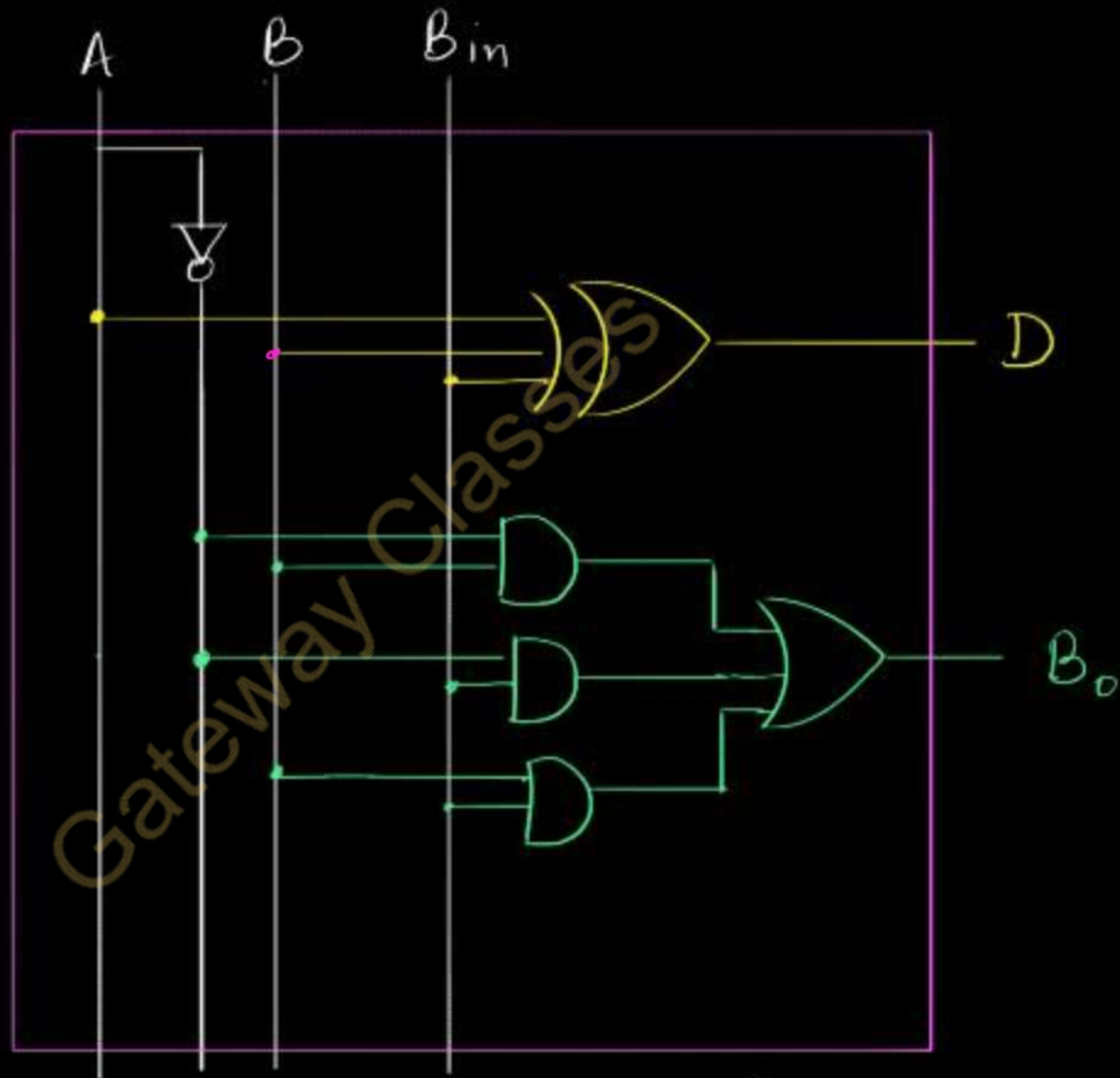
$$\begin{aligned}
 D &= X\bar{Y} + \bar{X}Y \\
 &= X \oplus Y \\
 &= B_{in} \oplus A \oplus B
 \end{aligned}$$

$$D = A \oplus B \oplus B_{in} \quad \checkmark$$

Now, $B_o = \bar{A}B + \bar{A}B_{in} + BB_{in}$

$$D = A \oplus B \oplus B_{in}$$

$$B_o = \bar{A}B + \bar{A}B_{in} + BB_{in}$$



(Full Subtractor)

FULL SUBTRACTOR USING HALF SUBTRACTORS

$$D = A \oplus B \oplus B_{in}$$

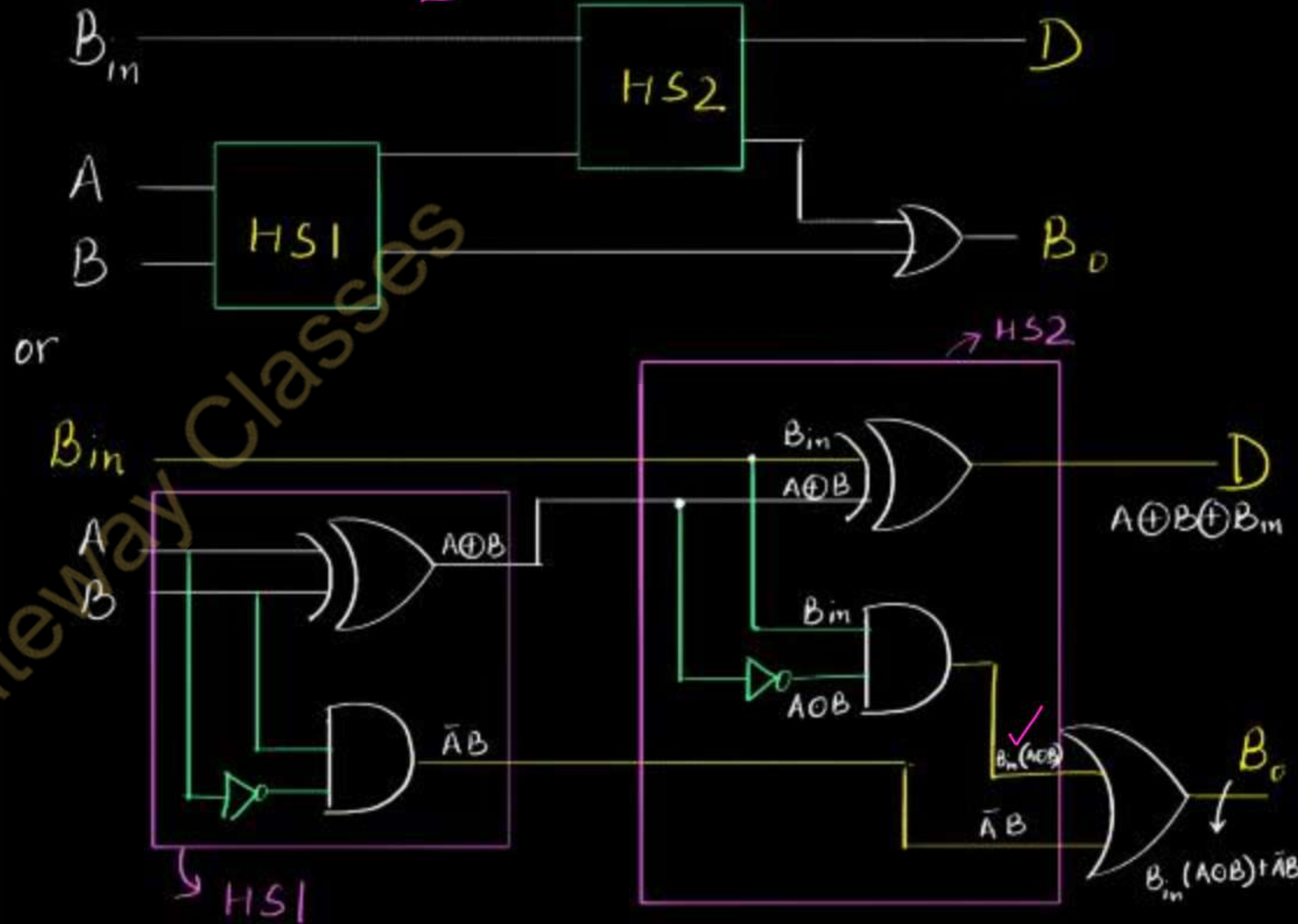
* For B_0 (Do not do pairing in K-Map)

$B_{in} \backslash AB$	00	01	11	10
0		1		
1	1	1	1	

$$B_0 = \bar{A}\bar{B}B_{in} + \bar{A}B\bar{B}_{in} + \bar{A}BB_{in} + ABB_{in}$$

$$= B_{in}(\bar{A}\bar{B} + AB) + \bar{A}B(B_{in} + \bar{B}_{in})$$

$$B_0 = B_{in}(A \oplus B) + \bar{A}B$$



FULL SUBTRACTOR USING NAND GATES

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$$D = \bar{A}B\bar{B}_{in} + A\bar{B}\bar{B}_{in} + \bar{A}B B_{in} + A B B_{in}$$

$$B_o = \bar{A}B\bar{B}_{in} + A\bar{B}\bar{B}_{in} + \bar{A}B B_{in} + A B B_{in}$$

$$\bar{A}B + \bar{A}B_{in} + B B_{in}$$

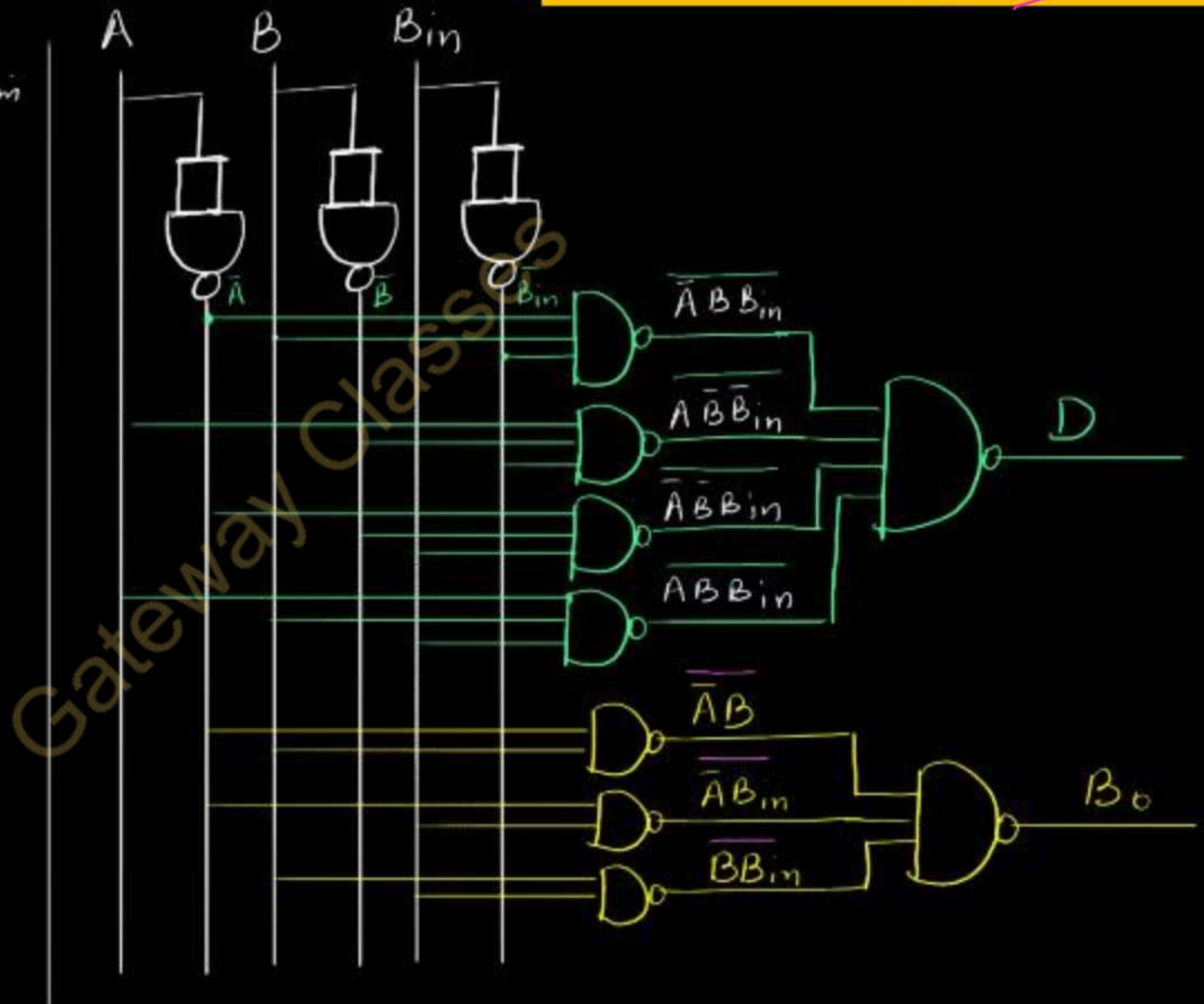
Double inversion

$$D = \bar{A}B\bar{B}_{in} + A\bar{B}\bar{B}_{in} + \bar{A}B B_{in} + A B B_{in}$$

$$D = \bar{A}B\bar{B}_{in} \cdot A\bar{B}\bar{B}_{in} \cdot \bar{A}B B_{in} \cdot A B B_{in}$$

$$B_o = \bar{A}B + \bar{A}B_{in} + B B_{in}$$

$$B_o = \bar{A}B \cdot \bar{A}B_{in} \cdot B B_{in}$$



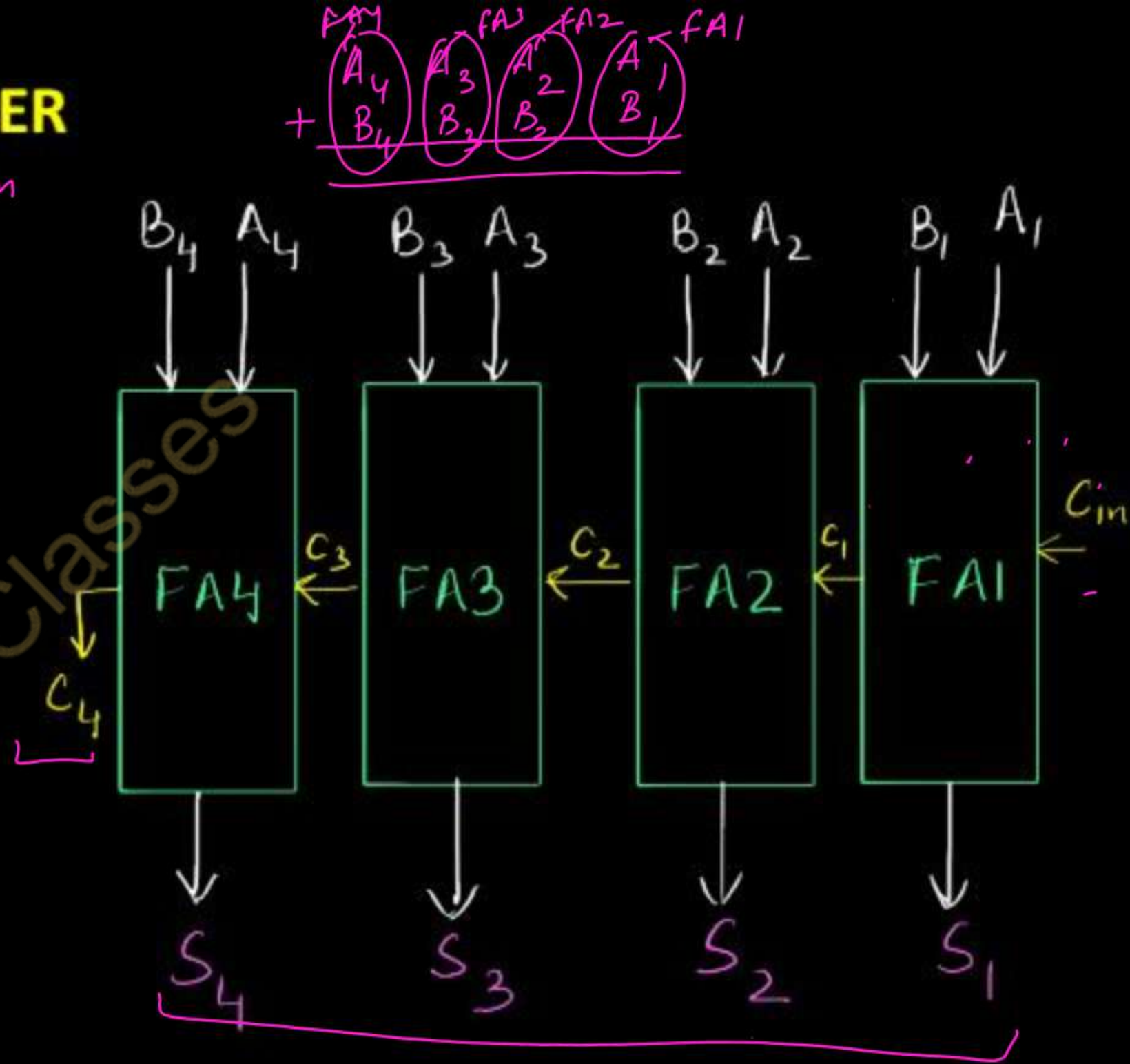
PARALLEL ADDER

A Full adder adds two single bits and a carry $\rightarrow C_{in}$ from the previous addition. It gives two outputs: a sum and a carry $\rightarrow C_{out}$.

A parallel adder adds two binary numbers that have more than one bit (like 4-bit or 8-bit numbers).

It adds all pairs of bits at the same time (in parallel) instead of one after another.

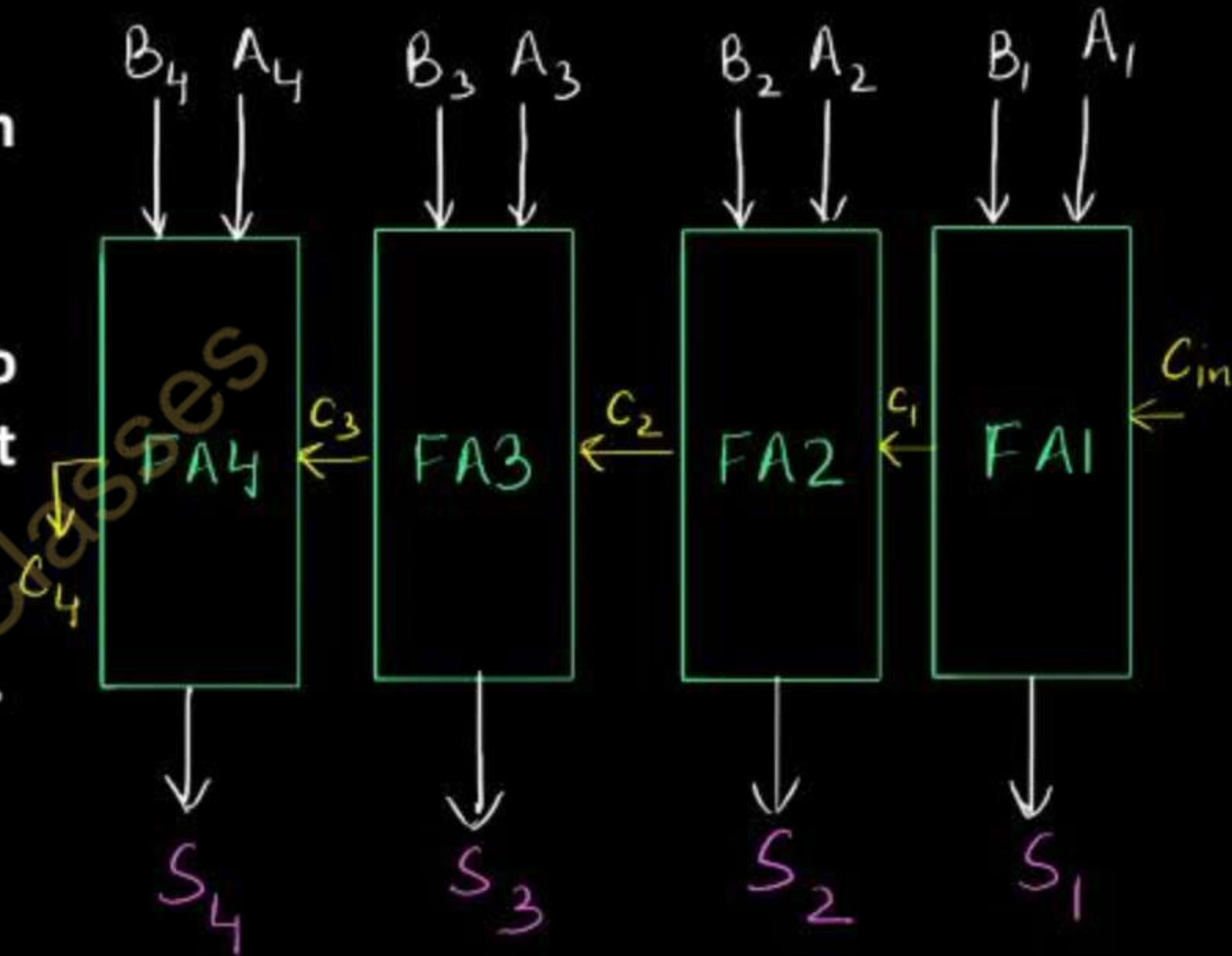
It is made by connecting many full adders in a row (chain).



Each full adder handles one pair of bits from the two numbers.

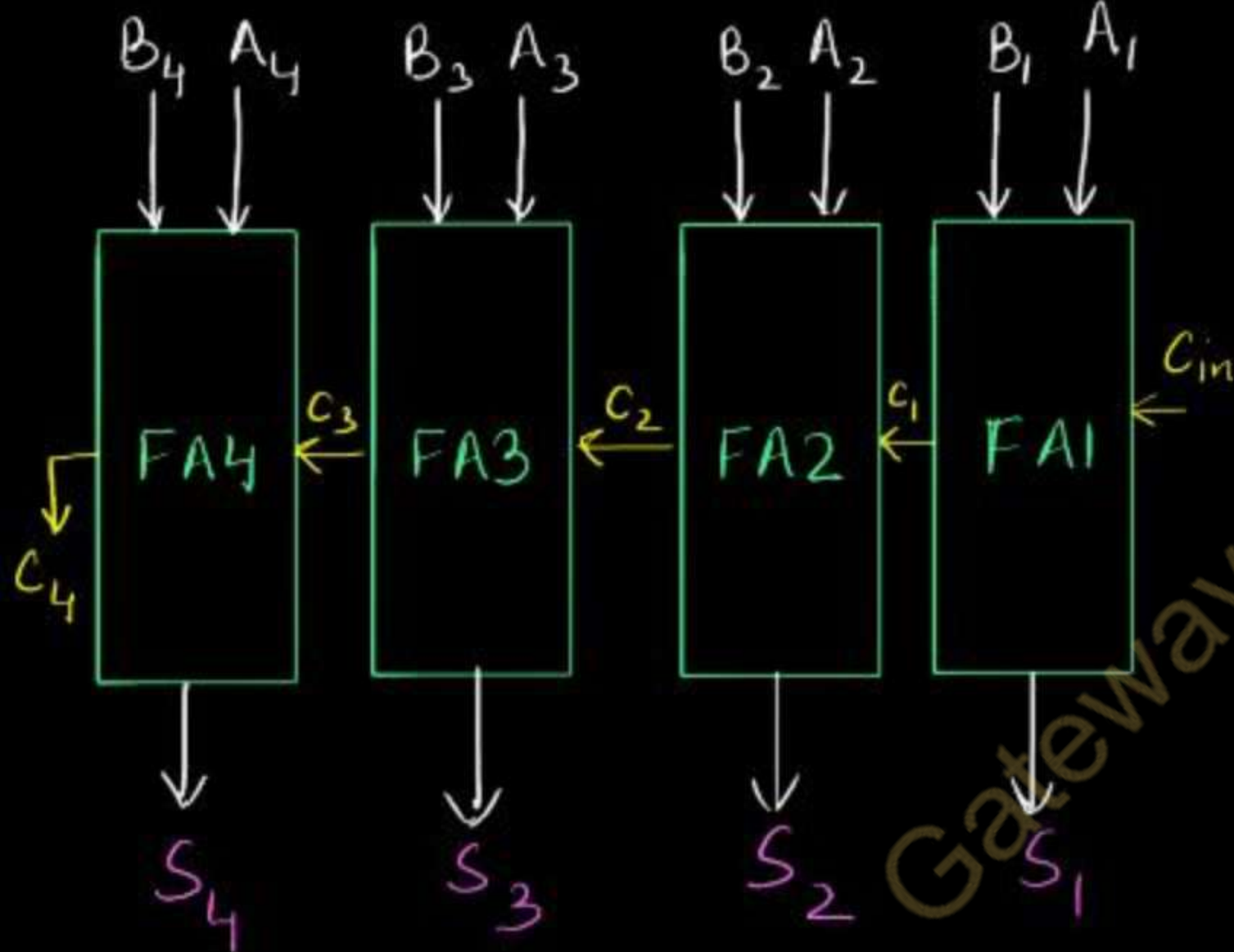
The carry output from one full adder goes into the carry input of the next full adder on the left (higher bit).

For an n-bit number, n full adders are required .



Design a 4-bit adder circuit using gates.

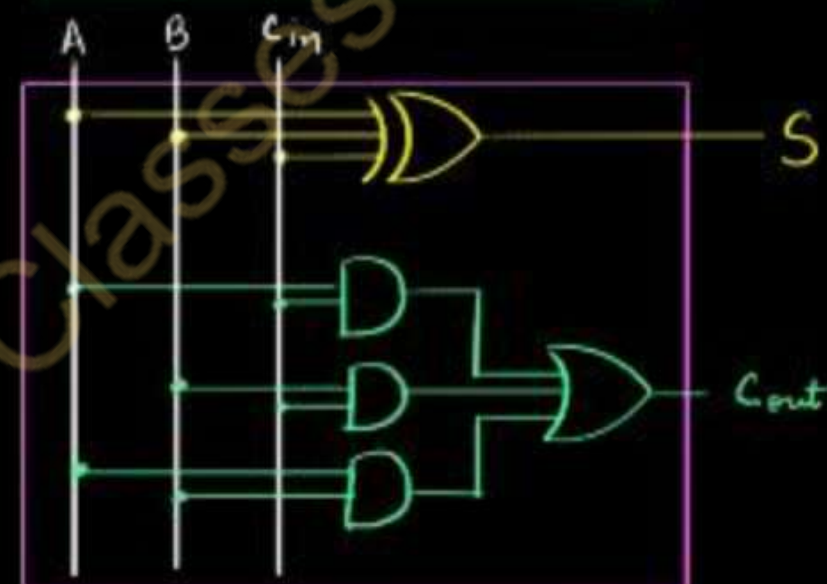
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For One Full Adder

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = AC_{in} + BC_{in} + AB$$



1 EXOR
3 AND
1 OR } are required

In total, for 4 bit adder

	Inputs	Outputs	EX-OR	AND	
FA1	A_1, B_1, C_{in}	S_1, C_1	1	3	1
FA2	A_2, B_2, C_1	S_2, C_2	1	3	1
FA3	A_3, B_3, C_2	S_3, C_3	1	3	1
FA4	A_4, B_4, C_3	S_4, C_4	1	3	1
			4	12	4

are required

SERIAL ADDER

Parallel Adder performs the addition at higher speed but requires a large amount of logic circuitry.

For example – If we wish to add 4 bits in a parallel adder then 3 Full adders and a Half adder of 4 Full adders will be required.

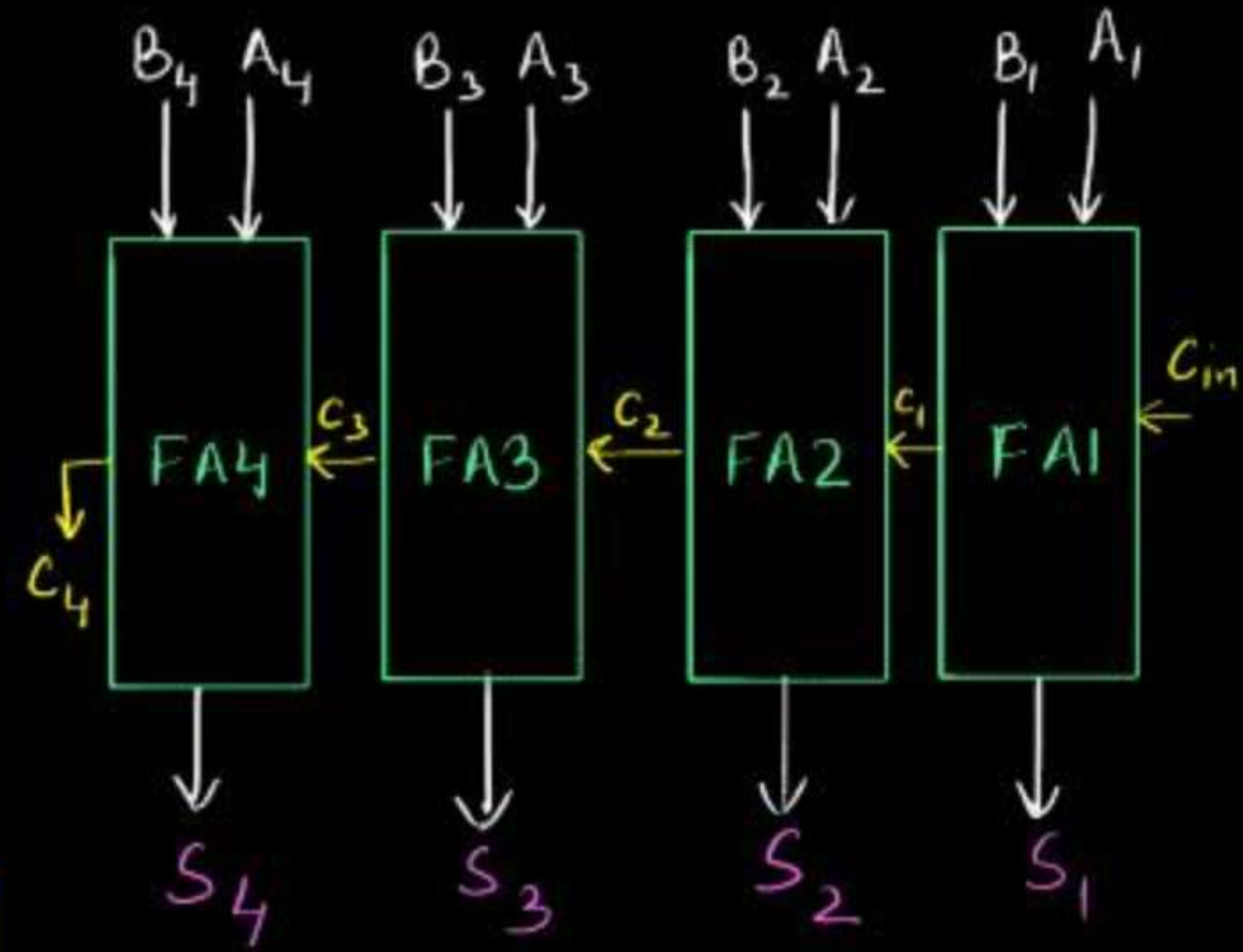
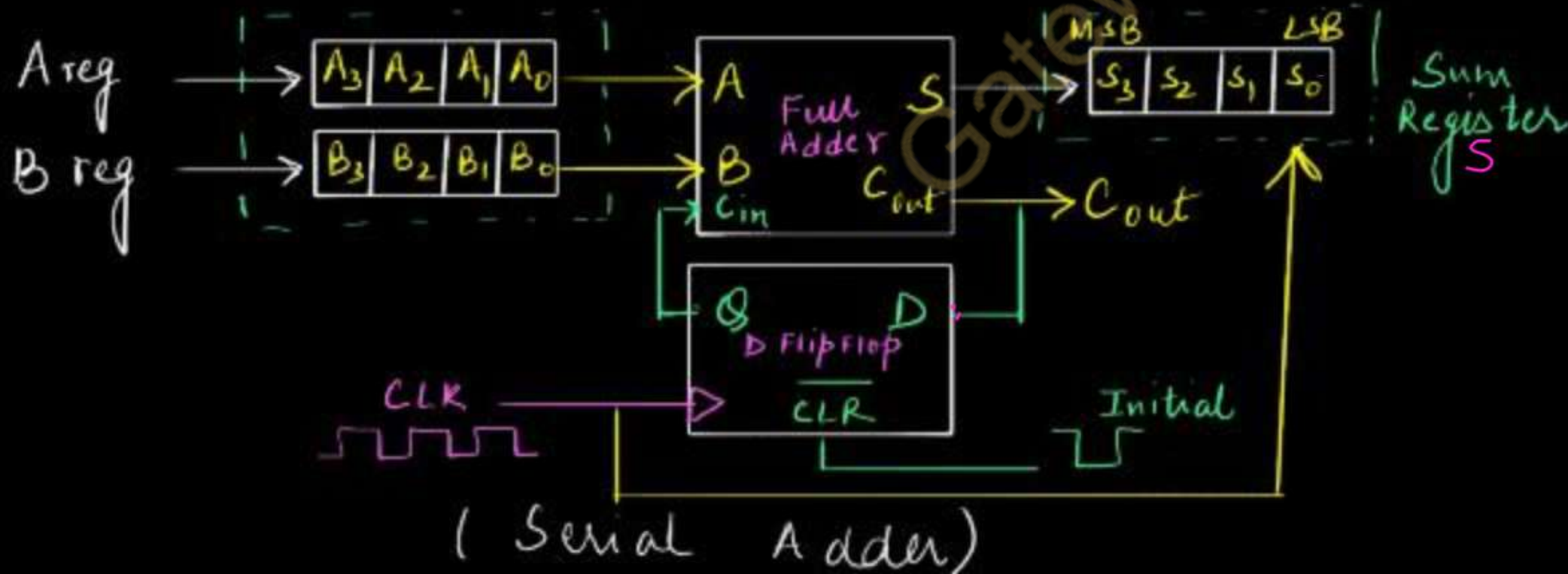
$A_3 \ A_2 \ A_1 \ A_0$
 $B_3 \ B_2 \ B_1 \ B_0$

On the other hand, Serial Adder performs bit-by-bit addition, so it requires simple circuitry than Parallel Adders. However it results in Low Speed of Operation.

For example – If we wish to add 4 bits in a serial adder then 1 Full adder will be required but speed will decrease.

Differentiate between Serial Adder and Parallel Adder.

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The two **Shift Registers A and B** are used to store the numbers to be added serially.

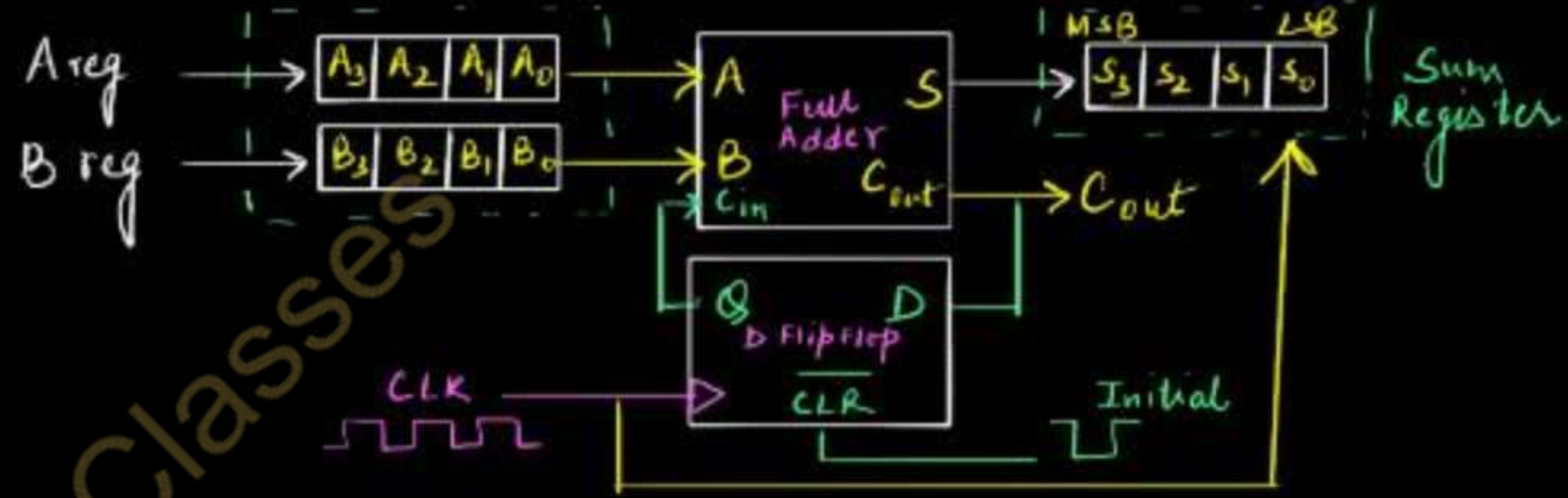
A single **Full adder** is used to add one pair of bits at a time along with the carry.

The contents of the Shift Registers A & B shift from left to right and their output starting from A_0 and B_0 are fed into a Single Full Adder along with the output of the carry Flipflop upon the application of each clock pulse.

The Sum output of the Full adder is fed to the MSB of the Sum Register .

The content of Sum register are also shifted to right when clock pulse is applied.

After applying four clock pulses the addition of two registers (A and B) contents are stored in Register S.



✓ V-imp ✓ BCD (Binary Coded Decimal) ADDER

AKTU PYQ-2024-25, 2022-23, 2021-22

The BCD-Adder is used in the computers and the calculators that perform arithmetic operation directly in the decimal number system.

It is also known as **Decimal Adder**.

The BCD-Adder accepts the binary-coded form of decimal numbers and requires a minimum of nine inputs and five outputs.

Logic 0-9 Binary and BCD numbers are same
 >9 BCD = Binary number + 6
= Binary number + 0110⁸⁴²¹

eg 18
Binary 10010^{16 8 4 2 1}

18 > 9

BCD 10010
+ 0110
11000

eg 11
Binary 01011^{16 8 4 2 1}

11 > 9

BCD 01011
+ 0110
10001

Truth Table

K	Binary Sum				BCD Sum					Decimal $\rightarrow$ (0-19)
	Z ₈	Z ₄	Z ₂	Z ₁	C	S ₈	S ₄	S ₂	S ₁	
0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	1	1
2	0	0	1	0	0	0	0	1	0	2
3	0	0	1	1	0	0	0	1	1	3
4	0	1	0	0	0	0	1	0	0	4
5	0	1	0	1	0	0	1	0	1	5
6	0	1	1	0	0	0	1	1	0	6
7	0	1	1	1	0	0	1	1	1	7
8	1	0	0	0	0	1	0	0	0	8
9	1	0	0	1	0	1	0	0	1	9
10	1	0	1	0	1	0	0	0	0	10
11	1	0	1	1	1	0	0	0	1	11
12	1	1	0	0	1	0	0	1	0	12
13	1	1	0	1	1	0	0	1	1	13
14	1	1	1	0	1	0	1	0	0	14
15	1	1	1	1	1	0	1	0	1	15
16	0	0	0	0	1	0	1	1	0	16
17	0	0	0	1	1	0	1	1	1	17
18	0	0	1	0	1	1	0	0	0	18
19	0	0	1	1	1	1	0	0	1	19

$$\begin{array}{r}
 01100 \\
 + 0110 \\
 \hline
 10010
 \end{array}$$

Binary Sum					BCD Sum					Decimal $\rightarrow (0-19)$
K	Z ₈	Z ₄	Z ₂	Z ₁	C	S ₈	S ₄	S ₂	S ₁	
0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	1	1
2	0	0	1	0	0	0	0	1	0	2
3	0	0	1	1	0	0	0	1	1	3
4	0	1	0	0	0	0	1	0	0	4
5	0	1	0	1	0	0	1	0	1	5
6	0	1	1	0	0	0	1	1	0	6
7	0	1	1	1	0	0	1	1	1	7
8	1	0	0	0	0	1	0	0	0	8
9	1	0	0	1	0	1	0	0	1	9
10	1	0	1	0	1	0	0	0	0	10
11	1	0	1	1	1	0	0	0	1	11
12	1	1	0	0	1	0	0	1	0	12
13	1	1	0	1	1	0	0	1	1	13
14	1	1	1	0	1	0	1	0	0	14
15	1	1	1	1	1	0	1	0	1	15
16	0	0	0	0	1	0	1	1	0	16
17	0	0	0	1	1	0	1	1	1	17
18	0	0	1	0	1	1	0	0	0	18
19	0	0	1	1	1	1	0	0	1	19

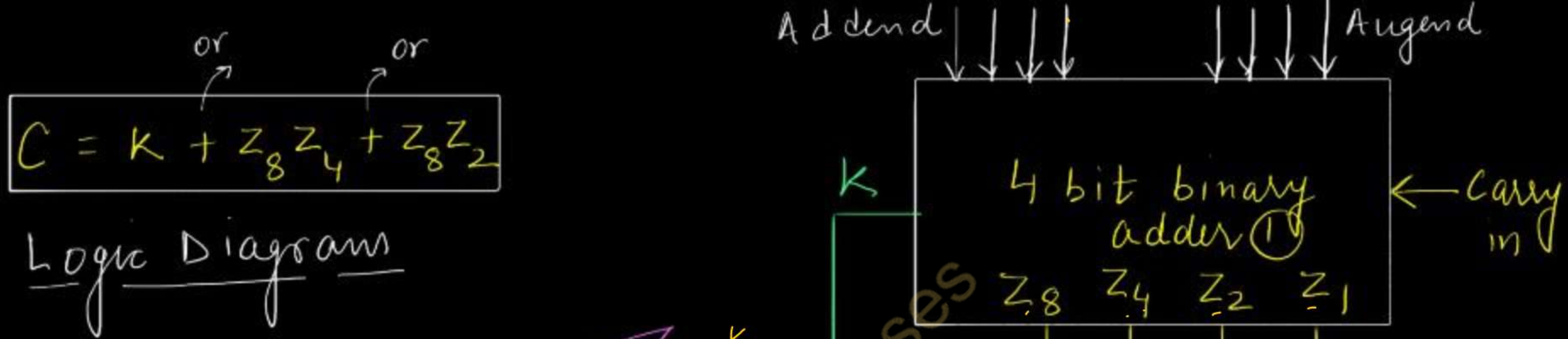
Analyze the table

C is 1 from 10 to 19

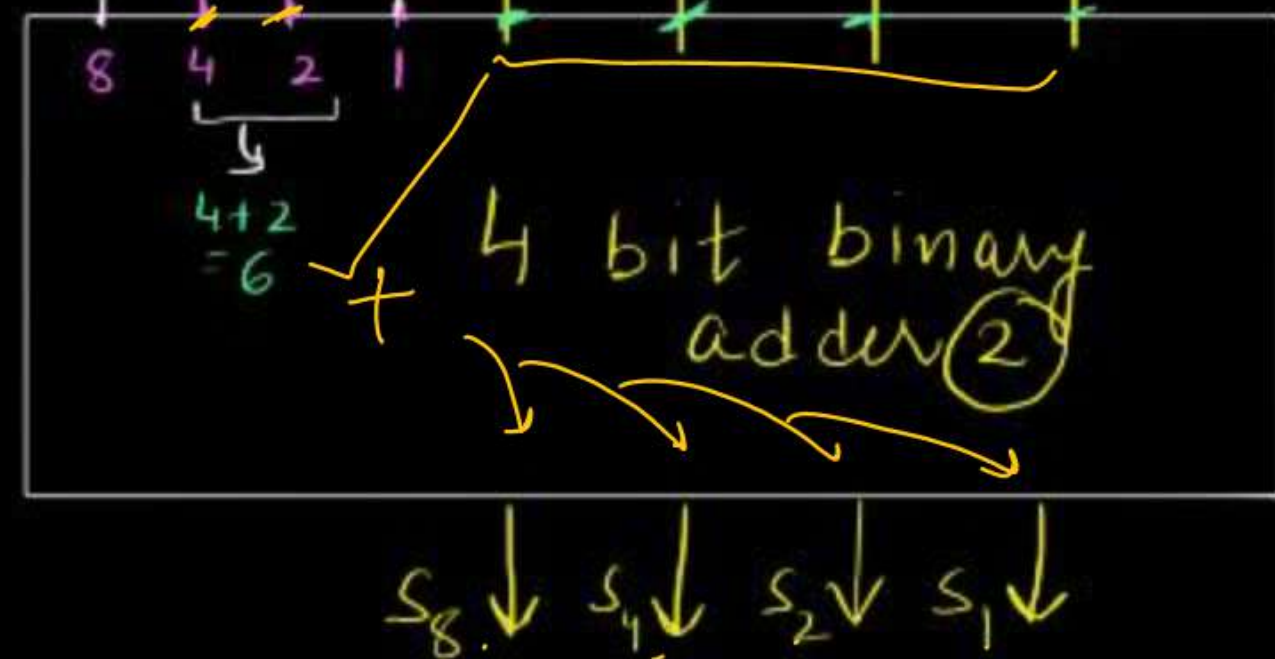
10-11 $\rightarrow Z_8 Z_2 = 1$
 12-15 $\rightarrow Z_8 Z_4 = 1$
 16-19 $\rightarrow K = 1$

or

$$C = K + Z_8 Z_4 + Z_8 Z_2$$



Whenever $C = 1$, 6 will be added to the sum from (1) to get BCD number at S_8, S_4, S_2, S_1 by (2)





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